

Gender-fair language in translation and post-editing

Insights and best practices

Manuel Lardelli

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Language Processing 26



Translation and Multilingual Natural Language Processing

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Manuel Lardelli

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Abbreviations

BLEU	Bilingual Evaluation Understudy
BLS	Bureau of Labor Statistics
CAT	Computer-Aided Translation
CERF	Common European Frame of Reference for Languages
DNL	Direct Non-Binary Language
DTS	Descriptive Translation Studies
ECC	Equality Evaluation Corpus
FAHQMT	Fully Automatic High-Quality Translation
GFL	Gender-Fair Language
HAMT	Human-Aided Machine Translation
INL	Indirect Non-Binary Language
ISO	International Organization for Standardization
LGBTQIA+	Lesbian, Gay, Bisexual, Transgender, Queer, Intersex, Asexual
MT	Machine Translation
MTPE	Machine Translation Post-Editing
MQM	Multidimensional Quality Metrics
NB	Non-Binary
NLP	Natural Language Processing
NMT	Neural Machine Translation
PE	Post-Editing
SD	Standard Deviation
SCT	Sexual Configuration Theory
TAP	Think-Aloud Protocols
TAUS	Translation Automation User Society
TER	Translation Edit Rate
TERF	Trans-Exclusionary Radical Feminist
TGBI	Translation Gender Bias Index
TM	Translation Memory
TNB	Trans and Non-Binary
TPR	Translation Process Research
TQA	Translation Quality Assessment
TR	Translation
TS	Translation Studies

1 Introduction

Gender bias in Machine Translation (MT) is a field of research that has gained popularity over the years, with a growing body of literature on issues of gender in MT (Savoldi et al. 2021). For instance, in 2018, examples of gender bias in Google Translate started to appear and be shared on different social media platforms. These images showed how Google’s MT engine systematically reproduced gender stereotypes when translating from Turkish, a genderless language, into English, a notional gender language, e.g. “o bir mühendis” became “he is an engineer”, while “o bir hemşire” became “she is a nurse” (Olson 2018). Studies on the topic can be mainly divided into two categories: The assessment of bias in commercial and academic systems, and debiasing methods that try to remove bias from MT outputs (Savoldi et al. 2021).

A common problem in this research field is that the term *bias* is rarely discussed and/or defined in publications. The term itself is polysemous (Crawford 2017) and is used differently across disciplines. Defining bias is not trivial because it implies a normative process for which something, in this case MT output, is deemed as harmful to certain groups of people (Blodgett et al. 2020). A literature review reveals how gender bias is often described as the reproduction of gender stereotypes, whilst masculine forms disproportionately occur in MT outputs (Savoldi et al. 2021). This application of the concept is influenced by assumptions on gender. For example, a survey of gender bias in the field of Natural Language Processing (NLP) demonstrates that gender is usually defined, explicitly or implicitly, as binary (Stanczak & Augenstein 2021). This is similar to MT research, where very few publications address non-binary genders (e.g. Dev et al. 2021, Burtscher et al. 2022).

The visibility of non-binary people has been increasing in the last few years. Numerous TV series, such as *One Day at a Time* and *Sex Education*, promote gender diversity. Celebrities like Sam Smith and Demi Lovato have come out as non-binary. Accordingly, the debate on gender-fair language (GFL) has attracted more and more attention, and strategies that were previously considered inclusive enough are now outdated. GFL usually subsumes two main approaches, i.e. gender-neutral and gender-inclusive (Sczesny et al. 2016, Hornscheidt & Sammla

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2021). Gender-neutral means that sentences are usually structured to avoid gendered elements, for example by using gender-neutral words, passive constructions and indefinite pronouns. Gender-inclusive describes the use of linguistic means, e.g. neopronouns (*hen* in Swedish), neomorphemes (*e* in Spanish), typographical characters (gender star (*) in German) and/or symbols (*ə* in Italian), to make all genders visible.

GFL poses great challenges for humans and MT alike. Strategies vary greatly among languages and cultures due to their grammatical structures, i.e. how they express and encode gender, and the dominant ideologies and beliefs in a society (see e.g. Lardelli & Gromann 2023b). In countries like Germany and Austria, where feminist discourse has a strong tradition, strategies have been, at least in part, proposed and used since the 1960s (Sczesny et al. 2016). Nowadays, the discussion around linguistic inclusivity also centres on genders beyond the binary. As a consequence, an increasing number of approaches and strategies have been proposed (see e.g. Hornscheidt & Sammla 2021). In other countries, for example Italy, where sexist attitudes are still dominant to a certain extent (Murgia 2021b), GFL is less actively used and often focuses on the representation of women. Nevertheless, non-binary language is gaining visibility, even though it is often harshly criticised (De Benedetti 2022).

With conceptualisations of translation shifting from being considered a simple process of cross-linguistic transfer to a complex process also involving culture and ideology, feminist work started to permeate Translation Studies (TS) from the 1970s onwards (von Flotow 1997). Recently, paradigms such as Queer Translation (Baer & Kaindl 2017a) have started to emerge. This is an interdisciplinary field that examines the intersections between translation and queer theory, focusing on how translation practices can both reveal and challenge normative understandings of gender and sexuality. It explores how translations can either obscure or highlight queer identities and experiences, and how the act of translation itself can be viewed through a queer lens. Such a queer lens questions fixed identities and categories, problematising the conventional notion of a clear-cut, hierarchical relationship between an original text and its translation, which influenced translation theory. Instead, it views translation as a dynamic and reciprocal process, where meanings are fluid and continuously negotiated (Spurlin 2014a, Baer & Kaindl 2017a). However, non-binary genders are still neglected in both TS and MT research. This is a highly problematic issue for various reasons.

First, misgendering, i.e. the act of referring to someone with gendered forms that do not correspond to their own gender identity, is a form of identity invalidation that can cause psychological distress and thus negatively impact the

mental health of trans and non-binary (TNB) people (McLemore 2015, 2018). Second, due to their recently gained visibility, TNB people are increasingly appearing in translations. Even with the best intentions, subtitled or dubbed TV series as well as media reports on non-binary individuals are brimming with incorrect and misgendering references (López 2022, Attig 2023, Lardelli & Gromann 2023b). Language has an impact on the visibility of genders and contributes to the (re)production of social norms and expectations. Misgendering or mistranslating leads to identity omission, invalidation and/or even ridicule because people might not understand what non-binary means. Given that MT is used on a large scale with millions of people accessing it every day for work, leisure, travel or understanding social media content, biased outputs can exacerbate the erasure and discrimination of non-binary people (Dev et al. 2021).

Most research on gender bias in MT tackles it from a purely technical perspective (Savoldi et al. 2021). This, however, ignores the social and (cross-)linguistic issues for the emergence of gender bias as well as their interrelation with technical aspects. A merely technical approach fails to account for the social impact of technology use, which exacerbates the oppression of marginalised groups (Bender et al. 2021, O’Neil 2017).¹ A more fruitful exchange between TS and the MT community would be beneficial because joint research endeavours would reveal a much more complex picture of gender bias in both human translation and MT.

The present work investigates the translation and post-editing (PE) of GFL beyond the binary. Twelve professional translators were recruited and asked either to translate or to post-edit machine translations of three brief English-language texts into German. The study was conducted online to aim at experimental realism, i.e. mirroring a real-life scenario in which participants could work in their usual environment. Participants completed a pre-study questionnaire with questions concerning socio-demographic information and their use of and previous experience with GFL. They were instructed to translate or post-edit each text using different strategies, namely (i) gender-neutral rewording, (ii) gender-inclusive characters, and (iii) neosystems.² Process data were collected by combining non-participant observation, screen recordings, and retrospective interviews. The translations and post-edited texts were also analysed. This multi-method approach provides insights into the cognitive processes involved in gender-fair translation and PE.

Similar studies in Translation Process Research (TPR) are usually conducted in a laboratory setting with strictly controlled variables (Carl et al. 2015, Jia et

¹The term *marginalised* is used to highlight that discrimination and exclusion occur because of unequal power distributions in society.

²Noesystems are new gender systems based on neomorphemes

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al. 2019a, Nitzke 2019). This was not the case with the present study. The main objective was to gain first insights into the translation and PE process but, more importantly, to put participants in a position to experiment with and gain insights into their personal experiences and points of view. The cross-linguistic expertise of translators is often neglected when tackling gender issues in MT. However, such expertise is fundamental to understanding the (cross-)linguistic implications of GFL use and investigating the feasibility and acceptability of different strategies. For this reason, while translation and PE times, as well as screen activity tracking data, provide interesting results, the core of the analysis is represented by the interviews during which participants expressed their opinions on the tasks and commented on the use of technology.

Attig (2023) calls for a community-informed translation practice where non-binary people are involved in the decision-making process concerning the selection of GFL strategies. This participatory model emphasises the importance of engaging with queer communities to ensure translations reflect authentic language practices, as demonstrated in the analysis of Spanish and French translations of non-binary pronouns in the sitcom *One Day at a Time*.³

Building upon this framework, the concept of *co-creation* offers a complementary perspective that deepens the collaborative nature of translation practices. For instance, Pérez-González (2013) explores how digital media and technological advancements have shifted the nature of subtitling, creating new collaborative and transformative practices. The article argues that fansubbing and other participatory subtitling activities demonstrate a move away from traditional, industry-controlled subtitling and towards a co-creative model where users actively participate in the creation and adaptation of media content.

Co-creation therefore involves stakeholders as equal partners throughout the design and decision-making processes, fostering a shared vision and iterative collaboration. In the context of TS, this approach aligns with participatory action research methodologies that include experiential experts, such as queer and non-binary individuals, in the translation process. Gromann et al. (2023) illustrate this through a case study focusing on German MT, highlighting the importance of context dependency and the desire for customizable solutions to avoid identity invalidation.

Integrating co-creation into community-informed translation practices ensures that non-binary individuals are not merely consulted but are active collaborators in the translation process. This approach acknowledges their expertise and lived experiences, leading to translations that are more inclusive and rep-

³[https://en.wikipedia.org/wiki/One_Day_at_a_Time_\(2017_TV_series\)](https://en.wikipedia.org/wiki/One_Day_at_a_Time_(2017_TV_series))

representative. By embracing co-creation, TS can move toward more equitable practices that honour the diversity of gender identities.

Although the involvement of non-binary people was marginal – as a matter of fact, only one non-binary translator volunteered to take part in the study – this work attempts to promote a change in research practices when tackling issues of gender bias and non-binary representation by involving a group whose expertise and role in circulating ideas are fundamental at a societal level, i.e. professional translators.

The main findings from this study show that neosystems were the most difficult strategy to use, with a considerable increase in screen activity tracking, perceived cognitive effort, and the number of mistakes made when they were applied. In terms of time, PE was found to be faster than translation, but no statistically significant difference emerged among assignments. Finally, participants clearly indicated a preference for a combination of gender-neutral rewording and gender-inclusive characters. The present work provides insights into the strengths and limitations of different strategies as well as into particularly challenging gender-specific terms, which can be of inspiration for MT debiasing and further research on gender-fair MT.

The present work is structured as follows:

- Chapter 1 presents the research questions, assumptions, motivation, and objectives underlying this research.
- Chapter 2 delves into the theoretical framework of the present work, which is highly multidisciplinary. First, this monograph is set in the context of feminist and queer TS, which are the result of shifts in the conception of translation from a mere linguistic transfer to a complex process of cross-linguistic, cross-cultural, ideological and political negotiation. Second, relevant concepts from gender studies, such as sex, gender, and gender identity, are introduced. The objective of this chapter is to create awareness that gender is a biopsychosocial construct and that there are numerous gender identities beyond man and woman. Finally, gender as a linguistic feature is introduced and the interplay between language, culture, and society is described.
- Chapter 3 is dedicated to gender issues in MT: Gender bias and its sociotechnical implications are discussed. Subsequently, the second section of the chapter focuses on PE as an increasingly common practice in the translation industry.
- Chapter 4 is dedicated to an overview of linguistic strategies in English, German, and Italian. Language-independent good practices for gender fair-

ness are also briefly mentioned. Finally, related work from TS and MT research, as well as work that combines both disciplines is summarised.

- Chapter 5 details the research methodology. It introduces the multi-method approach used for the investigation of the research object and focuses on the experiment preparation, data collection, and analysis.
- Chapter 6 provides an overview of the main findings. It briefly describes the participants' profiles, then concentrates on the analysis of translation and PE times, screen activity tracking data, participants' experience, the use of strategies during the experiment, and the bias in the machine translations post-edited in the study.
- Chapter 7 discusses the results. First, the different approaches, i.e. gender-neutral rewording, gender-inclusive characters and neosystems, and the translation and PE process are compared. Second, the implications for language professionals and MT research are addressed. Finally, the author's positionality and the limitations of this study are described.
- Chapter 8 presents a concise overview and a concluding analysis of the present work.

Following the introduction of the book topic, the research problem, and the structural outline of the work, the following section presents the research questions and the primary assumptions underpinning this study.

1.1 Research questions and main assumptions

This study represents the first endeavour to investigate GFL beyond the binary in human translation and MT. It is both a process and product-oriented study partially inspired by work in TPR. The following research questions were devised:

- Q1: How do translators apply different GFL strategies in translation and PE and what are the cognitive processes involved?
- Q2: To what extent is the integration of GFL in translation and PE successful?
- Q3: What are the differences in effort between gender-fair translation and PE?

The first question can also be divided into three sub-questions, namely:

- How do professional translators perceive the task of using GFL? What are the differences between strategies and what are the advantages and disadvantages of each?

- To what extent do translators' subjective impressions of and experience with GFL correlate with other indicators of cognitive processes, i.e. translation times, and screen activity tracking data?
- Which gender-specific terms are particularly challenging in the translation and PE process?

The main assumptions concerning the findings of this study are that language professionals use GFL differently, e.g. they select different neutralisation strategies or gender-inclusive characters. In terms of cognitive processes, neosystems are expected to be more challenging than the other strategies because they are largely unknown and actively change and disrupt traditional grammar rules. It is also expected that translators make more mistakes with them than when using gender-neutral rewording and gender-inclusive characters. Longer translation times should correspond to a higher perceived cognitive effort, and PE should be faster than translation. Different gender-specific terms are expected to be challenging based on the strategy applied, e.g. personal pronouns and professions when using gender-neutral rewording, and pronouns and articles when using gender-inclusive characters. Finally, terms specific to a gender, e.g. *Nanny* in German, are expected to cause greater difficulties in the translation process.

The present work is also influenced by more general assumptions that affected its realisation. Gender identity terms used in this book reflect Western ideologies, current views of gender, and an Anglocentric bias in research and academia. Identity terms and views on gender are different in non-Western languages and cultures (Lester 2018); these are not addressed in the present work. I also acknowledge that the views reflected in this work are not only the result of a social construction but they also change over time. Labels and language used here may become outdated in the future.

Lastly, I also recognise that GFL is an ideologically/politically charged and often controversial topic. Since one of its objectives is to undermine or overturn the status quo by making marginalised groups visible and prominent in language, GFL use tends to cause strong reactions of outcry and mockery (Vergoossen et al. 2020, De Benedetti 2022). While people who support GFL use tend to be accused of promoting a specific political and ideological agenda, its opposers are not. Therefore, I refuse to discuss non-binary representation in language from an ideological point of view, concentrating instead on challenges of a (cross-)linguistic nature. I maintain that using gendered forms which correspond to a person's gender identity is a matter of respect and not of ideology. Nevertheless, I will briefly touch upon my positionality as a researcher in Chapter 7.

1.2 Research relevance and objectives

The present work is highly relevant to both translation studies and practice as well as MT. Research on gender bias often neglects non-binary individuals. This represents a considerable research gap that this study tries to address. Queer TS predominantly focuses on sexuality, e.g. the translation of gayspeak, or on trans people who identify with either the male or the female gender (Baer & Kaindl 2017a). This is a significant limitation from a theoretical and empirical perspective.

From the point of view of the translation industry, audiovisual translation continues to grow as a field and non-binary characters are increasingly depicted in TV series that are dubbed and subtitled in many languages. In the next few years, it can be expected that translation professionals will be increasingly confronted with non-binary language and thus need to be able to reproduce the meaning of the source text without erasing gender identities beyond the binary. Unfortunately, translations are still brimming with misgendering (López 2022, Attig 2023, Lardelli & Gromann 2023b). As language technologies are increasingly integrated into translation workflows, language professionals should be aware of the biased outputs produced by MT and be able to adjust them.

Research on gender bias in NLP and MT often adopts an implicit binary conception of gender. While some studies acknowledge that gender is not binary, gender issues are tackled from a binary perspective due to the lack of groundwork (Sun et al. 2019, Stanczak & Augenstein 2021). TS and NLP should collaborate on gender issues that are common in human and MT. Integrating GFL in MT while also taking into account the multiplicity of the available strategies is not possible without the perspective of language professionals who have a greater awareness of (cross-)linguistic issues resulting from the adoption of specific strategies.

Finally, the present work is also relevant on a broader societal level. Greater awareness of gender and linguistic issues as well as the use of GFL can further promote gender equality (Sczesny et al. 2016). Dismantling old-fashioned gender norms, stereotypes, and expectations may have a positive impact on the lives of many people who daily experience forms of discrimination and oppression because they do not strictly adhere to said norms, stereotypes, and expectations (Barker & Iantaffi 2019). Academia and language professionals can become ambassadors for linguistic and societal change.

Having considered the research relevance, the present work pursues the following objectives: (i) gain insights into the translation and PE process concerning different GFL strategies, (ii) provide a methodological framework and materials to conduct further studies in other language pairs, (iii) create and share

novel datasets with non-binary language that can be used for further research endeavours, (iv) contribute to the exchange on gender bias between TS and computational linguistics/MT, and (v) produce a tentative best practice model for language professionals, which can also provide inspiration for debiasing techniques in language technologies.

1.3 Philosophical stance and research approach

Saldanha & O'Brien (2013: 10) advocate for critically examining one's worldview before embarking on a research project. This is fundamental in the selection of appropriate research methods, the validation of the findings, and the (self-)understanding of the researcher's motivation. Considerations of what knowledge is and how it is produced are therefore essential and relate to two complementary positions, the ontological and the epistemological.

Ontology is the branch of philosophy concerned with the nature and relations of being. In social research, ontology pertains to "the way the social world is seen to be and what can be assumed about the nature and reality of the social phenomena that make up the social world" (Matthews & Ross 2010: 23). Epistemology, on the other hand, concerns the theory of knowledge acquisition and production (Matthews & Ross 2010: 23). Three different ontological positions may be identified, i.e. objectivism, constructivism, and realism, which are broadly associated with three corresponding epistemological positions, i.e. positivism, interpretivism, and realism.

Objectivism posits that social phenomena exist independently of social actors (Bryman 2016: 33). This position is usually linked to positivism, which conceives science as value-free, objective, based on quantitative data and rigorous hypothesis-testing (Bryman 2016: 28). Constructivism "implies that social phenomena and categories are not only produced through social interaction but that they are in a constant state of revision" (Bryman 2016: 33). It is logically connected to interpretivism, which "prioritises people's subjective interpretations and understandings of social phenomena" (Matthews & Ross 2010: 28). The constructivist-interpretivist position acknowledges that research is a culturally and socially situated practice, privileging qualitative data and understanding (Bryman 2016: 30). Finally, realism, both as an ontological and epistemological position, recognises that social phenomena can have a reality independent of social actors, but the scientist's conceptualisation is just one way of knowing them (Saldanha & O'Brien 2013: 11, Bryman 2016: 29).

Finally, two basic orientations to research can be adopted, namely inductive and deductive. The inductive approach, often associated with qualitative

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research, involves deriving general principles from specific instances or observations. This method allows for the exploration of patterns and the generation of theories based on empirical data, offering a bottom-up perspective (Saldanha & O'Brien 2013: 14-15). Conversely, the deductive approach is characterised by a top-down methodology where hypotheses are formulated based on existing theories and subsequently tested through empirical observation. This approach is typically linked to quantitative research and aims to validate or refute theoretical constructs, providing a structured framework for testing specific translation hypotheses (Saldanha & O'Brien 2013: 14-15).

The present work and the author's personal beliefs are influenced by feminist theory, which asserts that knowledge and truth are situated in a specific societal, cultural, and historical context that inevitably shapes and influences them (Haraway 1998). This is in line with the social-constructivist stance and the conceptualisation of translation as a socially and culturally embedded practice. However, the methodological approach employed in this study is mixed methods. Even though quantitative and qualitative methods are often seen as mutually exclusive, their distinction is not always clear-cut and their combination might be fruitful to understanding the research object (Luker 2009: 5, Bryman 2016: 37). Mixed methods make it possible to analyse the translation process, the translation product and the social context, which in turn allows for a triangulation of data from different perspectives. The first perspective is the objective one and relates to the time needed to complete translation and PE assignments as well as the activities carried out, such as pauses and searches. The second is more subjective and it relates to the personal experience of language professionals. The analysis of data, for instance the coding process, followed an inductive approach. The result is a robust work that provides numerous insights into GFL translation and PE in the language pair English-German.

This chapter outlined the topic of the present monograph, namely the translation and PE of GFL to promote respectful representation of non-binary individuals. It also emphasised the underresearched nature of the topic in both TS and computational linguistics, underscoring the relevance of this work. Following an overview of the structural outline, the research questions, objectives, and philosophical stance underpinning this study were discussed. The theoretical background is presented in Chapters 2, 3, and 4.

2 Gender issues in translation studies

This chapter outlines the theoretical background of the monograph within TS, situating the work in the field of feminist and queer translation. Section 2.1 discusses the relevance of the present work for the disciplines and briefly conceptualises translation as a queer and social practice. The key concepts introduced here are further elaborated on in Section 2.2: A brief overview of the developments within TS that led to the rise of feminist translation is traced and activism in translation is discussed. In addition, key questions from the field of gender studies, such as sex, gender and gender identity, are also explored. Attention is paid to how categories of sex and gender are socially constructed and still often conceived as binary, even though research in different fields (e.g. Fausto-Sterling 2012, Richards et al. 2016) demonstrates that there are more than two sexes and genders. An overview of the nature and types of non-binary genders is therefore provided. In Section 2.4, the interplay between language, society, and gender is introduced. Languages are the result of cultural conventions, and differences in their structures can lead to social and cultural differences (see e.g. Prewitt-Freilino et al. 2012, Santacreu-Vasut et al. 2013) that both reflect and impact assumptions on sex and gender (see Boroditsky et al. 2003, Stahlberg et al. 2007). Thus, it is posited that language (re)produces norms, social expectations, and power relations and can be used both to discriminate as well as to promote equality (AG Feministisch Sprachhandeln 2015).

2.1 The social responsibility of translation

In his seminal work, Venuti (1995) uses the term “invisibility” to describe the situation and activity of translators within Anglo-American culture. Venuti illustrates how translated texts are judged by their fluency, implying that a translation is considered good when it is not perceived as such, but rather as an original work. This perspective carries significant implications. The translation is seen as a derivative, fake, and potentially false copy. Consequently, there has been a long-standing emphasis within TS on the notion of *fidelity* to the source text and its author (see Pym 2017). Furthermore, Simon (1996: 1) points out that the language

used to describe translation often “dips liberally into the vocabulary of sexism, drawing on images of dominance and inferiority, fidelity and libertinage”. For example, the phrase “les belles infidèles”, once commonly used to describe translations, reflects the expectation that a target text should either be beautiful or faithful, echoing societal perceptions of women (Simon 1996: 10).

While calling into question the position of discursive inferiority of women and translations, feminist TS understand language as having a performative nature (Simon 1996: 2). The idea of performativity is also at the core of Butler’s (1990) theorisation of gender as socially and discursively constructed. By asking how social, sexual and historical differences are expressed and how they can be transferred across languages, Simon (1996) supports sociolinguistic views of language as actively creating meaning and contributing to reality (Fairclough 2015). Being a socially and culturally embedded practice (Wolf 2007), translation can either uphold or challenge the dominant regimes of knowledge and power. Tymoczko (2010b), for example, outlines how evolving conceptualisations of translation from a mere substitution of linguistic code to an ethical, ideological, and political practice led to considering translators as agents for social change. Translators are sometimes even caught between the two aforementioned positions, representing both institutional power and people seeking empowerment. Translators therefore act “as a kind of double agent in the process of cultural negotiation” (Tymoczko & Gentzler 2002: xix) and their double role might be considered a strength that is indicative of the complexity of contemporary cultural hybridity.

From a theoretical point of view, feminist and queer translation contributes, together with other paradigms such as postcolonial translation, to the deconstruction of rigid binary oppositions prevalent in TS, such as original/copy and author/translator. Translation thus becomes “a site of struggle in the negotiation and production of meaning” (Spurlin 2017: 176) that shapes social and cultural identities. This perspective underscores the social responsibility of translators, defined by Drugan & Tipton (2017: 122) as “action-oriented and dynamic, encompassing value judgements and decisions that may lead as much to resistance as to acceptance and commitment to sustain a form of social consensus”. However, social responsibility remains an understudied concept in relation to the translation of GFL.

From an empirical perspective, feminist and queer translation aims to give visibility to the work of feminist and queer translators today and throughout history, address how the categories of sexual and gender identities are the results of social construction, and demonstrate how texts can be translated either to support or to question the status quo (Simon 1996, von Flotow 1997, Burton 2010,

Epstein 2017). For instance, feminist and queer translation practices can critically highlight hegemonic, sexist, homo- and transphobic views expressed in texts or “invert” (Burton 2010: 57) them.

The present monograph adopts a feminist and queer approach to translation, conceptualising it as a cultural, political, and ethical negotiation process. This understanding is rooted in the idea of translation as a *queer practice* characterised by “the study of borders, and their deconstruction and rearrangement” (Spurlin 2017: 172). According to Spurlin, translation disrupts and reconfigures the imaginary boundaries between concepts such as man/woman and author/translator, replacing dichotomies with continuums or spectrums. In other words, translation creates crossings across linguistic borders and social categories, producing new, hybrid forms of meaning and new knowledge through these very encounters, which may result in questioning the actual borders of language and cultures (Spurlin 2017: 173). This is exactly what language professionals do when they translate non-binary GFL.

This work aligns with the paradigms of feminist and queer translations for different reasons. First, it investigates the translation and PE of texts referring to non-binary individuals, a marginalised group often overlooked in TS research. Second, it demonstrates how translation, as a feminist and queer practice, engages in a cultural, political, and ethical process of negotiation. Language professionals have the power to either erase gender diversity and make it fit societal gender binary norms or to represent it. Such a decision implicates social responsibility. Misgendering translations may lead to identity invalidation, with detrimental effects on the mental health of transgender individuals (McLemore 2015, 2018).

While English is mostly gender-neutral, German requires constant gender inflection and it has no equivalent to singular *they* for reference to a hypothetical person, people of unknown gender, or non-binary individuals. Language professionals are thus required to experiment with different, even new and disruptive, GFL strategies. Their use carries ideological and political connotations in German that might not be present in the English source text. It might therefore be possible that language professionals have to explain and/or justify their linguistic choices. The translation process thus becomes visible, with professionals effectively engaging in an activist agenda by serving as spokespeople for a marginalised group. They mediate among and negotiate with numerous co-textual and contextual factors, such as clients’ specifications, personal and societal views and/or beliefs, and target audience amongst others.

Observing the translation and PE process and discussing it with language professionals contribute towards bringing translators to the forefront of the dis-

course around gender fairness in language and translation, showcasing their role as mediators and agents for social change. This is all the more important because this research expands the scope of activist, feminist, and queer translation beyond literary contexts to encompass fields like audiovisual and news translation, which have received less attention in the literature (Baer & Kaindl 2017b, von Flotow & Kamal 2020).

MT is also relevant to (queer) TS. Language professionals are constantly confronted with technology and increasingly receive PE assignments. MT systems, trained on large amounts of data that reflect a majority language use, exhibit gender bias (Savoldi et al. 2021). As a widely used translation technology, MT propagates hegemonic views on gender, discriminating against marginalised groups. It is therefore important that feminist and queer translation addresses this issue and engages in multidisciplinary endeavours to mitigate it. Furthermore, the propagation of bias in MT substantiates the idea that translators can act as agents for social change in a process of social, ethical, and political negotiation.

2.2 Shifting translation concepts: The emergence of culture and ideology

Gender issues have been of interest since second-wave feminism not only in the field of linguistics but also in TS (Simon 1996, von Flotow 1997). This is also due to the development of the discipline which has emancipated itself from linguistics and flourished in a well-established research area. In the following paragraphs, I present a brief overview of the changes in the conceptualisation of translation practice that led to the emergence of paradigms such as feminist and queer translation.

In Western Europe, translation has long been defined mainly as a simple process of cross-linguistic transfer. During World War II, translation was conceptualised as an instrument to decipher the linguistic and cultural codes of both enemies and allies and as an instrument of propaganda (Tymoczko 2010b: 3). This conceptualisation is also at the core of the first MT systems (Hutchins 2014) and it is specifically from the interest in automated translation (and its difference from human translation) that TS emerged as an academic discipline.

At the time, TS was strongly influenced by linguistics and the debate on translation revolved around the concept of equivalence (Munday et al. 2022: 59). The process of translation tended to be seen as a mere substitution of a linguistic code through another and the fundamental criterion to establish the quality of translations was their fidelity and equivalence to a source text (see Pym 2017

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for an overview of translation theories and their development). During this period, the work of scholars such as Jakobson (1959), Nida (1964), and Newmark (1981) was particularly influential. Jakobson was influenced by structuralism and focused on cross-linguistic differences in grammatical and lexical forms, noting that “[l]anguages differ essentially in what they must convey and not in what they may convey” (Jakobson 1959: 129). An example of such differences is gender. For instance, Romance languages must convey gender in nouns, whereas English is mostly gender-neutral. Nida (1964) borrowed concepts and terminology from semantics and pragmatics, and distinguished between formal and dynamic equivalence (Munday et al. 2022). The first focuses on the form and content of the message, whereas the second on the effect, thus introducing a reader-based orientation to translation theory. Newmark (1981) was influenced by linguistics as well and proposed two types of translation, semantic and communicative. They can respectively be compared to Nida’s formal and dynamic equivalence.

By the 1970s, two different paradigms gained prominence within TS, i.e. functionalist approaches and system theories. For instance, Reiß (1971) expanded the concept of equivalence to the text level and focused on the analysis of text types and genres. Furthermore, Reiß & Vermeer (1984) and Nord (1988) advocated for a more dynamic understanding of translation processes, emphasising the importance of context and purpose in translation and paving the way for a more nuanced analysis of translation practices. A further important shift in this direction was polysystem theory (Even-Zohar 1990), which considered literature as part of the cultural, literary, and historical system of the target language.

The 1980s witnessed the rise of Descriptive Translation Studies (DTS) as a paradigmatic shift within the field. Scholars such as Toury (1995) spearheaded this movement, advocating for an empirical approach to translation research that focused on describing translation practices as they occur in real-life contexts. DTS abandoned prescriptive approaches and aimed to describe translation products and texts according to their cultural and political coordinates (Toury 1995, Pym 2017: 64–70), paving the way for considering culture as a significant analytical category together with language in the 1990s.

At that time, Snell-Hornby (1990) attested to a cultural turn within the discipline, moving the conceptualisation of translation as text to translation as culture and politics (Munday et al. 2022: 198). Translation was accordingly redefined as a type of intercultural transfer (Bassnett & Lefevere 1990, Venuti 1992, 1995, Tymoczko 2010b: 5). This led translation theorists to reflect on the role of translation as an instrument of power and on its ideological aspects (Tymoczko 2010b: 3–4). Toury (1982: 25), for example, claimed that “translations are facts of one system only: the target system”, meaning that translation strategies, norms, functions,

and purposes are determined by the culture in which they are produced. Thus, translations have political and ideological implications.

Ever since, attention has been turned to issues of power, ideology, dominance, norms, and ethics in translation. The decision to translate (or not) entire texts as well as translation shifts would reveal historical and geopolitical patterns of target cultures as well as political constraints or initiatives underlying the choices made by translators (Tymoczko 2010b: 6–7). The decision to erase references to homosexuality could be interpreted as reproducing widespread homophobic views circulating in society. Conversely, the decision to translate literary works by women that are not considered part of the traditional literary canon may be seen as a political decision to oppose a patriarchal system that excludes women from many fields like art and literature.

This cultural turn, characterised by the dismissal of linguistic theories of translation (Bassnett & Lefevere 1990), led to a stronger image of translators as active agents. Translation started to be seen as an instrument of resistance and activism (Tymoczko 2010b: 7). Venuti (1992, 1995) called translators to action, urging them to become visible, i.e. to become involved in cultural and ideological matters. Translators were regarded as “agents for social change” (Tymoczko 2010b: 3), using translation to actively change societies, whilst translation began to be considered as an ethical, ideological, political practice – far more than a simple matter of linguistic transfer. This conceptualisation shift set up the basis for feminist TS, as explained in Subsection 2.2.1. Accordingly, from the late 20th century onwards, attention has been paid to the role of activist translators and their impact on society (Tymoczko & Gentzler 2002, Tymoczko 2010b).

Throughout the years, activist translators have dealt with matters such as European colonialism, imperialism, the Hispanic revolution in America (Tymoczko 2010a: 227), sexism (von Flotow 1991, 1997), and discrimination against the LGBTQIA+ community (Burton 2010). The last two points are of particular interest for the present work, as queer translation (Spurlin 2014a, Baer & Kaindl 2017a) can be seen as an extension of feminist translation, and it represents the positioning of the present work within the framework of TS. Translators have been coping with oppression brought about by dictatorships, colonialist regimes imposing social hierarchies, military oppression as well as with revolutions, nation-building, gender liberation, and supporting shifts in cultural values (Tymoczko 2010a: 229). To do so, they have adopted several different strategies.

Tymoczko (2010a: 229–235) posits that activism in translation can be achieved by a wide variety of strategies, also highlighting how these are situated in a specific culture, space and time. Struggles and protests can change rapidly, so translation must be constantly adapted to shifting cultural, historical, and political

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constellations. Translations and strategies are therefore valid in and specific to a certain period (Tymoczko 2010a: 234–235). To summarise, activism in translation can be performed through:

- Zero translation: This is the refusal to translate a text or parts of it, hence refusing to transmit cultural information, e.g. in Hawaii, translations have been refused to actively oppose colonial authority (Aiu 2010).
- Choice and manipulation of text types and/or genres, e.g. Monteiro Lobato inserted political messages in children's books during Getúlio Vargas' dictatorship in Brazil (Milton 2006).
- Text amplification: Translators add comments and paratextual materials so as to explain their political and ideological views. Such strategies have often been adopted in feminist translation (an overview can be found in von Flotow 1991).
- Text simplification: This is often done so that texts are fit for specific purposes.
- The insertion or the refusal of foreign elements into a translation.

There is no single activist strategy for all situations. Instead, the selection depends on the specific context of resistance (Tymoczko 2010a: 230).

2.2.1 **Feminist translation**

Before 1955, gender only described a grammatical feature of language (Haig 2004: 89). With second-wave feminism, gender began to indicate differences in men's and women's societal roles and opportunities that are the result of societal structures (von Flotow 1997: 5). The feminist movement sought to counteract sexism and achieve equality by concentrating on the analysis of gender and its manifestation in language (see for example von Flotow 1991, 1997, Pauwels 2003). The concept of patriarchy has been used in feminist scholarship to describe the system that privileges men based on alleged gender differences. Feminist scholars have tried to theorise the development and the history of patriarchy (Miller 2017). The literature produced in this field is vast (see again Miller 2017 for an overview) and goes beyond the purpose of the present work.

Feminists felt the urge to focus on language as a powerful tool developed by men which shapes society (von Flotow 1997: 8). Patriarchal language was seen not only as an expression of the system that oppresses women but also as playing an active role in their oppression. Therefore, it had to be studied and reformed in order for women to reach equality (Cameron 1985: 3). Linguistic analyses by feminists concentrated on factors such as (i) the study of sex/gender

differences in the use of language; (ii) the representation of women as well as sexism in language; (iii) women's feeling of alienation in language; and (iv) the relationships between language and power (Cameron 1985: 7, von Flotow 1997: 8, Kramer 2016: 73–75).

In Western Europe and North America, women started to reform language in the 1970s and 1980s actively. For example, they produced handbooks of non-sexist language, rewrote dictionaries, criticised sexist language, and introduced new literary forms to represent their experiences (von Flotow 1997: 9). In particular, feminist writers in Quebec were very active in producing experimental work that tried to challenge patriarchal language, hence giving birth to what is called *écriture féminine* (female writing) (von Flotow 1991: 72). These feminist writers tried to develop a new language for women, thus using new words, especially to describe women's bodies and experiences, spellings, grammar, and metaphors (von Flotow 1991: 72, von Flotow 1997: 15). Such reflections on patriarchal language, as well as experiments to create a new idiom that would make women visible and allow them to achieve equality, also had an impact on translation both as a practice and as a discipline (see von Flotow 1991, 1997).

For translators, this innovative work meant they had to experiment with new approaches. Female writing largely dealt with the topic of the female body. Derogatory vocabulary was reclaimed, and new words were coined to overcome objectified patriarchal descriptions and the lack of an everyday language to articulate them. Writings contained puns, cultural references, new grammatical forms, and played with sound associations (von Flotow 1997: 15), thus forcing translators to develop creative methods to reproduce the effect of the source texts. These included active intervention in translation after considering differences in linguistic structure and cultural situations, and the use of footnotes when strategies equivalent to the source texts were not found (von Flotow 1997).

Feminists also started to practise translation activism. There were two main approaches at the time: interventionist translation and the translation of female literature. The former saw translators actively shape target texts to remove or alter sexist elements of the source text (von Flotow 1997: 24–26). This process could occur more or less openly and could be more or less politically driven: de Lotbinière-Harwood (1991: 11), for instance, claims that translation is a political activity, and as such, her textual interventions are justified. von Flotow (1991: 74–76) identifies four strategies that characterise feminist translation. These are:

- **Supplementing:** This compensates for differences between languages or adds meaning to the text. For example, noun forms for people differ based on their sex/gender in French whilst they generally do not in English.

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When translating from the former into the latter, sex/gender can be specified.

- Footnoting: Translators stress their presence and try to convey with footnotes the multiple meanings of the source text that would otherwise be lost.
- Prefacing: Translators reflect and comment on their work, often explaining their political views and/or their translation strategies.
- Hijacking: Translators appropriate and manipulate source texts, using a mix of the above strategies as well as a didactic approach.

The second approach consisted of selecting and translating women's work that was not part of the traditional literary canon. The aim was to make their creativity available to the general public as it was maintained that patriarchal society defines aesthetics and literary values (von Flotow 1997: 30). For this reason, canons have mainly consisted (and in part still consist) of works produced by men. Attention was paid to issues such as censorship as well as the silencing and non-recognition of women's contributions to culture and literature (von Flotow 2011: 2). Though these early theories of feminist translation focused exclusively on women, TS have also been influenced by queer theories that seek to overcome binary categories such as those of sex/gender (von Flotow 2011: 3).

2.2.2 **Queer translation**

In order to understand queer translation, a brief explanation of the terms *queer* and *queer theory* is required. TS has only recently embraced queer theory and there is still often conceptual confusion within the discipline (Baer & Kaindl 2017a: 1). Queer can have different meanings: (i) it can be used as a synonym for homosexual or also to denote a person who does not fit normative ideas of gender and/or sexuality; (ii) it was originally a derogatory term, but nowadays it is often used as a form of linguistic reappropriation; (iii) it refers to a theoretical model that rejects binary oppositions like that of homosexual/heterosexual (Baer & Kaindl 2017a: 2). Queer theory questions the authenticity and universality of dominant structures of power and knowledge and emphasises their constructedness. As a consequence, queer can be used as a verb in order to describe such a critical process (Baer & Kaindl 2017a: 3).

From the 1970s onwards, TS focused on the issues of power, ideologies and ethics underlying the translation process while concepts such as identity and sexuality began to permeate the debate around translation (cf. Section 2.2 and Subsection 2.2.1). In the 1990s, LGBTQ studies also started to pay attention to

other discriminated minorities as well as their relationships with the majority culture. Sexuality also began to be conceptualised as fluid (Amenta 2020: 2f). This led to a queer turn in TS.

This new paradigm in TS has helped bring into question classical translation theories and approaches, also because it rejects binary categories such as those of source and target texts that have shaped the conception of the discipline (Amenta 2020: 3). Both queer theory and TS can benefit from their intersection, as Baer & Kaindl explain:

To the extent that queer theory problematizes the representation of otherness, and translation studies highlights the otherness inherent in representation, bringing together queer theory and translation studies should productively destabilize not only traditional models of representation, understood as mimesis, reflection, and copying, but also the authorial voices and subjectivities they project. (Baer & Kaindl 2017a: 1)

TS can finally overcome the dichotomy between the authoritative source and the subordinate target text, which led to the general view that translation must be faithful to the original, master text. Queer theory, as the study of borders and their deconstruction and/or rearrangement, promotes a radical rethinking of translation as a complex cultural, political, and ethical process of negotiation (Spurlin 2014a). At the same time, the Western-centric view of queerness in queer studies can be dismantled, showing how gender and sexuality are differently constructed across language, cultures, and history (Spurlin 2014a). These concepts are explored in more detail below, but first, a definition of queer texts and an account of the translation problems they present are provided.

Queer texts are characterised by the presence of non-normative sexual identities that deconstruct gender and sexual norms. The following challenges arise when translating them (Amenta 2020: 5):

- Sociolects: the use of camp talk by queer characters, or of female nicknames by gay men to refer to other gay men, neologisms, and semantic variations that are typical of the language of queer characters.
- Unusual grammatical and syntactical structures: often used to deconstruct the gender binary. For example, masculine and feminine gender markers may be alternated in a text to refer to the same person. When gender systems differ across languages, translators have to use synonyms or find compensating solutions, such as making gender explicit in other parts of the text.

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- Register and style: for instance, the use of double entendres and inflections in speech.
- Intertextuality: allusions to a character's or the author's sexuality can be made via reference to other texts, thus challenging the reader and the translator, who must recreate such allusions or substitute references to produce a similar effect.

According to Démont (2017), there are three different approaches to the translation of literary queer texts, namely, misrecognising, minoritising, and queer translation. The misrecognising translation ignores queer elements of the source text and suppresses its disruptiveness – this means that queer characters are often erased from texts and translations are “straightened”. The minoritising one reduces connotative elements to their denotative equivalents, thus making queerness explicit but flattening the power of queer texts which are often based on ambiguity, multiple meanings, and meaning associations. Finally, there is the proper queer translation – i.e. as intended for example by Burton (2010), which can consist of two different approaches. The first is the comparison and critique of a source text and its translation to investigate whether queer elements have been concealed or erased in the target texts. The second tries to find strategies to maintain the source text's queerness.

Epstein (2017) distinguishes between the queer translation of queer and homophobic texts. In the first case, similarly to feminist translation, strategies such as supplementing, prefacing, footnoting, or radical changes can be applied to reproduce the queerness of the source text. In the second case, texts can be translated to reveal the homophobic, biphobic, or transphobic views a text contains. Such a practice in particular is proposed by Burton (2010: 57), who uses the term *inversion*, meaning “a turning of the text against itself: inverting the hidden power relations of heterosexism by revealing and underscoring them through techniques borrowed from feminists”. Queer translation is therefore ideologically and methodologically related to feminist translation.

In addition, queer translation is “antihomophobic in motivation and practice, and destabilising and historicising of gender, sex, and sexuality norms” (Burton 2010: 55). Since the 1970s, discourses on discrimination and sexism have broadened and new identities have gained visibility. Debates on how translation can reinforce or counteract homophobic views started to emerge accordingly (Burton 2010: 54). Queer translation also adopts an intersectional lens (see 2.4.3 for a definition), i.e. it considers how queerness and other dimensions such as race and ethnicity interact with gender and sexuality (Burton 2010: 55).

Since translation allows the rendering of texts from different cultures and ages in which queerness is differently conceptualised, it can also foster the production of new knowledge in the field of identities and sexuality. Originally, the term *queer* is anglophone and as a concept it is related to Western cultures. However, queerness is differently conceived, experienced and lived in other cultures, and so it was in other ages. Therefore, it is also differently described. Many non-Western experiences of queerness are reduced to Western equivalents or completely erased in translation. Translation as a queer practice can thus help reduce “the monolingual, anglophone biases” of both TS and queer theory (Spurlin 2014a: 299): the analysis of other gender and sexual identities in translation may produce new knowledge that stimulates shifts in social identities and categories. Such an analysis shifts the attention to the way gender and sexuality are represented through language(s).

Analyses on how gender and sexuality differ across languages, on changes of meaning as well as on gaps between cultures can impact translation itself and how it is conceived. With regard to translation theory metaphors, Spurlin (2014a: 301f) notes how the translated text has historically been conceived as female, that is, subordinated to the authoritative original. In the field of queer translation, such feminisation of the target text is replaced by a (re)gendering that overcomes the source/target binary (see Chamberlain 1988 for metaphors used to describe translation). Since translation is an active process that involves language and culture as well as ideological and political aspects, the original is not reproduced but rather transformed. As a matter of fact, translation is a process of negotiation and meaning production, a site of struggle that questions the relationship between the original text, i.e. the dominant male, and the translated text, i.e. the submissive female (Spurlin 2014b: 206). Queering translation shows the power of translation to mitigate asymmetrical relationships between languages and cultures that reproduce existing power relations (Spurlin 2014b: 213). Hence, cultural and historical differences in source texts should be preserved as a way to oppose monolingualism and ethnocentrism (Spurlin 2014b), often intended as a tendency to adapt source texts to Western, and especially anglophone cultures.

2.2.3 Translating gender: Challenges

Subsection 2.4.2 addresses in more detail the extent of complexity in the relationship between gender and language. Here, it is sufficient to briefly highlight that (i) linguistic elements are used to refer to the extralinguistic reality, in this case gender identity; (ii) linguistic structures to do so vary across languages; (iii) linguistic gender has also social connotations. Accordingly, TS have analysed

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such issues in translation, highlighting for example how social gender influences translation choices (Jakobson 1959, Nissen 2002, Di Sabato & Perri 2020). This section presents a concise overview of some of the problems arising from translation among languages with different gender systems along with possible solutions.

Translation from a grammatical gender language, i.e. when every noun has a gender, into a notional gender language, i.e. when pronouns only and few nouns have a gender (see Subsection 2.4.2 for a classification of languages based on their gender system and an in-depth explanation), can be particularly difficult as their agreement structures differ greatly (Nissen 2002: 26). For instance, when translating adjectives and nouns describing a person from Spanish into English, gender information is lost. This can be compensated by lexically marking the terms in question by adding a word specifying the person's gender identity. However, especially in literary translations, such marking may shift the focus of the translation to gender in a way the source text did not require (Nissen 2002: 27).

In languages with a grammatical gender system, the same object or abstract concept can be assigned different genders and this can shape how it is perceived by speakers (see for example Boroditsky et al. 2003).¹ When translating, connotations conveyed by gender should be accordingly taken into account, especially in the case of metaphors and personifications. For instance, Jakobson (1959: 237) notes how death is masculine in German (*der Tod*), and depicted in books as an old man, but feminine in Russian, and therefore differently conceived. This issue can be tackled in translation by (i) looking for a synonym that has the same gender as the term to be translated; (ii) using a word from a third language that is also familiar to the readers of the translation; or (iii) providing a footnote (Nissen 2002: 29). These strategies are also relevant for multimodal translation, such as the localisation of advertisements. Animals are often used to convey particular messages through the characteristics they are associated with because of their grammatical gender, e.g. using a tiger to evoke athleticism and therefore to convey the idea of speed in a petrol advertisement in German (Di Sabato & Perri 2020: 368–369).

Similarly, animals are often personified in fables or tales (Di Sabato & Perri 2020: 368). When this occurs, as in the Italian fable *Il Ragno e l'Uva* (The Spider and the Grapes), the translation into notional gender languages is problematic. Even though in grammatical gender languages animals are assigned a gender, they are not perceived as having a male or female gender identity. For instance, even though the spider is masculine in Italian, the grammatical gender does not correspond to the animal's biological classification as male or female. However,

¹It should be noted that these results have been difficult to replicate (Mickan et al. 2014).

translators in English are likely to mark gender since the use of the pronoun *it* would weaken the anthropomorphic element of the text. This, in turn, adds a meaning for the target readers of the translation, i.e. the animal having a particular gender, that is not present in the source text (Di Sabato & Perri 2020).

Another issue concerns social gender. Social gender is the property for which a word, usually a profession, is generally assigned the female or the male gender. This assignment is arbitrary and mainly based on stereotypical associations that vary across time and societies (Nissen 2002: 31). For example, the word *nurse* is often associated with femaleness. Social gender represents a problem when translating from notional gender languages into grammatical gender languages, since the nouns' and, therefore, a person's gender must be specified. This often leads to the source language input being translated differently across languages (Nissen 2002: 33–34). For instance, a simple sentence such as “My *nurse* told me to take five pills a day” requires that the gender of the nurse is specified when translating into Italian or German. In such cases, when there is no cue about gender, the assignment occurs based on social assumptions (Nissen 2002: 30).

More generally, translation from notional gender languages into grammatical gender languages can force translators to opt for a gender (Di Sabato & Perri 2020: 367). In literary texts, authors can avoid gender markers so as to create a mysterious atmosphere, for example in Edgar Allan Poe's *The Tell-Tale Heart* (1843), where it is impossible to define the narrator's gender as they speak in the first person. Di Sabato & Perri (2020: 369) note how the Italian translator opted for masculine forms of nouns, adjectives, and participles, thus choosing a particular gender for the narrator.

Finally, Di Sabato & Perri (2020: 369) claim that the translation of occupations and/or institutional roles is related not only to semantics but also to ideological questions. For example, feminine forms of high-level professions and institutional roles in Italian are not particularly widespread although they are semantically and grammatically correct (see Gheno 2019). In Italy, their use is often associated with ideological matters rather than simple linguistic correctness (Gheno 2019: 112). Similar claims can also be made for GFL. Its use, for example in a translation in order to replace masculine generics, may be seen as a political and/or ideological statement rather than a simple matter of equality. In such a situation, translators can help build bridges between marginalised communities and their clients but they could also be denied an assignment because of their willingness to opt for GFL strategies.

The aforementioned issues show how gender in translation can be a complex and ambiguous phenomenon and every solution adopted has different connotations and implications. There are no clear, one-size-fits-all solutions on how

to handle gender appropriately and translators should, while being creative and experimenting with language, consider a wealth of co-textual and contextual factors. The translation of gender represents a challenge for MT as well. This is thoroughly discussed in Subsection 3.1.1.

2.3 Sex/Gender: Introduction

This section introduces some key concepts that permeate the field of gender studies. It focuses on sex and gender, more specifically their meanings and interconnections, their importance, and the role of society in shaping assumptions concerning them. Attention is paid to how the concepts of sex and gender are, at least partially, socially constructed and still conceived as binary, i.e. male/female, man/woman. Since GFL in the present work is intended as a language that includes all or no genders, an overview of non-binary² gender identities is provided.

2.3.1 Sex vs gender

Gender identity must be indicated to access numerous goods and services, e.g. to open a bank account or to enrol at a university (Fae 2016). What is asked in forms or on websites, however, is not always clear since the terms sex and gender are often used synonymously in different contexts (van Anders et al. 2017: 194f, Acanfora 2021: 105). It is generally assumed that a person with certain physical characteristics (i.e. sex) will identify as a certain gender. For instance, it is taken for granted that a person born with female genitalia will identify as woman (Richards et al. 2016: 95, Morgenroth & Ryan 2020: 1). In such a view, gender follows from sex. It is very difficult to find an uncontentious and clear definition of both terms as well as to understand the difference between them (Lester 2018).

Sex is generally described as referring to biological characteristics, such as chromosomal makeup, hormones, and primary and secondary sexual characteristics, whereas gender is defined as a historical, cultural, and social construct that refers to roles and behaviours associated with the equally socially constructed concepts of masculinity and femininity (Barker 2018a: 122, 2018b: 32, Barker & Iantaffi 2019: 56). In the heterosexist Western society, both sex and gender are thought to be binary and this binarism not only dictates how sex and gender are linked – with the latter supposedly following from the former – but also

²The term non-binary is used here to refer to all sexes and genders beyond male and female, also including intersex people.

how they influence sexuality (Butler 1990: 23–24, Barker 2018a: 123). As shown in Figure 2.1, it is generally assumed that a person is born having one sex (either male or female) and that they will congruently grow up being a masculine man if male or a feminine woman if female. According to their sex/gender, they will also be attracted to a person of the opposite sex/gender (Barker 2018a: 123). This is, however, an essentialist view of gender.

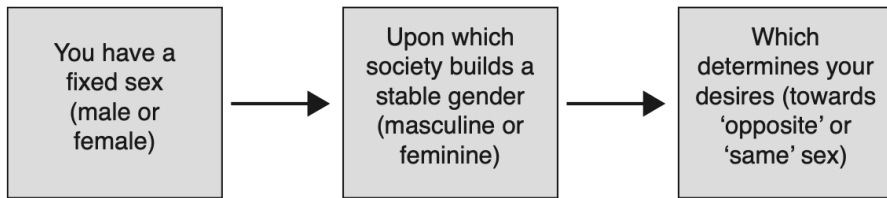


Figure 2.1: Interconnections among sex, gender and sexuality in a heteronormative society (Barker 2018a: 123)

2.3.2 Defining sex

The term sex, often interpreted as biological sex, is used to refer to biological features. However, when referring to a person's sex, the term sex assigned at birth is preferred and even recommended (Fausto-Sterling 2012, 2018, ProNa 2015: 24, Richards et al. 2016: 95, Serano 2017, Barker 2018b: 30, Lester 2018, Barker & Iantaffi 2019: 57, and see also recommendations by APA Style 2019a). At birth, sex is assigned based on the sole inspection of the newborn's genitalia provided that they are dimorphic, i.e. they can be clearly categorised as female (clitoris) or male (penis). This has historically erased the identity of intersex people.³ Individuals who were born with mixed genitalia have usually undergone a surgical operation to fit dimorphism (Fausto-Sterling 2012, 2018, 2020b, ProNa 2015, Richards et al. 2016, Barker 2018a, Lester 2018, Barker & Iantaffi 2019). Three different arguments can be advanced for preferring the expression sex assigned at birth over the simplistic term sex when referring to a person:

1. If human biology in its real complexity is taken into account, then five layers of sex should be examined. They are not always binary and sometimes also develop independently of one another (Fausto-Sterling 1993, 2000, 2012, 2018).

³Intersex people, however, can have perfectly dimorphic genitalia. This is further explained in the present section.

2. Science is always produced and interpreted within a specific sociocultural context. In this case, biologists produced or interpreted facts, namely the distinction between male and female, based on their assumptions on the gender binary (see Fausto-Sterling 2020a). Sex can therefore be considered in part as socially constructed as gender (Butler 1990: 10–11, Fausto-Sterling 2016: 199, Serano 2017).
3. Biological features are actually malleable so that if a person chooses to undergo hormonal and/or medical treatment, their biology as well as their primary and secondary sex characteristics change (Serano 2017, Lester 2018).

Fausto-Sterling (2012: 66, 2016: 190) argues that the human body is extremely complex and many traits such as voice pitch or breast size are not dimorphic. Rather, they overlap and make it difficult to provide precise answers about sex differences. As regards sex and its development, the biologist distinguishes five layers (Fausto-Sterling 2012: 3, 2018). The first layer of sex develops at conception and (i) is the chromosomal sex: a sperm bearing X or Y chromosomes normally fuses with an egg, which bears X chromosomes. From this, the common misconception originates that a person with XX chromosomes is female and a person with XY is male. Other combinations, such as XXY, XYY or even XO, are possible (Fausto-Sterling 2012: 25, 2018). After eight to twelve weeks the embryo develops a pre-gonadal structure that can develop either in a male or female direction, but unexpected developments can also occur, which is the (ii) foetal gonadal sex. At this stage, primary sex characteristics (testes and ovaries) are determined. Hormones developed by the embryo, this is the (iii) foetal hormonal sex, push the development of testes and ovaries further, determining their differentiation and leading to the formation of (iv) the internal reproductive sex, i.e. uterus, cervix and so forth in females or prostate, epididymis and so forth in males. Finally, in the fourth month, (v) the external genital sex is developed (Fausto-Sterling 2012: 5, 2018). Fausto-Sterling (1993: 20, 2000: 20, 2012: 11, 2016: 190, 2018) has more than once claimed that different combinations at every layer of sex are possible and it would be better to refer to sex as a continuum rather than as binary. This complexity is usually reduced to an essentialist model and Fausto-Sterling (2016: 190) underlines that the term sex is used in various different ways in both science and everyday language. To summarise, if sex is used to account for psychological, physiological or sexual behaviours and differences, it is not a useful and precise concept.

Discoveries in the field of sex development and how they were interpreted led to different theories. Butler (1990: 10–11), Fine (2017) and Fausto-Sterling (2020a, 2020b) claim that the production as well as the interpretation of scientific facts

is always influenced by cultural assumptions. For instance, gender is believed to be binary and to follow from sex as well as alleged (stereotypical) differences between men and women. The scholars conclude that sex too is, at least in part, socially constructed like gender. Fine (2017) analysed extensive research in the fields of biology, psychology, and sociology. She noted, for example, how testosterone and other hormone levels were once thought to be genetically determined as well as fixed, leading to the use of such arguments to support claims on the nature of the different roles of men and women in society. The human brain too was believed to differ based on biological sex, but dimorphism is actually rare (see Fine 2017 and Hyde et al. 2019: 172 for an overview of the existing literature on the topic). Nowadays, scholars are unanimous on the position that biological features are not fixed, but are rather influenced by social and environmental conditions. For example, fathers, and especially those who invest more time in caregiving, show decreased levels of testosterone (Gettler et al. 2011). Such levels increase in female actors who enact power roles, thus reproducing stereotyped roles (van Anders et al. 2015). Differences in hormone levels or human brains are a product of sex/gender differences rather than a cause thereof since we live in a highly gendered society (Fausto-Sterling 2016: 200, 2020a, Fine 2017, Barker 2018a: 142, 2018b: 36, Lester 2018, Hyde et al. 2019: 174, Barker & Iantaffi 2019: 58). This shows how research is sometimes biased towards social assumptions that affect how experiments are carried out and results interpreted (see Fine 2017 and Hyde et al. 2019 for an overview in the field of sex/gender-related differences as well as Fausto-Sterling 2020b for a very detailed account of how changing societal assumptions shaped interpretations of discoveries in the field of sex development).

In addition, transgender activists such as Serano (2017) and Lester (2018) explain how biological makeup, primary (e.g. genitalia) and secondary characteristics (e.g. facial hair) can change through hormonal or surgical intervention. This holds also for cisgender people (see Subsection 2.3.4 for a definition of this term), who may have their breasts removed after cancer or take the birth control pill, for example. Lester notes:

The body of a trans woman who has pursued hormone therapy and surgery might combine XY chromosomes, higher levels of estrogen and progesterone with concurrent lower levels of/lower sensitivity to androgens, no testes, a prostate, a vulva and vagina, little body hair, no facial hair, breasts and curves. When looking at all the different parts of her physical makeup, what counts as “biological sex”? All of these categories are sexed, and all are “biological.” (Lester 2018: 60)

The process of sex assignment can be further detailed by looking at intersex movements. A newborn with a penis is not only assigned male at birth but is also recognised as such by society and law (Serano 2017, Barker & Iantaffi 2019: 56). However, if in the past one's reproductive anatomy did not fit dimorphism, i.e. the external genitalia could not be clearly labelled as clitoris or penis, genital surgery was readily performed for genitals to fit a perfect dimorphism. The children were then raised according to the surgically assigned gender (Fausto-Sterling 2000: 19–20, 2020b), thus systematically erasing the identity of intersex people to make them conform to binary social and cultural expectations (Barker & Iantaffi 2019: 57–58). As intersex movements rose, intersex people started to advocate for more ethical treatments that should include surgery only as a means to save lives; they also demanded that intersexuality be defined as “normal” (Fausto-Sterling 2000: 21, Barker & Iantaffi 2019: 57–58).

Some intersex people may not find out about their intersex status until adolescence; other may never find out at all. For example, some XX babies can be born with a penis, but as long as the external genitalia are clearly distinguishable, no further test is carried out at birth (Fausto-Sterling 2018, Barker & Iantaffi 2019: 57). Also, people have their chromosomal makeup examined quite rarely (Lester 2018). This means that they could be assigned the wrong sex at birth. Acknowledging all the aforementioned points is a step towards dismantling transphobic arguments. Fausto-Sterling claims, describing the treatments given to intersex people, that:

labeling someone a man or a woman is a social decision. We may use scientific knowledge to help us make the decision, but only our beliefs about gender – not science – can define our sex. (Fausto-Sterling 2020a: 3)

This quotation confirms that labels, as explained in Subsection 2.4.1, are influenced by culture and society. Therefore, sex is categorised differently in other, non-Western cultures. The question of how sex and/or gender identities are represented is a crucial matter not only for gender studies but also for TS (Spurlin 2014b), highlighting its interdisciplinary character.

Finally, Serano (2017) argues that efforts to discriminate against trans people, especially trans women, resort to the alleged distinction between sex, hence biology, and gender, hence social construction. According to such distinction, transphobic people would argue that a trans woman is biologically male. Similar arguments are used by Trans-Exclusionary Radical Feminists (TERF) to exclude trans women from feminism (see Lester 2018). Understanding how the category of sex is interpreted by social assumptions, how people are required and supposed to

fit a perfect dimorphism and how sex is diverse helps dismantle such arguments. To this end, transgender movements also ask that their opinion be considered by the scientific community and that trans people be included in the process of diagnosis or classifications about transsexuality or gender dysphoria.⁴ They claim that medical language pathologises them and that science is often imprecise, delivering ill-founded conclusions (Fausto-Sterling 2020a). In this regard, Acanfora (2021: 76–77) affirms that diagnoses are made in a pathologising, static, normalising, and fixing language that is then often also used in everyday life, thus the diagnosed person tends to identify with their diagnosis.

2.3.3 Defining gender

Before 1955, the term *gender* was almost exclusively used to refer to grammatical gender (see Subsection 2.4.2) (Haig 2004: 89). In 1955, Money (1955) introduced the concept of gender role to refer to social behaviours and characteristics that are associated with masculinity and femininity (Haig 2004: 89). From the second half of the 1960s, feminists then began to use the term *gender* to account for differences in men's and women's societal roles and opportunities that were the results of social processes and could not be ascribed to biology (von Flotow 1997: 5). Nowadays, in public discourse, gender is still often used as a synonym for sex, and there are at least two views of gender: essentialist and constructionist (Acanfora 2021: 106).

The essentialist view of gender posits that some gender features are innate, universal, and biologically determined (Acanfora 2021: 106). The constructionist view posits that gender is a historical, cultural, and social construct, tied to the notions of masculinity and femininity (Barker 2018a: 122, Barker 2018b: 32, Barker & Iantaffi 2019: 56, Acanfora 2021: 106–107). A third view was proposed by Butler (1990: 25) who introduced the concept of gender performativity. The concept is based on the speech act theory proposed by Austin (1975), who describes explicit primary performative utterances, i.e. utterances that simultaneously perform an act and describe its performance, e.g. "I promise I will call you tomorrow". The actions in the utterances are therefore produced through their description. Similarly, through discourse, gender performatively constitutes itself and it is treated as something that already exists. Gender is therefore performed by being enacted and, at the same time, its categories are (re)created and presented as natural (Butler 1990: 25). In this process of (re)creation of gender, language and translation

⁴The negative feeling that a person's body and experiences do not fit with how they perceive themselves (Barker & Iantaffi 2019: 65).

play an essential role because they can uphold or challenge societal views (Simon 1996, AG Feministisch Sprachhandeln 2015, Fairclough 2015) as discussed in Section 2.1 and Subsection 2.4.3.

In Western societies, gender was (and often still is) believed to be binary and to follow from sex (Richards et al. 2016: 95, Matsuno & Budge 2017: 116–117, Barker 2018a: 122, Barker 2018b: 28, Lester 2018, Barker & Iantaffi 2019: 56–57). Butler (1990: 174) tried to dismantle such a view, describing different forms of what she defined as *gender troubles*, i.e. people that challenge the performative binary gender system, such as drag queens. The binary system is still believed to be a valid model, with the consequence that men and women are encouraged to take on specific roles and act in specific ways.

For instance, Heilman (2012: 114) defines gender stereotypes as “generalizations about attributes of men and women” that include their main features as well as inclinations for particular roles and/or responsibilities. They can be divided into descriptive and prescriptive stereotypes. The former describe what men and women are like; the latter, what they should be like. Stereotypes thus dictate how people have to act and devalue whoever does not fit them (Heilman 2012: 114). As shown in Table 2.1, Heilman (2012) argues that women are described as having communal traits, and men agentic ones.

Table 2.1: Gender stereotypes according to Heilman (2012: 115)

Agency	Communality
achievement-orientation (competent, ambitious, task-focused)	concern for others (kind, caring, considerate)
inclination to take charge (assertive, dominant, forceful)	affiliative tendencies (warm, friendly, collaborative)
autonomy (independent, self-reliant, decisive)	deference (obedient, respectful, self-effacing)
rationality (analytical, logical, objective)	emotional sensitivity (perceptive, intuitive, understanding)

Gender stereotypes are often used by the media, especially for marketing reasons, to get the attention of the public (Barker 2018a: 124). They are also perceived as rules and negatively affect men, women, and non-binary people (Barker 2018a: 132). Men have to be rational, dominant, and protective of other people such as their partners or children. They must not be intimate with other men and feminine men are discriminated against since femininity is despised. Women are

often defined by their relationships with other people, especially with men, discriminated against in the workplace, or seen as mere caregivers or sex objects (Heilman 2012, Grover 2015, Barker 2018a: 132, Murgia 2021b). Non-binary people are subject to even more discrimination: they are misgendered, denied access to public facilities such as lavatories, verbally and physically harassed or their existence is even denied (Herman 2013, McLemore 2015, 2018, Matsuno & Budge 2017: 118, Lester 2018). Gender roles and stereotypes originating from a binary and outdated understanding of gender thus prove harmful to many people. The discourse around gender stereotypes is relevant for TS as well, e.g. because marketing advertisements playing with them are translated into other languages (Di Sabato & Perri 2020), once again showing the social relevance and impact of language and TS.

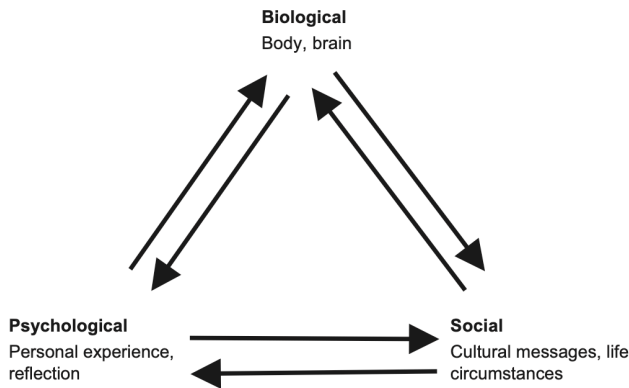


Figure 2.2: Biopsychosocial processes and their interaction (Barker 2018b: 36)

A newer understanding of gender consists in defining it as a biopsychosocial construct, hence biological, psychological, and social aspects interacting with one another, as shown in Figure 2.2 (Barker 2018a: 99f, 2018b: 35, Barker & Iantaffi 2019: 58). Biological aspects may involve brain functionality and hormone production as well as their interaction. Psychological aspects may involve the way people experience gender, e.g. how they identify, and how they think about gender. Social aspects involve how gender is enacted in a particular historical, geographical, and cultural context, hence expectations about genders, gender roles and norms, e.g. how one has to dress, which activities one takes up, etc. (Barker & Iantaffi 2019: 58). These three aspects are all interrelated and cannot be considered separately as they influence one another. Thus, gender is a construct,

but it influences biology and also other tangible aspects of human life. Since the binary, static, and fixed gender system is deeply rooted in Western cultures, people have to locate themselves in relation to it. They are deeply influenced by it, e.g. in the way they dress, act and speak, and cannot completely abandon this model (Barker & Iantaffi 2019: 81, Acanfora 2021: 108f). Besides, when talking about a person's gender, reference could be made to their: (i) gender identity: the gender they identify as, like woman, man, genderfluid, non-binary, genderless, etc.; (ii) gender role: how feminine and/or masculine they act according to gender norms; (iii) gender expression: how they express their gender through clothes, hairstyle, etc.; (iv) gender experience: how they face the world based on their gender (Barker 2018a: 129, Barker & Iantaffi 2019: 60). Finally, gender also interacts with sex.

2.3.4 Gender identities: Cisgender, Transgender, and other terms

In Subsection 2.3.2, it was argued that sex is assigned at birth based on certain physical characteristics, namely the form of the external genitalia:

If people are assigned a certain sex at birth and identify with a gender that is considered congruent with that sex assigned at birth culturally, they are cisgender—that is their sex assigned at birth and gender identity are on the same side. (Barker & Iantaffi 2019: 59)

When people identify with a gender that is not congruent with the sex assigned at birth, they might be considered transgender (Barker & Iantaffi 2019: 60). More specifically, by the fourth month of pregnancy, genital sex is developed and therefore sex is often assigned before birth through inspection of the baby's genitalia. A process of genderisation begins: medical professionals and parents expect the baby to identify as the gender which is socially held as congruent with their genital sex, and parents buy clothes and toys accordingly, e.g. dolls, skirts, or other pink clothes for female babies, cars and blue clothes for male children. Through clothes and colours, children's sex assigned at birth is made visible so that adults can make assumptions about their genitalia (Fausto-Sterling 2012: 7, 2016: 195, Acanfora 2021: 107). Hence, when a person is assigned a sex before or at birth, e.g. female, they are also assigned a specific gender based on social expectations, e.g. woman. If the newborn identifies as the gender assigned at birth, they might be defined as cisgender. Otherwise, they might be defined as transgender.

This distinction may seem straightforward; however, it is also a binary concept, which is problematic. For example, according to Barker & Iantaffi (2019:

72), cross-dressers, such as drag queens, who perform by temporarily impersonating another gender, would fall under neither the cisgender nor the transgender umbrella. On the contrary, Lester (2018) argues that cross-dressers fall under the transgender umbrella. Even though there is no unanimity, the terminology presented in this subsection helps move away from the category of biological sex.

Moreover, there are intersex people, i.e. people who are born with variations in sex characteristics. As explained in Subsection 2.3.2, these include chromosomes, gonads, hormones, and genitals. In this case, the use of the term cisgender or transgender may be controversial. Since the use of labels, such as transgender, is also a question of personal preference, one should always ask for the terms people identify with and use them.

That said, it is useful to define the term non-binary. Non-binary is both a term describing a gender identity and an umbrella term that includes all genders other than man and woman (Richards et al. 2016: 96, Barker 2018a: 150, 2018b: 28, Lester 2018, Barker & Iantaffi 2019: 61–62). It can be abbreviated to *enby* or NB, but some people also use the term genderqueer (Richards et al. 2016: 96, Barker & Iantaffi 2019: 61–62). Labels for gender identities are numerous and include different experiences of gender (Richards et al. 2016: 96, Barker 2018a: 150, Barker & Iantaffi 2019: 61):

- No sense of gender (e.g. agender, genderless, etc.).
- Muted/partial sense of gender (e.g. demi boy or demi girl).
- Both binary genders, i.e. male and female (e.g. androgynous, mixed gender, etc.).
- Changing over time (e.g. genderfluid, bigender).
- A third gender (e.g. third gender, other gender).
- Multiple genders (e.g. pangender).
- People who challenge the ontology of gender (e.g. genderqueer, genderfuck, etc.).⁵

At this point, it is useful to sum up the different meanings of the terms cisgender and transgender. Cisgender people's gender identity is aligned with the gender that is socially expected based on their sex assigned at birth. They can be either (cis-)men or (cis-)women. Transgender people's gender identity is not aligned with the gender that is socially expected based on their sex assigned at birth. They can have a binary gender, hence being (trans-)men or (trans-)women. The former were assigned female at birth and identify as men, while the latter were

⁵Note that this list does not claim to be exhaustive and representative of all gender identities and experiences. Also, note that some terms may overlap and one must be aware of people's identified labels.

assigned male at birth and identify as women. Transgender people may also be non-binary, which is an umbrella term for gender identities outside the binary. However, not every non-binary person defines themselves as transgender. Finally, people who cross-dress may or may not be considered transgender (see Figure 2.3 for a simplified illustration of such umbrella terms) (Richards et al. 2016: 96, Barker 2018a: 150, Lester 2018, Barker & Iantaffi 2019: 61, Acanfora 2021: 109). Transgender people may or may not transition – this is a long and complex process to move from living in the sex assigned at birth to living in one's gender identity. The transition may be social, e.g. it may involve coming out as transgender, changing one's name, pronouns and documents, and/or physical, e.g. it may consist of hormone treatments and different types of surgery to align the body with one's perception of how this should be. Transition and the representation of transgender people are also a research object of queer translation: for instance, Rose (2016) examines how to preserve the gender identity of trans people in the translation of two memoirs from French, where both masculine and feminine gender markings are used in reference to the same transgender person, into English.

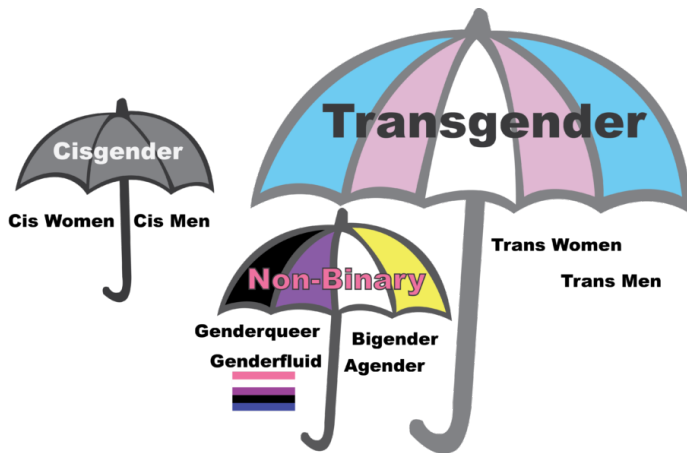


Figure 2.3: Simplified illustration of the gender umbrellas with gender identity terms

In Subsection 2.3.3, it was argued that when talking about gender, reference could be made to different aspects, such as gender identity, role, expression, and

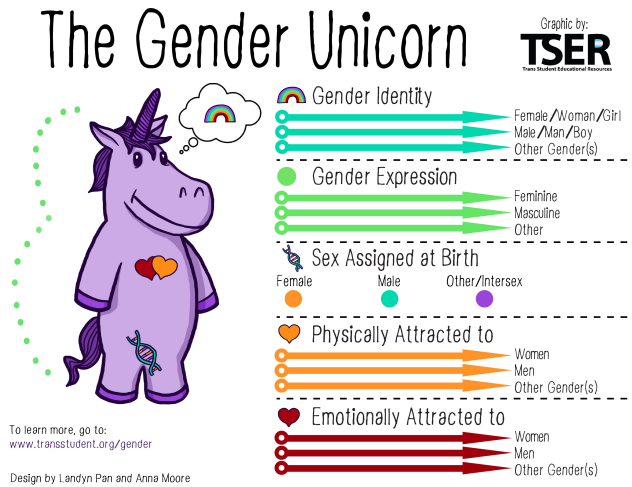


Figure 2.4: Simplified illustration of sex/gender identity, sexual and romantic orientation (Trans Student Educational Resources 2015)

experience (Barker & Iantaffi 2019: 60). In this subsection, labels for different gender identities were explained. The interaction of different aspects of gender as well as of sex and sexuality has been summarised and illustrated in some models such as in Figure 2.4 by Trans Student Educational Resources (2015). This is an adjustment of an outdated version of the model proposed by Killermann (2017)⁶ to more accurately depict the differences between gender (identity and expression), sex assigned at birth, and sexuality. Note that physical and emotional attraction are regarded as separate, so a person could be emotionally attracted to more genders but physically attracted to one gender only (see for example van Anders 2015: 1200f, Barker 2018a: 110, Barker & Iantaffi 2019: 40). Such models, however, are extremely simplified and not representative of our gender and sexual identities.

2.3.5 Beyond sex and gender

van Anders (2015) and Iantaffi et al. (2018) proposed Sexual Configurations Theory (SCT) because, they maintain, the classical model of sexual orientation poses several problems: (i) it is based on a binary and confused conception of gender

⁶Even though Killermann (2017) updated his model in 2017, the re-elaboration proposed by Trans Student Educational Resources (2015) is to be preferred as it is more terminologically precise.

(and sex) and it is not clear what is meant by the term; (ii) other aspects of sexuality are ignored, gender being only one of them, e.g. levels of sexual attraction, number of partners involved in the sexual act, and the roles played (see Figure 2.5 for a simplified overview); (iii) it is a static and fixed model that does not account for sexual fluidity and the existing variety of experiences of sexuality (van Anders 2015: 1178f).

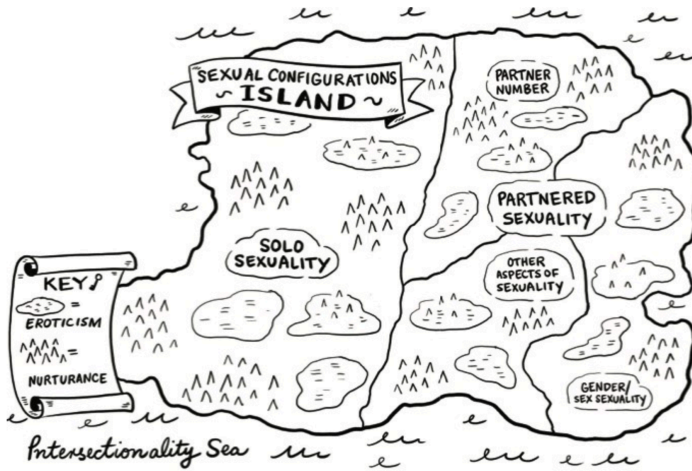


Figure 2.5: Simplified overview of the defining elements of sexuality according to SCT (Iantaffi et al. 2018: 12–13)

van Anders (2015: 1178) stated that she conceived “Sexual Configurations Theory (SCT) as a way to address the complexities of actual people’s sexualities”. In the context of this book, it is important to underline that van Anders chose to use the term gender/sex, as she claims that sex and gender cannot always be easily distinguished and some phenomena cannot be sorted into one or the other, e.g. the question is whether hormones, which are a biological feature whose level nevertheless changes based on the social context, are related to sex or gender (van Anders 2015: 1181–1182, Lester 2018, Barker & Iantaffi 2019: 59).

Once again, it should be noted that the gender, and also sex, binary is the cause of different types of discrimination. The existence of individuals whose identities or bodies fall outside the binary, as well as that of people whose gender identity is binary but not coincident with the sex assigned at birth, is hardly acknowledged. They are discriminated against by being misgendered, denied access to public facilities such as lavatories, and verbally and physically harassed (Herman 2013, McLemore 2015, 2018, Matsuno & Budge 2017: 118, Lester 2018, Hyde et al. 2019:

184). Cis-men and cis-women are then believed to be very different from one another and have to fit old gender roles and stereotypes that affect them in various ways,⁷ for example preventing women's access to leadership positions (Grover 2015, Barker 2018a: 132, Hyde et al. 2019: 183–184). Such beliefs negatively affect one's self-perception, so that one thinks about people's identity in a stereotyped manner (Hyde et al. 2019: 184). This also shapes legal and social policies. Single-sex schooling was introduced based on alleged differences between male and female brains. Hormone levels, and in the past also genetic testing and the examination of genitalia, were and are used to regulate women's access to sports competitions (Fausto-Sterling 2020a). Sex or gender – it is not always clear which criteria the policies employ – are used as a basis to manage access to public spaces (van Anders et al. 2017, Hyde et al. 2019: 184, see Fausto-Sterling 2020a for some accounts of the history of sports competitions and the discrimination of inter-sex people). Finally, the gender binary has proven to be an obstacle to advances in research. Since gender/sex includes a wider variety of phenomena, the use of simple variables, such as female/male or woman/man, fails to acknowledge the causes (e.g. hormones, gender roles expectations, socioeconomic status) of differences for example in depression rates among women, men and individuals of other genders (Hyde et al. 2019: 184). Gender and sex are hence intertwined, biological and social/cultural processes interacting and influencing one another and causing a multitude of differences.

2.4 The societal impact of gender in language

The concept of gender was introduced in the previous section with reference to its biopsychosocial nature. Gender also interacts with language. It is expressed through linguistic structures which reflect and create assumptions about it (Stahlberg et al. 2007). The following sections deliver insights into the interplay of language, society, and gender, providing a better understanding as to why gender issues have received great attention in the field of TS as well. Languages are the result of cultural conventions, thus they express the same events or concepts in different ways. Such differences, in the case of gender, not only complicate the translation process as shown in this book (see Chapter 6) but they also may impact people's perceptions. Since language – and therefore translation (Wolf

⁷This is also true for transgender people: for them to access medical treatments for transitioning, they are required to fit the stereotypical characteristics of femininity or masculinity. This is especially discriminating for transgender non-binary people as they could be more easily denied hormone therapy and/or surgery (ProNa 2015: 28, Lester 2018).

2007) – is socially situated, the reverse also happens: societal norms and power relations, such as sexism, are (re)produced through language and translation (Simon 1996, von Flotow 1997). This means that language can also be used to foster change. The chapter concludes with a discussion of empirical research that shows how the use of GFL positively impacts gender equality.

2.4.1 Language and thought

Language is a cultural convention (Deutscher 2010: 9f) and it can be used to express abstract and concrete concepts. Concepts are named differently across languages, e.g. happiness is *felicità* in Italian but *felicidad* in Spanish. The distinction among concepts may be based on natural differences. For example, the difference among animals, such as cats and dogs, is the same in every language. In other cases, the distinction among concepts may be the result of cultural conventions. For instance, the differentiation among body parts, such as arm and hand, is not the same in every language and culture (Deutscher 2010: 13).

Languages differ considerably in their structures as well. In Tagalog, a language spoken in the Philippines, different pronouns are used for *we*, depending on the different meanings, i.e. (i) you and I; (ii) someone else and I; and (iii) someone else and I but not you (Deutscher 2010: 15). Other languages, such as Mian, spoken in Papua New Guinea, use different tenses specifying whether a past event took place in an immediate, recent or distant past, whereas others, such as Indonesian, do not (Boroditsky 2001: 64). These differences lead people to use different linguistic structures and hence can be assumed to think differently of the same event to express it (Boroditsky 2011: 64).

This idea originates from Whorf (1956). He was a linguist who posited that language affects cognitive reasoning (see also Pederson 2007: 1014). Such a notion became famous as the Sapir-Whorf hypothesis. It is also known as the theory of linguistic relativity and has long been debated in the field of cognitive linguistics (see Pederson 2007 for an overview). Kramer (2016: 72) argues that a version of this theory, namely a strong linguistic relativism or linguistic determinism, initially became popular. According to this idea, language influences logical reasoning and people are not able to understand concepts foreign to their language and perceive reality in a completely different manner (Deutscher 2010: 21–22, Kramer 2016: 72).

This version is nowadays rejected, with linguists putting forward a moderate version (see for example Levinson & Haviland 1994, Boroditsky 2001, Boroditsky et al. 2003, Phillips & Boroditsky 2003, Casasanto 2008, Prewitt-Freilino et

al. 2012, Santacreu-Vasut et al. 2013), investigating whether differences or similarities in linguistic categories such as time, space, and gender are correlated with cognitive and behavioural differences (Kramer 2016: 72). For instance, time is conceived differently based on linguistic metaphors used to describe it, e.g. when saying “move a meeting forward” instead of *up* or *right* (Boroditsky 2001: 7). Or again, speakers of numerous aboriginal languages like Guugu Yimithirr use only absolute directions to refer to the position of objects and people and develop a better sense of orientation than speakers of languages like English that use relative directions (Levinson & Haviland 1994, Boroditsky 2011). Thinking for speaking, i.e. cognitive processes involved in the selection of words, their use in grammatical structures, and speech planning amongst others, has been shown to affect the habits of mind, impacting memory, attention, perception, and associations (Deutscher 2010: 22, Kramer 2016: 72). Such concepts are relevant for the present work, inasmuch as one of the differences in language on which scholars have concentrated is gender. This will be further explained in the next section.

2.4.2 Gender in language

The relationship between language and gender is quite complex. It can be distinguished as linguistic (grammatical, referential, lexical) and extralinguistic gender (gender identity) (Corbett 1991, Unterbeck et al. 2011, Cao & Daumé III 2020). Moreover, there is a distinction between linguistic and social gender (Unterbeck et al. 2011). However, in the literature on language and gender, the latter term is used with different meanings (see for example Nissen 2002). Accordingly, this monograph opts for *extralinguistic* gender that indicates the referents’ gender identity.

Grammatical gender is defined by Hockett (1958: 231) as “classes of nouns reflected in the behavior of associated words”. It concerns the classification of nouns based on agreement between several words and the nouns. Word classes, such as articles, adjectives, and determiners, display inflection (see Corbett 1991). Lexical gender refers to the semantic properties of words (lexical items) that are used to denote femaleness or maleness as in the case of nouns related to kinship, such as mother and father. Lexical gender is thus a property of the noun itself and not of its extralinguistic referent (Fuertes-Olivera 2007). Referential gender relates to linguistic structures used to refer to the extralinguistic reality, i.e. to the gender identity of speakers and/or referents (Corbett 1991, Fuertes-Olivera 2007).

Linguistic gender finds different expression in languages, which can be classified into (i) grammatical gender; (ii) notional gender; and (iii) genderless lan-

guages (see Table 2.2) (Stahlberg et al. 2007). Originally, the second category was named natural gender languages; however, due to the potential semantic overlap of natural with biological/anatomical sexual categories, the term notional is here preferred as in McConnell-Ginet (2013). In grammatical gender languages, e.g. German and Italian, every noun has a gender, i.e. even inanimate objects and abstract concepts. Notional gender languages, e.g. Danish and English, are marked by lexical gender (girl/boy), which also influences compounds, e.g. chairman/chairwoman, derivational nouns, e.g. actor/actress, and gendered pronouns, e.g. she/he/it. In both (i) and (ii), gender is also referential. Human referents should thus be referred to with linguistic structures that are congruent with their gender identity. The relationship between linguistic gender, referential gender, and gender identity is not one-to-one (Cao & Daumé III 2020). For instance, many European languages have binary structures, e.g. Italian (Deutscher 2010: 99) that hinder the representation of non-binary people. Accordingly, new linguistic forms must be introduced (Hornscheidt & Sammla 2021, Italiano Inclusivo 2021a). Finally, in genderless languages, e.g. Turkish and Finnish, neither nouns nor pronouns express gender – with a few exceptions such as references to kinship, e.g. *sisko*, Finnish for sister.

Table 2.2: Expression of gender/sex in different language types (Stahlberg et al. 2007: 166)

	Grammatical gender languages	Notional gender languages	Genderless languages
Frequency	High	Middle	Low
Necessity	Often	Sometimes	Rare
Linguistic Forms	Lexical, pronominal, grammatical	Lexical, pronominal	Lexical

Additionally, gender in language (re)produces social expectations and norms. Although the representation of sex/gender varies across languages, Stahlberg et al. (2007: 164) note that maleness and femaleness can be always marked, thus concluding that sex/gender is perceived as an essential social category by speakers. The scholars also underline how such representations do not communicate a simple distinction between the sexes/genders. Rather, they reflect the underlying gender belief system, revealing evaluations and stereotypes in the way languages refer to the sex/genders. This is the reason why the question of gender

in language goes well beyond grammar and is of great interest to linguists as well as to psychologists, sociologists and many others (Corbett 1991: 163). For instance, as further described in Subchapter 2.4.4, female forms undergo *semantic derogation* (Lakoff 1973: 63), i.e. they take on an additional, negative meaning (e.g. mister/mistress). Gender-neutral forms, such as bigender nouns, can also trigger bias. In these cases, gender assignment usually occurs based on social gender, namely stereotypes by which a doctor is thought to be a man and a nurse a woman (see for example Nissen 2002). As already explained in Subsection 2.2.3, differences among languages in gender structures and connotations are particularly challenging for translation and have therefore been one of the main focuses of feminist TS (Simon 1996, von Flotow 1997).

Studies from psycholinguistics and cognitive linguistics on such a relationship between language and gender are reported in the next sections. Note that such research has been carried out quite exclusively focusing on European languages, i.e. in systems where the binary distinction between male and female is dominant (Deutscher 2010: 199). Additionally, the differences among language structures and social connotations of gender are amongst the factors that contribute to issues of gender bias in MT along with, e.g., bias in the training datasets and in MT's functioning (algorithmic bias) (see Subsection 3.1.1).

Probably the first experiment to study the influence between language and gender was conducted at the Moscow Psychological Institute in 1915. Fifty participants were asked to visualise the days of the week as people and describe them. Participants described Monday, Tuesday, and Thursday as men but Wednesday, Friday, and Saturday as women. The only reason researchers could find for this is that the former are masculine and the latter are feminine in terms of grammatical gender in Russian (Deutscher 2010: 209).

Konishi (1993) focused on two languages with grammatical gender, namely German and Spanish. To find whether grammatical gender carries connotations of femininity and masculinity, the researcher provided 40 German and 40 Mexican participants with a list of 54 words of opposed gender, such as bridge, i.e. *die Brücke*, feminine in German, and *el puente*, masculine in Spanish, in the two languages. Half of the words were masculine and half feminine. The participants were asked to give judgements using three semantic differential scales for evaluation, potency, and activity. The researcher found that masculine words received higher judgements on potency than female words in each language, concluding that grammatical gender influences perception and carries societal connotations of maleness and femaleness.

To understand whether participants indeed imagine objects as having stereotypically masculine or feminine qualities or whether the results are influenced

by the naming of the object with the corresponding article, Phillips & Boroditsky (2003) conducted a similar experiment. They asked German- and Spanish-speaking participants to describe objects that have opposite genders in the two languages. This time the study was carried out in English to test the influence of the first language even when not actively used. It resulted in participants using different adjectives according to the gender of the object. The word *key* is masculine in German and feminine in Spanish. German speakers provided adjectives such as *hard* or *heavy*, whereas Spanish speakers preferred *beautiful* or *elegant*. This tendency was observed throughout the whole experiment, confirming that if objects are provided with gender, then people tend to mentally represent that object by highlighting its stereotypically masculine or feminine qualities (Boroditsky et al. 2003: 65).

Another experiment by the scholars consisted of testing the memory of a group of Spanish and German speakers to see if gender exerts some sort of influence. Objects were given a masculine or feminine proper name. The gender of the name was consistent with the grammatical gender only half of the time. They found that participants tended to remember the object's name when its grammatical gender and the gender of its proper name corresponded, e.g. Germans were more likely to remember Patrick than Patricia as a name for an apple since apple is masculine in German (Phillips & Boroditsky 2003: 929, Boroditsky et al. 2003: 68). Results of this last experiment are considered particularly promising as, in this case, associations are not forced by asking participants to describe objects, thus bypassing the use of subjective judgements (Deutscher 2010: 212–213).

Prewitt-Freilino et al. (2012) conducted an entirely different study in which they compared gender equality across 134 countries. Their purpose was to understand whether there is a correlation between gender equality and the typology of the language spoken, namely with grammatical gender, notional gender or genderless. Based on the results provided by Boroditsky et al. (2003), the scholars wondered if the influence of gender was to be extended to the relationships between women and men as well (Prewitt-Freilino et al. 2012: 269). The scholars compiled a list of 134 countries and compared the levels of gender equality by using the gender system of their official languages as metrics. They found that (i) higher levels of gender equality are found in countries where a notional gender language is spoken; (ii) countries where a grammatical gender language is spoken show lower levels of gender equality, and (iii) countries with genderless languages are to be found in the middle of this ranking. The authors hypothesise that the use of symmetrical gender forms, where men and women are equally visible, may promote gender equality. Grammatical gender languages often have asymmetrical gender forms, e.g. German where feminine nouns are derived from

the masculine form by adding the suffix *in*, and/or numerous gender suffixes, e.g. Italian where suffixes span from *o/a* to *tore/trice* and *ore/essa*. This complicates the creation and use of symmetrical gender-fair forms (Prewitt-Freilino et al. 2012: 279).

Santacreu-Vasut et al. (2013) created an index to measure how strong and common female/male distinctions are in languages using grammatical gender. Based on this, the scholars compiled a list of 84 countries to assess the effect of female/male distinctions on the political participation of women in such countries. Their findings were that (i) the more a language emphasises the aforementioned distinctions, the more the country in which it is spoken is likely to adopt a gender quota in politics; (ii) these very same countries also show a larger increase of women participating in the political agenda after the adoption of the quota. The explanation for (i) is that distinction in gender could trigger gender stereotypes, thus representing a barrier for women who would like to actively participate in politics.

As explained in this section, languages can be classified as grammatical gender, notional gender or genderless based on their representation of sex/gender. Society and language seem to influence one another as regards gender marking: not only do speakers associate objects with stereotypically feminine or masculine characteristics according to the objects' grammatical gender, but different gender systems can also be correlated with more or less gender equality. For instance, countries in which grammatical gender languages are spoken show lower levels of gender equality because, as research suggests, the constant distinction between masculine and feminine seems to foster gender stereotypes. This point is of particular interest. Through linguistic expressions and behaviour, gender identities can be (mis- or under-)represented to different extents in languages, hence reflecting and contributing to the prevailing sexism of Western societies. It emerges that language is a socially situated phenomenon that can contribute to discrimination. This aspect is further illustrated in the next section.

2.4.3 Language as a social practice

In the previous sections, it was posited that linguistic differences can impact thought – more specifically, memory, attention, perception, and associations. The focus was on gender in language, showing that gender differences in languages are often believed to reflect gender differences in society and vice versa (e.g. Boroditsky et al. 2003). Besides, gender marking can impact gender equality (e.g. Santacreu-Vasut et al. 2013). Language, as well as translation, is thus a form of social practice, both part of and conditioned by society (Fairclough 2015: 55f).

2.4 The societal impact of gender in language

Consequently, it is not a neutral communication instrument, but rather a concrete action that shapes reality and reflects certain views of the world (AG Feministisch Sprachhandeln 2015: 7, Kramer 2016: 79). Language reflects social processes, power relations, and norms, but it can also create, normalise and reinforce them (Fairclough 2015: 55f, AG Feministisch Sprachhandeln 2015: 11, Kramer 2016: 71). This is also true for translation (Tymoczko & Gentzler 2002) as explained in Section 2.1. People can be named, classified, categorised, evaluated and excluded through linguistic means. This effect is so profound that many of the existing naming possibilities seem evident and clear, and their construction, as well as their power, is not questioned (Hornscheidt 2008: 11).⁸ Discrimination is hence (re)produced and normalised (AG Feministisch Sprachhandeln 2015: 11f).

This can be exemplified by analysing how assumptions on sex/gender are made in everyday life. A person's gender identity and their identified pronouns are often assumed based on their voice, behaviour, clothing and so forth (ProNa 2015: 32). This means that people tend to instinctively categorise a person as a woman or a man just by looking at them or by hearing their voice on the phone. In written communication, one's gender identity is also assumed by reading one's name. However, using such clues often leads to false assumptions and consequently to misgendering. Studies on misgendering show that it may cause feelings of stigma and psychological distress in transgender and non-binary people (McLemore 2015, 2018). This demonstrates the importance of using linguistic forms that include and represent people of all genders in discourse. Moreover, reading gender through the physical aspect reinforces the gender binary, cisgenderism⁹ as well as norms and assumptions on how people and sexes/genders should be. Such norms and assumptions are reinforced by the use of binary gendered forms and also stereotypical expressions, e.g. *both genders*, constantly implying that people identify either as women or as men.

Gender is continuously produced through discourse in so many different ways that its production and conceptualisation are unquestioned. It is hence perceived and experienced as something natural (Hornscheidt 2008: 24f). This happens even before birth, when the doctor assigns sex after the inspection of the foetus' genitalia via ultrasound and pronounces sentences such as "It's a girl". Butler explains:

Consider the medical interpellation which (the recent emergence of the sonogram notwithstanding) shifts an infant from an "it" to a "she" or a

⁸Actually, Hornscheidt (2008: 11) refers to German and Swedish in the passage, but such claims can be generalised.

⁹The assumption that everyone is cisgender.

“he,” and in that naming, the girl is “girled,” brought into the domain of language and kinship through the interpellation of gender. But that “girling” of the girl does not end there; on the contrary, that founding interpellation is reiterated by various authorities and throughout various intervals of time to reinforce or contest this naturalized effect. The naming is at once the setting of a boundary, and also the repeated inculcation of a norm. (Butler 2003: 7–8)

This process by which sex is assigned through the pronunciation of the sentence “it is a girl” has far-reaching consequences for the subject in question. It is expected that they will identify as a woman and people around them will behave accordingly, for example by choosing a stereotypically feminine name, using feminine pronouns, buying stereotypically feminine clothes for them and so forth. Language reproduces, confirms and creates social norms (AG Feministisch Sprachhandeln 2015: 11).

Through generalisations and assumptions, linguistic discrimination can be explicitly or implicitly practised (AG Feministisch Sprachhandeln 2015: 7). For this reason, when talking about people and sex/genders, one’s language must be carefully analysed in terms of which sexes/genders are named, how people are categorised or evaluated based on their sexes/genders, and how they are referred to, i.e. with which linguistic form and/or expressions. Such linguistic decisions and actions are, more or less consciously, based on assumptions that (re)produce gender norms and expectations (AG Feministisch Sprachhandeln 2015: 11). Through such analyses, language use can be questioned and modified to change the ruling norms, assumptions and expectations (AG Feministisch Sprachhandeln 2015: 12). Language can ultimately become an instrument of resistance. In the same way, translators can decide whether to uncritically reproduce the sexism of source texts, openly highlight and criticise it, or remove it from target texts (Burton 2010).

From a historical perspective, the interrelation between language and society has been a point of interest for feminist linguistics and TS since the 1960s (see for example Cameron 1985, Simon 1996, Kramer 2016, von Flotow 1997). During the second wave of feminism, feminists started focusing on language. Their starting point was that language is not only part of society, but that society is shaped by patriarchal notions. Language analysis should therefore help understand the system that oppresses women (see Subsection 2.4.4) and their liberation might be achieved through language reform (Cameron 1985: 3). Feminist language theory has focused on a variety of questions including: (i) the differences in language use between men and women; (ii) the representation of women in language through

the use of stereotypes and metaphors; (iii) the reinforcement of gender norms and relationships through language (von Flotow 1997: 8). Such work shows the strong link between linguistics and TS, focusing on language use as a social practice and representing the theoretical background of the present work. In the next section, studies on gender in the fields of cognitive and psycholinguistics are presented to provide concrete evidence for the impact of using gender-exclusive language.

To conclude, sexism is not the only form of discrimination that is constantly (re-)produced in language. Others include genderism, racism, ableism, and classism, amongst others (AG Feministisch Sprachhandeln 2015: 11). These forms of discrimination cannot be regarded as distinct; they interact and overlap cumulatively. Crenshaw (1989) called this phenomenon *intersectionality*. An illustrative example of how intersectionality works is the lawsuit by Emma DeGraffenreid against General Motors (GM) in 1976. GM was accused of discriminating against black people but won the lawsuit. When the judges verified that the company hired black people and white women, they believed there was no discrimination against black women, ignoring that black women face discrimination precisely because they are both black and women (Crenshaw 1989: 142f). One analogy that can be used to understand intersectionality is that of a crossroads. Every road represents a line of identity, for example, gender, sexuality, race, and class. The unique combination (or intersection) of these roads describes the multiple oppression faced by a single person.

2.4.4 Linguistic sexism

As mentioned in the previous section, interest in sexism in language spread with second-wave feminism, i.e. by the mid/late 1960s. During that time, gender was introduced as an analytical category to account for the differences in societal roles and opportunities between men and women (von Flotow 1997: 5). The feminist movement focused on the analysis of gender, which also manifests itself in language. Language as a powerful tool, used by institutions ruled by men, had to be reformed to free women from patriarchy and was therefore documented to reveal inequalities between men and women (von Flotow 1991: 73, Pauwels 2003: 552f, von Flotow 1997: 17).

As Stahlberg et al. (2007: 167) point out, a language proves to be sexist not because of the expression of gender, but rather because of how feminine forms are marked. In many languages the feminine form derives from the masculine one, hence being more complex (Stahlberg et al. 2007: 167). An example is German, a grammatical gender language, where feminine nouns are often obtained by adding the suffix *in* to masculine nouns, e.g. *Nachbar* – *Nachbar-in* to indicate

the male and the female neighbour. In such cases, Stahlberg et al. (2007: 167) argue, men and women are not treated equally on a linguistic level. One could also suggest that derived feminine forms imply that women are somehow dependent on or subordinate to men. Menegatti & Rubini (2017: 8) cite other examples from Italian too. Although in Italian noun forms are often symmetrical with suffixes *-o* and *-a* referring to male and female, some others are not – like the feminine suffix *-essa* in *dottoressa* as the feminine form of *dottore* (doctor in English, a high-level profession). This phenomenon can also be found in English, as in the pairs *hero*-*heroine*, and *actor*-*actress*.

Another main feature of a sexist use of language is the invisibilisation of women (Weatherall 2002: 13). Women are underrepresented in language due to the use of masculine generics. Masculine generics are linguistic forms used to refer to both men and to people in general (Stahlberg et al. 2007: 169). This is especially common in idioms with grammatical gender like Italian or German, (e.g. *studenti* or *Studenten* is often used to refer to a mixed group of students) but is also common in notional gender languages like English with the use of the pronoun *he* as generic (Menegatti & Rubini 2017: 9). Stahlberg et al. (2007: 175f) summarise different studies to confirm that the use of masculine generics tends to evoke masculine associations, thus discriminating against women – some of these can be found in Subsection 2.4.5. The use of masculine generics has been an object of discussion in the last few years with different strategies proposed to avoid it. In German, generic masculine forms are nowadays less common in written language and usually gender-fair alternatives are chosen, e.g. *Student*innen* or *Student:innen* (see En et al. 2021). German-speaking countries have always been sensitive to gender issues and language policies have been implemented by institutions (Sczesny et al. 2016: 5). In English too, the generic *he* is often avoided, and language authorities recommend the use of singular *they* (APA Style 2019b). In Italian, the debate on GFL has always been less lively, focusing instead on the feminisation of agent nouns. For example, high-level professions such as *minister* or *president* are still often used in the masculine form even with reference to women (see Gheno 2019).

Another issue of sexism is the conceptual asymmetry in languages (Litosseliti 2014: 15). Several words describing women have no counterpart and vice versa. Menegatti & Rubini (2017: 5f) remark how this phenomenon relates to gender stereotypes according to which women are pure and family-oriented. Terms or expressions like *working mother* or *career woman* have no male counterpart. This asymmetry is also found in the field of high-level or male-dominated professions, i.e. job titles like policeman, fireman and so forth (Menegatti & Rubini 2017: 6). As Pauwels (2003: 553) underlines, the phenomenon of lexical gaps in this field

is very telling. There is a lack of words describing women in certain roles and occupations whereas masculine lexical gaps in female-dominated professions are usually filled quickly.

Women are also often referred to in terms of their relationships to men (Lakoff 1973: 65). They are characterised as someone's wife or girlfriend. Besides, as Lakoff (1973: 65f) underlines, in journalistic writing men are more likely to be referred to just by their name, whereas women by relationship to a man. Forms of address even used to change depending on marital status. An unmarried woman was called *Miss* while a married one *Mrs.* This distinction was and sometimes is still made in other languages such as German with *Frau/Fräulein* and Italian *signora/signorina*. This difference is crucial because it seems to imply that marital status was somehow more important for a woman than for a man because English only has one form for addressing a man, namely *Mr.* (Weatherall 2002: 21). Since the 1960s, *Ms* has been introduced to avoid such asymmetry with men, although the other forms of address are sometimes still used. Similarly, when telling stories about women in journal articles, reference is often made to their being mothers, even though this is irrelevant to the context of the story, to their womanhood (Murgia 2021b), or, in general, to their appearance (Weatherall 2002: 19).

There are several further elements of a sexist use of language. These include metaphors. Metaphors used to describe women relate for example to immaturity, animals, clothing, food, vehicles, and furniture (Weatherall & Walton 1999). This means that women are often referred to as *babe*, *bitch*, *sweetie pie* and so forth. In particular, the animals named in such metaphors are usually domesticated or hunted for sport (Weatherall & Walton 1999). Other forms of linguistic sexism are the lack of expressions to refer to the experience of women, especially in the field of sex and sexuality, giving compliments by comparing them to men, e.g. "she thinks like a man", insulting men by comparing them to women, e.g. "he's a real old woman" (see Weatherall 2002), and the use of gender stereotypes (see APA Style 2019a).

2.4.5 The societal impact of gender-exclusive language

Feminist analyses of sexist language use show how women are linguistically disregarded in a multitude of ways. Many claim that this also has an impact on society (see for example Gasparrini 2019 and Murgia 2021b). In particular, research in psycholinguistics and/or cognitive linguistics has concentrated on the use of masculine generics and its impact on (i) the perception that the prototypical human is a man; (ii) on the presence of women in the labour market; and

(iii) their visibility. Some studies in this field are reported in this section. This overview is by no means complete, but it demonstrates the concrete societal impact of gender-exclusive language. Accordingly, it is time to adopt GFL strategies to make all (or no) genders visible and hence foster linguistic and social equality.

Based on previous findings, Hamilton (1988) wondered whether the use of masculine generics also had an impact on speakers rather than only on listeners. The scholar asked 120 female and male college students to complete a set of sentences in two different conditions, namely, using masculine generics or gender-fair alternatives. The participants then had to describe the images evoked in their minds. Bias was present in both conditions but to a far greater extent in the first one, i.e. when completing sentences with masculine generics. Moreover, men were generally found to be more biased than women.

Bem & Bem (1973) discovered that women hesitated to apply for jobs if the job adverts were presented solely in masculine forms. They carried out an experiment asking female and male students to indicate how likely they were to apply for twelve stereotypically male or female jobs. The advertisements were written (i) in gender-exclusive language; (ii) in (binary) gender-inclusive language; (iii) by exclusively appealing to the gender that rarely takes such jobs. Gender-exclusive language affected both female and male students, who were not interested in jobs if the advertisement targeted the opposite gender. In addition, women were more likely to apply for a job if the advert addressed them explicitly.

Stahlberg & Sczesny (2001) focused on the effect of masculine generics on German native speakers in different studies. Participants (i) were asked to fill out a questionnaire about personal preferences, in which they had to name their favourite athlete, musician and so forth. Questions were phrased using either only masculine forms, paired forms, or gender-neutral forms. Similarly, (ii) other participants were asked about their media consumption and, among the questions, had to name three athletes, singers, politicians, etc. they knew. In this case, the questionnaire was written using masculine forms, paired forms or the nowadays outdated *Binnen-I*.¹⁰ Furthermore, (iii) Stahlberg & Sczesny (2001) asked participants to name possible candidates for the office of chancellor in Germany, presenting questions either using the masculine form or paired forms. In every study configuration, women were named less often when masculine forms only were used.

Stout & Dasgupta (2011) conducted three studies assessing the reaction of women and men to the use of gendered language in a professional context. Par-

¹⁰The so-called *Binnen-I*, i.e. a capital I separating the masculine from the feminine form of the nouns, is nowadays considered as exclusive as it implicitly reproduces a binary conception of gender (see for example Hornscheidt & Sammla 2021: 49).

ticipants were asked to read (study #1) a job description or participate in a mock interview where they were given a description of the role and the work environment (study #2 and #3). The descriptions were delivered using (i) only masculine forms (all studies), (ii) paired forms (study #1), and (iii) also neutral forms (study #2 and #3). Participants were asked about perceived sexism, feelings of inclusion or exclusion in the work environment, motivation to pursue the job, and identification with it. The researchers found that, although both men and women perceived the jobs described in gender-exclusive terms as more sexist, women felt generally more ostracised, less motivated and identified less with such jobs than men. In a nutshell, women were more affected by the use of gender-exclusive language than men.

Horvath & Sczesny (2016) carried out a study to investigate the effect of masculine generics on the evaluations of women in hiring simulations. They asked participants to evaluate male and female applicants for two different job positions, namely a low-status role and a high-status leadership position. The advertisements for the positions were written using masculine forms, masculine forms with the addition of “m/f” in brackets, or paired forms. Horvath & Sczesny (2016) found that women were perceived as similarly suitable for the lower status position in each case, but were perceived as less suitable for the higher status position when masculine forms, with or without “m/f” in brackets, were used.

Noting that white men are often leaders in different domains, Bailey & LaFrance (2017) wanted to find whether this is supported by an androcentric belief system. The researchers conducted an experiment and asked participants to first pre-select three faces out of a set of 20 photos and then choose a single face that could represent humanity as a whole. The task was phrased using either a gender-exclusive, a gender-neutral, or a binary expression, namely (i) mankind, (ii) human, or (iii) man or woman. Bailey & LaFrance observed an androcentric bias in each case, i.e. participants generally selected a male person as representative of humanity. However, language had an impact on the selection, since in (iii), participants’ choices were more balanced.

These studies represent only a very limited selection of research on masculine generics in the field of cognitive linguistics and psycholinguistics. The consequences of gender-exclusive language are nevertheless quite clear. Masculine generics foster the idea that the prototypical human is a man, who is often a straight, white, middle-class, able-bodied cis-man, hence men being the norm and women being rather the exception. Women are less visible, with even ‘real-life’ consequences such as their exclusion from specific job positions. Non-binary individuals and non-binary GFL are still underresearched. Some first studies are presented in Chapter 4.1.5.

This chapter has elucidated the close connection between TS and sociolinguistics. Language does not exist in isolation; it is shaped by societal assumptions and norms, which influence language use. These norms and assumptions are also reflected in translation practices, as demonstrated by feminist and queer translation scholars. Consequently, translators bear social responsibility for ensuring respectful representation of marginalised groups, such as non-binary individuals. This social responsibility of translators underscores that translation extends beyond a mere process of linguistic transfer, highlighting its potential to foster societal change.

3 Gender issues in machine translation

The previous subsections (see Subsection 2.2.3, 2.4.2, and 2.4.4) showed how (i) languages differ in the way they refer to gender and gender conveys social connotations; (ii) language use can be sexist and disregard women and other sexes/genders; and (iii) human translators are confronted with difficulties when translating gender in language. It is well-known that technology reflects societal prejudices or biases (e.g. Bolukbasi et al. 2016). Therefore, this chapter introduces gender bias in MT starting with a definition of bias itself. The term is complex and has overlapping and/or contrasting definitions (Crawford 2017). A precise and exhaustive definition was provided by Crawford in the field of machine learning, so this broader field is briefly taken into account in the first section. The second section focuses on gender bias in MT, presenting an overview of the literature in the field, while the third section analyses its sociotechnical implications. Finally, PE is introduced as a common practice within the translation industry. Gender bias remains one of the greatest issues in MT, but little research addresses the PE effort to adjust biased output, i.e. Savoldi et al. (2024) and the present work.

3.1 Gender bias in machine learning: definition and classification

Machine learning can be defined

as a set of methods that automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision-making under uncertainty. (Murphy 2012: 1)

Typical machine-learning applications may include document classification, email spam filtering, image classification and face recognition (Murphy 2012: 5) as well as deep learning applications like the recently popular ChatGPT. According to Crawford (2017), the term *bias* has overlapping and also partially contradictory meanings.

It is worth focusing on (i) the common meaning of the term as an undue prejudice; (ii) statistical bias, since statistics is used in several machine-learning

3 Gender issues in machine translation

applications; models may be skewed and select more frequently some members of a population instead of others, e.g. facial recognition models are mostly trained with images of white males and perform much worse on black people or women (see Buolamwini & Gebru 2018); and (iii) the normative legal meaning, based on which judgments are the results of preconceived notions or prejudice (Crawford 2017).

Crawford (2017) underlines how a system can be statistically unbiased but nevertheless produce legally biased outputs. The researcher reports that 83% of the 4.4 million people who were stopped and questioned based on the Stop-and-Frisk programme run by the New York City Police Department in 2012 were Black or Hispanic. This reflects the long history of discrimination against these groups in policing and the programme was discontinued. Bias is therefore to be considered both a social and a technical issue, i.e. sociotechnical.

Bias is to be found in machine-learning applications because the training data used can be incomplete, skewed, drawn from non-representative samples, or constructed in non-transparent ways (Crawford 2017). In 2015, Amazon realised that the recruiting tool used to review the CVs of candidates applying for programming jobs was discriminating against women. The system had been trained with resumes sent by applicants over a 10-year period and a vast majority of them were men – thus confirming male dominance in this particular sector. The tool learned to penalise CVs including the word *women* leading Amazon to discard it (Dastin 2018). Data are not neutral but produced in a specific societal and cultural setting, so they encode societal biases like sexism, racism, and classism amongst others. O’Neil (2017) explains, for instance, how bias is encoded in algorithms that impact the life of many marginalised individuals and reinforce inequality.

Zou & Schiebinger claim that algorithms also play an important role in the propagation of bias:

A typical machine-learning program will try to maximize overall prediction accuracy for the training data. If a specific group of individuals appears more frequently than others in the training data, the program will optimize for those individuals because this boosts overall accuracy. (Zou & Schiebinger 2018)

Bias can be mainly classified into allocative and representational bias (Crawford 2017). The former is related to decision-making and is found when applications deny resources to a certain group, e.g. a woman is not considered for a job or is denied a mortgage just because she is a woman. The latter relates to identity and reinforces the subordination of certain groups (Crawford 2017), e.g. MT

3.1 Gender bias in machine learning: definition and classification

systems rarely generate feminine and gender-fair forms (Prates et al. 2020, Dev et al. 2021). As shown in Table 3.1, there are different types of representational bias.

Table 3.1: Representational harms in different applications proposed by Sun et al. (2019) based on Crawford (2017); the columns represent denigration (D), stereotyping (S), recognition (R), and underrepresentation (U)

Task	D	S	R	U
Machine Translation	x	x	x	x ¹
Caption Generation	-	x	x	-
Speech Recognition	-	-	x	x
Sentiment Analysis	-	x	-	-
Language Model	-	x	x	x
Word Embedding	x	x	x	x

These are denigration, stereotyping, recognition, and underrepresentation. Specifically, denigration refers to the use of culturally or historically derogatory terms, e.g. gendered identity terms are sometimes translated with slurs into other languages; stereotyping relates to the use of societal stereotypes, e.g. word embeddings associate nurses more strongly with women and doctors with men; recognition concerns the inaccuracy of algorithms in recognition tasks, e.g. speech detection works better for standard varieties of a given language; and underrepresentation describes the low representation of a specific group, e.g. neopronouns are mistakenly machine-translated.

Since language technologies have found wide use in everyday life, their societal impact has raised concerns (see for example Hovy & Spruit 2016, Blodgett et al. 2020). Hovy & Spruit (2016: 593) concentrated specifically on NLP applications, noting how the use of language data always implies demographic bias, intended as “latent information about the demographics in it”. Such bias can have severe consequences, namely (i) exclusion, (ii) overgeneralisation, and (iii) underexposure. The first means that an NLP application developed in the USA is usually based on a standard language that may be easier to use for white males than for citizens of Latino origins, rendering it less user-friendly. The second may cause,

¹Sun et al. (2019) do not categorise underrepresentation as one of the harms caused by MT systems. However, this is the case of women but also of non-binary individuals, who precisely according to Sun et al. (2019) are ignored in the field of NLP. Savoldi et al. (2021) also argue that MT systems misrepresent women and erase the identity of non-binary people.

amongst others, users’ attributes such as gender, age, sexual orientation or religious views to be wrongly inferred, e.g. an email could address people with the wrong gender. The third refers to the underexposure of language varieties for economic reasons – English is the most frequently used language in NLP tools. The literature on bias in NLP applications is vast (see Sun et al. 2019 for an overview of gender bias, and Hutchinson et al. 2020 for an overview of other societal biases in general).

3.1.1 Gender bias in machine translation

The previous section provided a definition of bias in machine learning. Blodgett et al. (2020: 5454) underline how the analysis of bias is actually an “inherently normative process” that requires establishing “what kinds of system behaviours are harmful, in what ways, to whom and why”. Such a claim has also been endorsed by Hardmeier et al. (2021) who propose writing a bias statement when submitting papers on gender bias in NLP. Consequently, based on the definition by Friedman & Nissenbaum (1996), biased MT models are here defined as those “that *systematically* and *unfairly* discriminate against certain individuals or groups of individuals in favor of others” (emphasis as in original).

The term *discrimination* is occasionally used in this chapter. However, in this context, it does not refer to explicit legal discrimination but to systematic unfair treatment in algorithmic decisions based on gender. This aligns with the concept of normative bias (Crawford 2017), and with Friedman & Nissenbaum’s (1996) formulation of unfair system behaviour toward underrepresented groups.

Similarly to human translation, gender bias in MT systems arises because linguistic gender is complex and ambiguous. First, its implementation differs across languages. For example, sentences in English such as “I am happy” are neutral, but a gender must be assigned when translating into Italian or French. Moreover, gender has social connotations; doctors are usually thought of as men and nurses as women. MT systems must therefore simultaneously handle various gender-related tasks with reference to three different areas, namely (Monti 2020: 457):

1. Anaphora resolution: the process by which MT must identify references back to a previously introduced entity. (Mitkov (1999) provided an overview of the literature in this field back at the end of the 1990s). This is especially problematic when pronouns refer back to entities in another sentence because some NMT systems mostly translate texts sentence by

3.1 Gender bias in machine learning: definition and classification

sentence or part by part without taking the context of the whole document into account.²

2. Named-Entity Recognition (NER): the correct identification and classification of proper nouns, such as those referring to people and institutions. Research has focused on approaches to improve MT system performances on NER (see for example Babych & Hartley 2003).
3. Agreement between word classes such as verb-noun or adjective-noun. Saunders & Byrne (2020) claim that such errors particularly affect translation into grammatical gender languages, as they are more inflected than notional gender (or genderless) ones.

As a result, MT systems rarely generate correct gendered inflections or proforms (Sun et al. 2019, Monti 2020: 457), rely on gender stereotypes (Prates et al. 2020) and have a strong male default (Monti 2020: 457).

Friedman & Nissenbaum (1996: 333) divide bias into three different categories, namely (i) pre-existing, (ii) technical, and (iii) emergent bias. Bias (i) exists independently of the system and is rooted in institutions, social practices, and attitudes. In MT, it arises from both an overrepresentation of men and a sexist use of language (see Subsection 2.4.4) in human-produced datasets used to train such systems. For instance, Vanmassenhove et al. (2018) and Choubey et al. (2021) found that sentences with gender-neutral or female gender inflections occur much less frequently in the data used for training.

Bias (ii) is related to the technical design of the system. In this case, models not only reproduce bias, but they also amplify it due to their functioning and their architecture³ (Vanmassenhove et al. 2019, Vanmassenhove, Shterionov & Gwilliam 2021, Costa-jussà et al. 2020a, Roberts et al. 2020). For instance, Vanmassenhove et al. (2019) found that both statistical and neural MT systems are biased towards the most frequently occurring words in the language corpora used to train them, hence overgeneralising and presenting loss of linguistic richness and diversity. The scholars refer to such MT outputs as “machine translationese”. This tendency was confirmed by Roberts et al. (2020), who found that some models are biased towards more recurring words, and therefore also masculine pronouns. This leads MT systems to perform poorly on linguistic diversity. In addition, Costa-jussà

²While much of the early work in MT, and consequently in gender bias, focused on the sentence level, there is growing recognition of the importance of context beyond individual sentences. MT systems are increasingly addressing discourse-level phenomena and document-level contextual information to improve translation quality and reduce bias (Jin et al. 2023)

³In this context, architecture refers to how components of a computer system are combined.

et al. (2020a) found that the architecture of NMT impacts performance on gender diversity. When assessing gender bias or proposing debiasing methods, the specific architecture and training settings employed should be considered.

Bias (iii) arises when a system is used in a context other than the one it was designed for. For example, Rabinovich et al. (2017) demonstrated that gender traits are weakened in translation, leading Hovy et al. (2020) to find that MT systems make authors sound averagely male in translation. This means that models are not equipped to adapt to user diversity.

Finally, another issue in this research area concerns Translation Quality Assessment (TQA). Bentivogli et al. (2020) and Monti (2020: 465) claim that metrics used for this task, such as the Bilingual Evaluation Understudy (BLEU) (Papineni et al. 2002), Translation Edit Rate (TER) (Snover et al. 2006) or the Multidimensional Quality Metrics (MQM) (Lommel et al. 2014), do not specifically consider the issue of gender bias. Consequently, using these metrics after debiasing MT does little or nothing to confirm whether the debiasing approach was effective. Moreover, Zaranis et al. (2025) found that quality estimation (QE) metrics – metrics that automatically assess the quality of machine-translated text, without relying on human reference translations – are prone to bias. For instance, when a human entity’s gender in the source text is undisclosed, masculine-inflected translations score higher than feminine-inflected ones, and gender-neutral translations are penalised. Monti (2020: 465) proposes that corpora be annotated with gender information and used as reference material when evaluating MT models, whereas Bentivogli et al. (2020) created the MuST-SHE Corpus, a specific benchmark for the assessment of gender bias in (speech) translation technologies.

Research on gender bias in MT has been going on for some time. It initially focused on the translation of grammatical and pronominal gender in the aforementioned areas of anaphora resolution, named entity recognition and agreement between word classes (see for example Mitkov 1999, Babych & Hartley 2003). More recent contributions, however, have consisted of comparative studies assessing MT performance across different commercial and/or academic systems, approaches, and language pairs, starting with a study by Frank et al. (2004). These newer studies demonstrate how all MT paradigms, which are mainly rule-based, statistical and neural, are subject to gender bias. In the case of NMT, gender bias in the training corpora is also exacerbated and systems tend to overgeneralise. An overview of this research field is presented below.

As already mentioned, gender issues in MT systems have been studied for decades (see for example Mitkov 1999, Witthoft et al. 2003, Frank et al. 2004). However, interest in this field has dramatically grown over the last few years. Olson (2018) explains how, at the beginning of 2018, examples of gender bias

3.1 Gender bias in machine learning: definition and classification

found in Google Translate had been shared via social media. Since then, numerous studies on gender bias in Neural MT (NMT) have appeared. Based on Savoldi et al. (2021), gender bias in MT is discussed by grouping studies according to different conceptualisations: (i) benchmarks created to assess gender issues in MT systems; (ii) studies on contrastive linguistic phenomena to assess whether gender is correctly rendered in translation; (iii) studies on the perpetuation of gender stereotypes in MT; and (iv) gender preservation in MT (Savoldi et al. 2021; an overview of the studies can be found in Table 3.2.)

Sun et al. (2019: 1632f) refer to datasets specifically created to assess gender bias in an NLP system as “Gender Bias Evaluation Testsets (GBETs)”. For instance, Stanovsky et al. (2019) produced WinoMT, a challenge test for the evaluation of gender bias in NMT models that consists of 1,826 masculine, 1,822 feminine and 240 neutral sentences respectively. Sentences are structured as follows: “The doctor asked the nurse to help her in the procedure”. They contain two human entities and one of them is referred to with a pronoun specifying their gender. Entities are featured both in pro- and in anti-stereotypical gender roles. This allows researchers to evaluate if NMT systems can correctly resolve pronouns or if they perpetuate stereotypes when translating. However, a shortcoming of WinoMT is that in examples such as the aforementioned, *her* could refer to either the doctor, nurse, or a third party. Therefore, WinoMT may not necessarily indicate gender bias. The metrics used for assessment are (i) gender accuracy; (ii) the difference in score between correctly translated masculine and feminine forms; and (iii) the difference between pro- and anti-stereotypical translation. A neutral version of WinoMT was produced by Saunders et al. (2020) in which 1,826 entities are referred to with singular *they* to assess the translation of gender-neutral entities in NMT systems. To the best of my knowledge, this is the first research study to investigate the translation of gender-fair forms in MT, even though this was not its main purpose.

Escudé Font & Costa-jussà (2019) developed an occupation test set to evaluate the effect of the debiasing methods tested on their NMT models. The test consists of a sentence pattern featuring human entities in pro- or anti-stereotypical gender roles and is structured as follows: “I’ve known [her/him/<proper Spanish noun>] for a long time, my friend works as [a/an] <occupation>”. The list of occupations is retrieved from the U.S. Bureau of Labor Statistics as in Prates et al. (2020). Since the noun *friend* has to be gendered in Spanish and the sentence includes context, this benchmark allows for the assessment of gender bias, more specifically the perpetuation of gender stereotypes.

Habash et al. (2019) created The Arabic Parallel Gender Corpus. The corpus consists of 12,238 first-person-singular Arabic sentences with their English trans-

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Table 3.2: Overview of the studies conducted to assess gender bias in MT. Based on Savoldi et al. (2021), the table provides information on the typology of studies as well as the genders (binary or non-binary) and the harms addressed.

Study	Typology	Gender	Harms
Stanovsky et al. (2019)	benchmark: WinoMT	B	underrepresentation, stereotyping
Escudé Font & Costa-jussà (2019)	benchmark: Occupation test set	B	underrepresentation, stereotyping
Habash et al. (2019)	benchmark: The Arabic Parallel Gender Corpus	B	underrepresentation
Costa-jussà et al. (2020b)	benchmark: GeBioCorpus	B	underrepresentation
Saunders et al. (2020)	benchmark: gender-neutral WinoMT	NB	recognition, underrepresentation
Frank et al. (2004)	contrastive	B	underrepresentation
Abu-Ayyash (2017)	contrastive	B	underrepresentation
Rescigno et al. (2020)	contrastive	B	underrepresentation, stereotyping
Prates et al. (2020)	perpetuation of stereotypes	B	underrepresentation, stereotyping
Cho et al. (2019)	perpetuation of stereotypes	B	underrepresentation, stereotyping
Stanovsky et al. (2019)	perpetuation of stereotypes	B	underrepresentation, stereotyping
Gonen & Webster (2020)	perpetuation of stereotypes	B	underrepresentation, stereotyping
Vanmassenhove et al. (2018)	gender preservation	B	allocative: reduced quality of service
Hovy et al. (2020)	gender preservation	B	underrepresentation
Vanmassenhove, Emmery, et al. (2021)	gender preservation	B	underrepresentation
Currey et al. (2022)	gender preservation	B	underrepresentation

3.1 Gender bias in machine learning: definition and classification

lation. The scholars retrieved data for the corpus from the OpenSubtitles Corpus (Lison & Tiedemann 2016), selected sentences including first-person-singular pronouns and annotated gender (F as feminine, M as masculine and B as ambiguous). Sentences containing masculine entities were copied and modified to change the gender into feminine and vice versa to obtain a balanced corpus. This was later used to train different NLP systems.

Costa-jussà et al. (2020b) developed the GeBioToolKit, a tool to collect parallel sentences retrieved from bibliographical entries on Wikipedia that are balanced in reference to gender (see Figure 3.1 for an overview of its architecture). They created the GeBioCorpus, comprising 2,000 sentences in English, Spanish and Catalan, that allows the evaluation of gender accuracy in translation. The languages were selected as they belong to different linguistic subfamilies, namely Germanic and Romance, and English-Catalan represents a low-resourced pair.

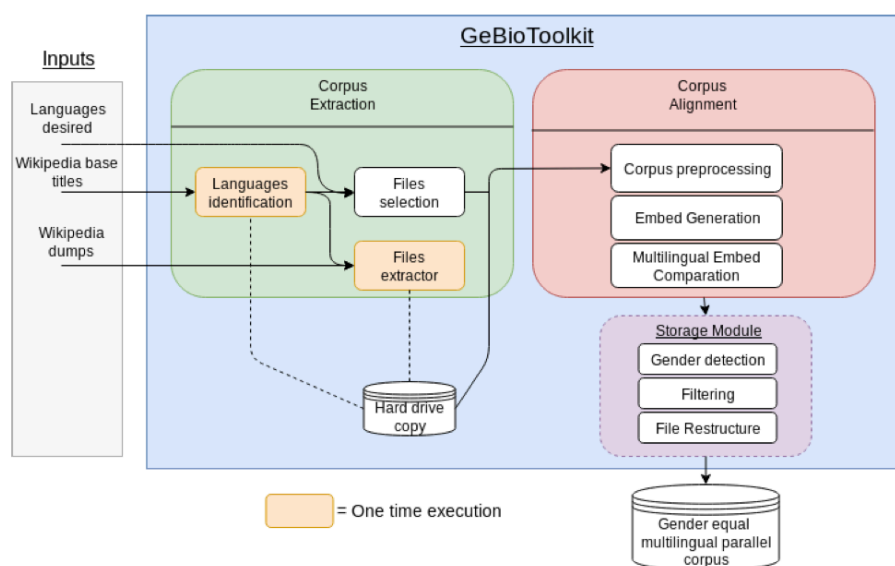


Figure 3.1: GeBioToolKit architecture (Costa-jussà et al. 2020b)

Bentivogli et al. (2020) posit that the use of artificial datasets in gender bias assessment may produce unreliable results. The researchers released the MuST-SHE Corpus, a multilingual, natural benchmark built on TED talks and composed of about 1,000 audio, transcript and translation triplets for each of the three available language pairs (EN-IT, EN-FR, and EN-SP). The balanced, annotated corpus features two main gender phenomena: gender can be recovered (i) through the

person speaking and (ii) through context. While MuST-SHE was used to assess gender bias in speech translation technologies, it can also be used in text-to-text MT.

Vanmassenhove & Monti (2021) introduce gENder-IT, an annotated English–Italian parallel challenge set specifically designed to examine how MT handles natural gender phenomena, with a focus on cross-linguistic ambiguity. Their dataset enhances the English source with word-level gender tags, and on the Italian side they offer multiple alternative translations where gender distinctions are relevant. By enabling fine-grained evaluation of MT’s ability to correctly resolve gender when translating into a grammatically gendered language, gENder-IT offers both a benchmark and diagnostic tool for identifying and improving gender-aware translation behaviour.

Currey et al. (2022) present MT-GenEval, a multilingual benchmark for evaluating gender accuracy in MT across eight languages. The dataset includes both counterfactual sentence pairs, differing only in gender, and contextual segments requiring discourse-level gender inference. They propose metrics for gender accuracy and the gender quality gap, revealing that most MT systems exhibit significant gender bias and perform inconsistently across genders. Their findings underscore the importance of gender-balanced data and contextual modeling for mitigating bias in MT.

Other research endeavours consist of contrastive analyses of linguistic phenomena relating to gender. As shown in Subsection 2.4.2, languages express gender in different ways, i.e. some languages are not or are rarely marked for it, while others are morphologically richer. This complicates the translation process, as MT systems lack extralinguistic information to handle such differences (see Subsection 3.1.1 for deeper explanations on the topic). For example, Frank et al. (2004) selected a set of 20 sentences to be translated from EN into DE and from DE into FR. The sentences contained peculiar gender phenomena of each target language, such as pronominal reference, appositions, and predicate constructions. They were used to assess gender bias in four different commercial Statistical MT (SMT) systems, e.g. Prompt Online Translator and Systran. The researchers found that when translating from EN into DE the majority of the tested systems could not correctly process feminine forms, even though the gender of the entity in the source sentences was clear. For example, in “Our manager, Mary, arrived today”, the profession was translated as *Manager* instead of *Managerin* (Frank et al. 2004: 5).⁴ As regards the language pair DE-FR, SMT systems exhibited most of the analysed phenomena.

⁴Note that this study, like many others, is based on a binary conception of gender and it is supposed that gender can be inferred from the proper name.

3.1 Gender bias in machine learning: definition and classification

Abu-Ayyash (2017) translated four technical texts from English into Arabic in order to study errors made by three commercial MT systems, namely Systran, Google Translate, and Bing Microsoft Translator. The scholar provided a qualitative analysis of errors concerning (i) subject-verb agreement; (ii) adjectival-noun agreement; and (iii) pronoun-antecedent agreement. The gender systems of the two languages differ considerably, Arabic being morphologically richer than English. Hence, automated translation struggled to produce correct inflections. The area that showed more issues was (iii). While in English *you*, *they*, and *it* have single a form, in Arabic they differ based on number and gender. The tested systems lacked extralinguistic information in English in order to correctly transfer them.

Rescigno et al. (2020) created a simple test set of English sentences composed of the first-person singular *I*, the verb to be, and the indefinite article, mixed with 107 professions or 30 epicene nouns and 136 personality adjectives. The researchers produced two different types of sentences: (i) minimal clauses like “I am an assistant” and (ii) clauses with an adjective as well, for instance “I am a beautiful assistant”. This test set was then translated into Italian, French, and Spanish with Google Translate, DeepL Translator and Bing Microsoft Translator. The researcher found that in the first case, the systems adopted a male default, translating sentences much more often in the masculine than in the feminine. In the second case, they found that systems perpetuated gender stereotypes. For example, when sentences contained the adjective *beautiful*, they were more often translated in the feminine than in the masculine form.

Prates et al. (2020) compiled a list of short and simple sentences in 12 genderless languages, among others Turkish and Hungarian. As shown in Figure 3.2, these sentences contained different typologies of professions. The test set contained 1,019 professions altogether that were obtained from the U.S. Bureau of Labor Statistics (BLS). The test set was translated into English with the Google Translate API. Based on the produced target sentences, statistics were collected to prove that the translation system is generally biased. Target sentences included a male subject more often than a female or a neutral one. More specifically, if the sentences contained male-dominated professions, gender bias was higher. In addition, a comparison was drawn between these results and data on how many women are to be found in each job position. By doing so, the researchers demonstrated that Google Translate did not reproduce the actual ratio of female to male workers. They also created another set of sentences with 21 adjectives in order to understand whether stereotypical gender roles can affect MT’s outputs as well. They found that sentences containing certain adjectives

3 Gender issues in machine translation

were translated with masculine pronouns (e.g. proud, guilty, brave) but others with feminine ones (e.g. shy, desirable, dumb).

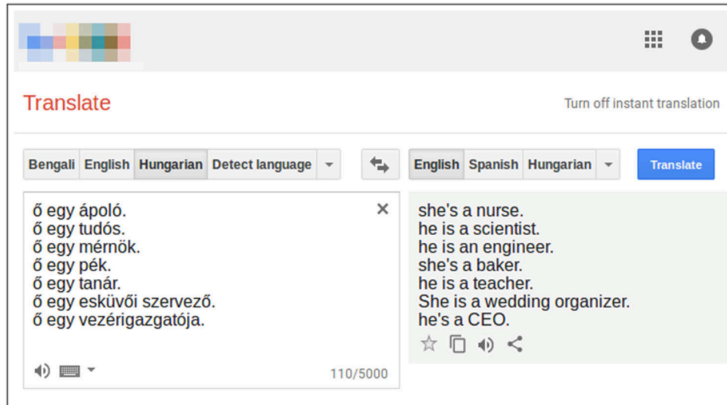


Figure 3.2: Example of gender bias based on stereotyping in the language pair Hungarian-English (Prates et al. 2020: 6365)

Cho et al. (2019) conducted similar and yet wider research. They focused on the creation of a test set in Korean, another genderless language, which was translated into English. The corpus was called the Equality Evaluation Corpus (ECC) and was composed of 4,236 sentences of three different kinds: 324 containing a positive or a negative judgment towards a person and 735 containing an occupation. To these standard 1,029 sentences, a formal version of the pronoun as well as a politeness suffix were added in order to reach the final size. The purpose was to understand whether sentences were translated with different subjects depending on (in)formality, (im)politeness, (negative and positive) judgements, and occupations. The scholars also developed an index, namely the translation gender bias index (TGBI), to evaluate three different translation tools – Google Translate, Naver Papago, and Kakao Translate – and they did a qualitative analysis of the collected data. The study confirmed that occupations are the most biased field. Formal and negative sentences also had a stronger male default than the informal and positive ones. In contrast, politeness did not seem to affect the MT outputs.

Stanovsky et al. (2019) created the WinoMT test set to assess gender bias in four commercial MT tools, namely Google Translate, Bing Microsoft Translator, Amazon Translator and Systran, and two academic models. WinoMT was translated from English into eight other languages with grammatical gender, for instance

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French, German, and Italian. The researchers found that NMT systems often ignore context and are inclined to a certain translation because they reproduce gender stereotypes.

Gonen & Webster (2020) designed an automated system to identify gender issues in MT. This pipeline filters gender-neutral sentences in English that include a human entity (e.g. “so is that going to affect my chances of becoming a *counselor*?”) which is then replaced to enlarge the dataset and detect gender issues (e.g. “so is that going to affect my chances of becoming a *nurse*?”). Such sentences are translated into French, German, Spanish, and Russian (all grammatical gender languages that require gender assignment) with Google Translate, aligned and the gender of the entity is tagged with a morphological analyser. As shown in Figure 3.3, the pipeline finally identifies sentence pairs in which entities have been assigned a different gender and includes them in a final dataset used to assess gender bias in MT. On analysing the resulting dataset, the researchers found that classical gender stereotypes are reproduced by MT and that *fighter* is associated with maleness.

	Source Sentence (En)	Translation	M/F
Fr	also should i ask the manager what the pay would be if i got the job prior to flying out?	De plus, devrais - je demander au <i>gestionnaire</i> quel serait le salaire si je obtenais le poste avant de prendre l'avion?	M
	also should i ask the secretary what the pay would be if i got the job prior to flying out?	De plus, devrais - je demander à la <i>secrétaire</i> quel serait le salaire si je obtenais le poste avant mon départ?	F
De	currently thinking about learning a trade (mostly a electrician).	Derzeit über das Erlernen eines Gewerbes nachdenken (meistens <i>Elektriker</i>).	M
	currently thinking about learning a trade (mostly a cook).	Derzeit über das Erlernen eines Gewerbes nachdenken (meistens eine <i>Köchin</i>).	F
Es	- decided to become a teacher : spent a year working 2 jobs and doing prerequisites for a masters in education.	- Decidí ser <i>maestra</i> : pasé un año trabajando en 2 trabajos y haciendo requisitos previos para una maestría en educación.	F
	- decided to become a lecturer : spent a year working 2 jobs and doing prerequisites for a masters in education.	- Decidí ser <i>profesor</i> : pasé un año trabajando en 2 trabajos y haciendo requisitos previos para una maestría en educación.	M
Ru	i read about a psychologist who upgraded into becoming a m.d.	Я читал о <i>психологе</i> , который превратился в Мд.	M
	i read about a nurse who upgraded into becoming a m.d.	Я читал о <i>медсестре</i> , которая превратилась в доктора медицины.	F

Figure 3.3: Examples from the dataset produced with the pipeline designed by Gonen & Webster (2020: 1994)

Research on gender preservation builds on previous findings that demonstrate how MT weakens specific author traits, such as gender (see for example Rabinovich et al. 2017). From this starting point, Vanmassenhove et al. (2018) proposed to morphologically tag corpora used to train NMT with speakers’ gender and eventually compared their baseline model with their debiased systems. As often claimed, when translating simple sentences such as “I am happy” from less into

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more inflected languages, for example from English into French (the output may be either “je suis heureux” or “je suis heureuse”), NMT systems lack extralinguistic information to define the speaker’s gender and therefore often adopt a male default or rely on gender stereotypes learned during training. This claim was confirmed by Vanmassenhove et al. (2018), who found that their baseline model struggled to render the correct speaker’s gender, whereas the debiased systems showed improved performance.

Hovy et al. (2020) collected reviews containing information about user age and gender in English, German, French, and Dutch from Trustpilot and translated them (from or into English) with Google Translate, DeepL Translator, and Bing Microsoft Translator. They used an age and gender classifier to investigate whether such author traits are preserved in translation. They found that when translating into English, the systems were generally biased towards men, i.e. making authors sound generally more male. When translating from English into the other languages, systems showed bias towards women in French and German. Generally, the age classifier overestimated the age for translations.

Finally, beyond identifying bias, recent work focuses on developing technical debiasing techniques, emphasising interpretability and control in MT systems. Interpretability aims to understand why a model makes a particular gendered translation, shedding light on the underlying biases in its training data or architecture. This allows for more targeted interventions rather than relying solely on post-hoc corrections. For instance, Attanasio et al. (2023) conducted the first thorough study of gender bias in instruction-fine-tuned models using the WinoMT benchmark (English → German/Spanish). They find that these models default to male-inflected translations, even when feminine pronouns or anti-stereotypical contexts are present. Through interpretability analysis, they reveal that misgendering tends to occur when models ignore pronouns in input utterances. Leveraging this insight, the authors propose a few-shot debiasing strategy, selecting in-context examples where pronouns were previously overlooked. This method substantially improves gender fairness with as few as four exemplars.

As shown in Table 3.2 and by Lardelli & Gromann (2023a), most of the studies in this field have been conducted based on the assumption that gender is binary. Nevertheless, there is a multitude of genders (see Chapter 2.3.4) which are often neglected in NLP and MT (see also Sun et al. 2019, Cao & Daumé III 2020 and Dev et al. 2021). Similarly, very few studies in TS address non-binary genders, this monograph being one of them. An overview of such research is provided in Section 4.2.

The present work was developed prior to the popularisation of Large Language Models (LLMs) in NLP research after the introduction of ChatGPT (Wu et

al. 2023), yet it is important to acknowledge this shift. LLMs introduce new opportunities for prompt-based control over gendered forms but simultaneously raise concerns regarding reproducibility, consistency, and ethical grounding. Current research (e.g. Sant et al. 2024) shows that base LLMs exhibit stronger gender bias than traditional NMT systems. However, prompt engineering can significantly reduce bias, improving gender accuracy up to 12% on standard benchmarks. Such work highlights prompting as an effective, lightweight strategy for mitigating gender bias without model retraining. Similarly, Sánchez et al. (2024) show that decoder-only LLMs can be prompted to produce gender-specific translations, offering greater control over gender forms.

3.1.2 Sociotechnical implications of gender bias

MT is ubiquitous. Many online platforms such as social media and e-commerce services apply automated translation to make their content available to the widest possible audience (Monti 2020: 457). Since the introduction of neural networks, MT's quality has dramatically improved, leading to claims of human parity (e.g. Hassan et al. 2018). Although this was shown not to be true (see e.g. Läubli et al. 2018), NMT outputs can be deceptive because of their fluency (Läubli et al. 2018, Toral et al. 2018, Läubli et al. 2020). Hence, users might lack awareness of being exposed to automatically translated texts (Martindale & Carpuat 2018). This may be extremely problematic because MT is known to suffer from gender bias.

The analysis of bias is a normative process that requires decisions on “what kind of system behaviours are harmful, in what ways, to whom and why” (Blodgett et al. 2020: 5454). In the field of MT, research has mainly concentrated on representational harms, such as underrepresentation, stereotyping, and recognition (see Savoldi et al. 2021 as well as the previous section for an overview). In their survey on gender bias in the broader field of NLP, Stanczak & Augenstein (2021) note how gender is explicitly or implicitly defined as binary (see also Sun et al. 2019, Cao & Daumé III 2020). Harms affecting non-binary people are thus generally ignored, with the exception of Savoldi et al. (2021) who briefly mention them, and Dev et al. (2021).

One of the main causes of gender bias in MT is that training data are biased (Vanmassenhove et al. 2019, Vanmassenhove, Shterionov & Gwilliam 2021). Bias can be qualitative, i.e. in which terms the genders are represented, as well as quantitative, i.e. how often they are represented (Savoldi et al. 2021). This is mostly a societal issue. There is a general lack of awareness that gender is not binary and not all people identify either as men or women. In addition, trans

and non-binary individuals frequently experience discrimination; for instance, their existence is questioned and/or ridiculed. The main consequence is that they are underrepresented or presented in negative and/or stereotypical ways in the media (McLaren et al. 2021). This leads NMT systems to (i) not recognise non-binary language, and (ii) learn spurious associations for pronouns and terms (Dev et al. 2021: 1970). Moreover, there is not a one-size-fits-all solution to representing non-binary people. As discussed in Section 4.1, there is a multitude of gender-fair strategies in languages such as German and Italian. This complicates both the human and MT process which, with the exception of the present work, has not been investigated yet.

At a technical level, MT systems lack contextual information, which complicates the translation from languages less marked for gender into morphologically richer languages. Furthermore, there are few gender-fair datasets available and there is no de facto standard for referring to non-binary people in numerous languages, such as German and Italian. This complicates MT debiasing in the extreme. Savoldi et al. (2021) call for a heuristic approach to gender bias which recognises that the issue is of a sociotechnical nature. Participatory research as put forward by Burtcher et al. (2022) and Gromann et al. (2023) may be a first step towards gender-fair MT beyond the binary. This particular research design allows for a variety of stakeholders, e.g. MT experts, translation professionals, and non-binary people, to critically engage with the topic considering the different interests involved.

Drawing on the definition of allocative and representational harms by Crawford (2017) presented in Section 3.1, the harms faced by non-binary people in MT are described here. If a non-binary person wants to translate their biography for their personal website, they may need a more considerable effort in post-editing the outputs than men and, partly, women due to mistakes in gendered forms (Savoldi et al. 2021). In the field of medical and legal translation, incorrectly gendered forms may lead to invalidation or incorrect care, e.g. a patient who was assigned female at birth may not be asked about their pregnancy status when a new medication is prescribed (Dev et al. 2021). The fact that non-binary language is not processed by MT leads to misgendering – which can cause psychological distress, emotional pain and a feeling of identity invalidation (Matsuno & Budge 2017, Zimman 2019) – and omissions. Finally, NMT systems learn spurious associations from the training data. Terms that are accepted within a language and culture might be translated to offensive, derogatory terms, such as in the case of the word *kathoe*y used to refer to trans people in Thailand and translated as *ladyboy* in English.

Since users might lack awareness of being exposed to MT because of its fluent output (Martindale & Carpuat 2018), stereotypical assumptions and prejudices may be reinforced (Levesque 2011, Régner et al. 2019). Representational harms are, as discussed here, not unique to MT or machine learning because they pertain to wider societal issues. They can also be observed in human translations, e.g. of subtitles (Misiek 2020) and news reports (Lardelli & Gromann 2023b), with reference to non-binary individuals.

3.2 Machine translation PE (MTPE)

The previous section presented an overview of gender issues in MT. Despite the great advances made in terms of quality due to the use of neural networks, MT systems are not only gender-biased, but they still produce several mistakes, e.g. grammatical, syntactic, lexical, and terminological errors. The type and number of errors may depend on various factors, including the language pair and direction, the content of the source text, and the data used to train the MT engine (O'Brien 2022: 106).

To the best of my knowledge, there is little research concerning the correction of gendered (or gender-fair) language in PE. Only Lardelli et al. (2022) conducted a brief investigation into the choices of non-professional post-editors during an online hackathon. Savoldi et al. (2024) conducted a human-centred study quantifying the tangible impact of gender bias in MT. By analysing post-editing tasks across multiple languages and user profiles, they demonstrate that feminine translations require significantly more time, effort, and financial cost to correct than masculine ones. Crucially, their findings reveal that existing automatic bias metrics do not correlate with these human-centered costs, highlighting the need for evaluation methods that capture real-world harms.

The present work aims to contribute to the field by comparing the translation and PE process of GFL with the aim of investigating, for example, whether MT PE (or simply PE) may be a suitable instrument to produce gender-fair translations and datasets. To this end, a brief overview of PE as an established work practice in the language industry is presented in this section – for a more comprehensive introduction to the topic, see Nitzke & Hansen-Schirra (2021)).

There are different definitions for PE. One that is frequently adopted was offered by O'Brien (2011: 197f), who describes it as “the correction of raw machine-translated output by a human translator according to specific guidelines and quality criteria”. PE introduced a new perspective to both TS and the translation profession as translators are to deal with half-finished texts that must be improved

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(Allen 2003: 1). Its adoption dates back to the infancy of MT, also attracting the attention of translation scholars (Vieira 2019: 319) (see also Allen (2003) who provided a first overview on the topic). However, its use and the interest around it have dramatically increased only in the last few years (Vieira 2019: 319).

While translators have always integrated technology in their workflow, e.g. word processing (O'Brien 2012, Läubli & Green 2019), Computer-Aided Translation (CAT) tools and especially MT have become widely adopted in the last fifteen years (O'Brien 2012). The history of MT is crucially linked to PE and therefore relevant to the topic. Initial attempts by linguists, computer scientists, and mathematicians focused on fully automatic high-quality MT (FAHQMT), which Bar-Hillel (1952) had already labelled as unrealistic back in 1951. A similar statement was also made in the well-known ALPAC report (ALPAC 1966), where it was argued that MT requires PE (Läubli & Green 2019). The focus thus shifted from FAHQMT to human-aided MT (HAMT) (see also Figure 3.4 with the dimensions of human and MT). In such a scenario, MT remains at the centre of the translation process, but professional translators have to improve its output (Nitzke & Hansen-Schirra 2021: 9f).

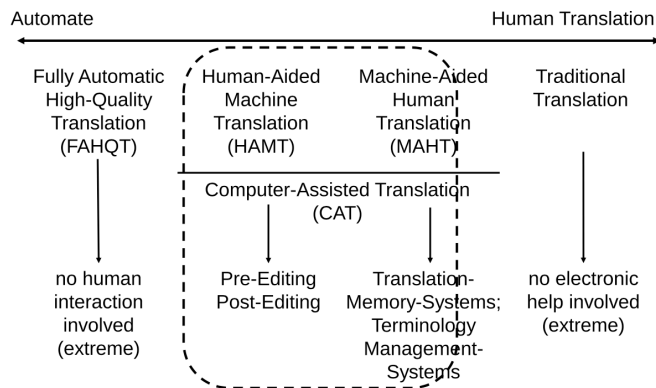


Figure 3.4: Dimensions of human and MT as in Nitzke & Hansen-Schirra (2021: 9)

Raw MT outputs can nevertheless serve specific purposes, e.g. internal communications in a company. In similar cases, MT can be used for gisting, i.e. grasping the meaning of the original content. In others, e.g. online help or support documentation, lightly or heavily post-edited forms may be preferred (Way 2018) – see Subsection 3.2.2 for the distinction between light and full PE. Bowker (2019) advocates for *fit-for-purpose* translations, highlighting how language professionals do not have to produce the highest-quality translation possible but rather adapt

to client specifications. A central question concerns the purpose of a particular text. On this basis, raw MT outputs could suffice, whereas sometimes light, full PE or even human translation might be preferred.

The terms *PE* to describe the activity as well as *post-editor* to describe the person who performs it are debatable (O'Brien 2022: 106): translation professionals usually work in CAT tool environments, recycling previously translated text chunks in the translation memory (TM) and also integrating MT in their workflow. In such a scenario, a translator might translate a sentence, edit a fuzzy match for the next and post-edit another. Hence, the lines between translation and PE are increasingly blurring (Vieira 2019), especially when it comes to interactive and adaptive PE (see next section).

PE has become a well-established activity because the translation market is increasingly growing as a result of globalisation (see Nitzke & Hansen-Schirra 2021: 1). Advances in the quality of MT outputs due to the application of neural networks (Castilho et al. 2017) additionally increased the use of and research on PE (O'Brien 2021: 177). Since the topic is vast, various aspects are briefly discussed in the next section, namely different approaches to PE, guidelines, and factors that should guide its adoption. Finally, a brief overview of research in the field is also presented.

3.2.1 PE approaches

PE can be carried out manually, e.g. in Excel sheets in which a column is used for the source and another for the automatically translated text, or simply in a text document. However, MT can be and is increasingly integrated into CAT tools (Nitzke & Hansen-Schirra 2021: 45, O'Brien 2022: 111).

There are different ways to do so. The translator can resort to MT when there is no useful match from the TM. For instance, when fuzzy matches show a similarity of less than 75%, MT might be preferred. The translator could decide whether to use the MT output, edit it, or discard it in favour of a translation. Based on different factors such as the language pair and the MT tools, this 75% threshold could also be increased. The translator can otherwise customise their CAT tool settings to automatically receive MT suggestions when certain conditions are met (O'Brien 2022: 111f). Though different, these approaches can be referred to as *static PE* because the generation and correction of the MT output are performed as separate steps (Vieira 2019: 322). In terms of human agency, static PE is in the middle between automatic PE and PE with adaptive/interactive MT (see Figure 3.5).

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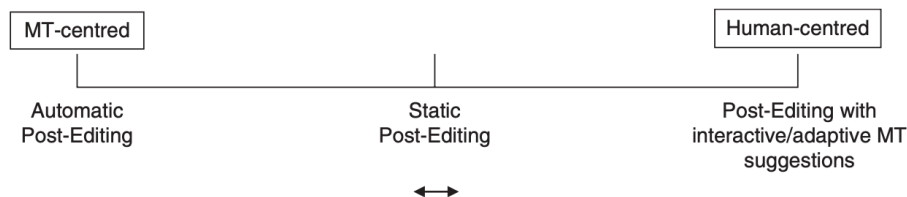


Figure 3.5: Spectrum of human agency in the PE process (Vieira 2019: 328)

Automatic PE (APE) describes systems that learn from human post-edited data to improve MT outputs (do Carmo et al. 2021). This approach is MT-centric and headed towards FAHQT because humans only provide the system with data which are used to learn patterns in their edits and automatically correct MT outputs. At the other end of the spectrum are more human-centred approaches. These include adaptive and interactive MT tools that have been developed to make PE less repetitive (Nitzke & Hansen-Schirra 2021: 46).

Adaptive MT refers to a feature whereby the system learns from the post-editor feedback (O'Brien 2022: 113). When the post-editor finds an error in the MT outputs, they fix it so that it does not occur when the same sentence is translated again. This feature has been implemented, for example, in the widely adopted CAT tool Trados Studio.⁵

Interactive MT describes systems that adapt their outputs on the fly based on the post-editor changes (O'Brien 2022: 113), as explained by Daems & Macken:

An interactive system tries to autocomplete the text the user is going to type; it either predicts the text the user is going to type or changes the MT suggestion on the basis of what is typed, whereas an adaptive system is an MT system that learns from corrections on the fly and is continuously trained. (Daems & Macken 2019: 118)

An example of this is the LILT translation and localisation tool.⁶

In the present study, as explained in Chapter 5, participants were divided into two groups. The first received a translation assignment, whereas the second had to perform static PE (see Figure 3.5), i.e. the author machine translated the study texts and delivered them to the participants.

⁵<https://www.trados.com/blog/adaptivemt-sdl-trados-studio-2017-transformational-mt-technology.html>

⁶<https://lilt.com>

3.2.2 PE guidelines

The main reasons behind the adoption of PE are to increase the volume of translated content while saving time and money. The translations need not necessarily be of high quality, and quality criteria that need to be met must be defined in advance (Nitzke & Hansen-Schirra 2021: 29). This led to a distinction between the different extents to which PE might be performed, i.e. light and full PE (O'Brien 2022: 107).

The International Organization for Standardization (ISO) defines light PE as the “process of PE to obtain a merely comprehensible text without any attempt to produce a product comparable to a product obtained by human translation” while full PE is the “process of PE to obtain a product comparable to a product obtained by human translation” (ISO 2017). O'Brien (2022: 107f) criticises these definitions because they are very vague and the differences between the two modes are unclear. The Translation Automation User Society (TAUS) created specific guidelines and the main difference between full and light PE is that the former involves stylistic changes (Massardo et al. 2016). This section focuses on the TAUS guidelines, though there are others such as ISO 18857:2017 (ISO 2017).

Light PE should be preferred when *good enough* quality is required. Translations should be comprehensible and accurate, but style is of marginal importance. Furthermore, grammar and syntax may be incorrect as long as the meaning is comprehensible. The guidelines for light PE by TAUS (Massardo et al. 2016) include:

- Aim for semantically correct translation.
- Ensure that no information has been accidentally added or omitted.
- Edit any offensive, inappropriate or culturally unacceptable content.
- Use as much of the raw MT output as possible.
- Basic rules regarding spelling apply.
- No need to implement corrections that are stylistic only.
- No need to restructure sentences solely to improve the natural flow of the text.

The objective of full PE is to achieve quality that is similar or equal to human translations. In this case, style is an important aspect and syntax, grammar, and punctuation must be correct. The guidelines for full PE by TAUS (Massardo et al. 2016) include:

- Aim for grammatically, syntactically and semantically correct translation.
- Ensure that key terminology is correctly translated and that untranslated terms belong to the client's list of “Do Not Translate” terms.

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- Ensure that no information has been accidentally added or omitted.
- Edit any offensive, inappropriate or culturally unacceptable content.
- Use as much of the raw MT output as possible.
- Apply basic rules regarding spelling, punctuation and hyphenation.
- Ensure that formatting is correct.

O'Brien (2022: 110f) highlights that the link between PE levels and levels of quality is problematic because the underlying assumption is that human translators always produce high-quality texts. She maintains that the level of editing required depends rather on the MT output quality, i.e. the lower the quality of the machine-translated text, the higher the number of edits needed to adjust it. O'Brien also highlights that professional translators do not normally want to produce translations of *good enough quality*. This view is supported by Bowker who advocates for a change

of the way that translators are educated and indoctrinated into the profession [...] where tool use is typically associated with poor quality work, to a continuum where it is recognized that translations can be commissioned for a diverse range of purposes, and a translator's job is to choose the strategy best suited to producing a target text that fits the specified purpose. (Bowker 2019: 454)

While PE can be distinguished based on the quality criteria that need to be met, another distinction concerns the language(s) involved. PE can be bilingual, i.e. source and target texts are compared, or monolingual, where the source text is not taken into account and the person performing it may not know the source language at all (Nitzke & Hansen-Schirra 2021: 32) – in which case there is no assessment of whether the meaning of the source text has been adequately transferred.

Within the context of the study, participants engaged in bilingual PE, which required them to compare the source and target texts. Bilingual PE was preferred because it is a more common practice, and certain linguistic strategies, such as using the pronoun *they* to refer to a single person, may only be recognized in contrastive scenarios. Participants were also provided with the full PE guidelines outlined in this subsection.

3.2.3 Factors guiding the adoption of MTPE

Several factors must be taken into account when deciding whether to use MT and/or light or full PE. To this end, Nitzke et al. (2019) propose the decision tree

in Figure 3.6 with criteria that guide the decision-making process. These include text type, risk considerations, data security, MT quality, turnaround time, life span of translations, and available resources. A brief overview of these factors is provided below.

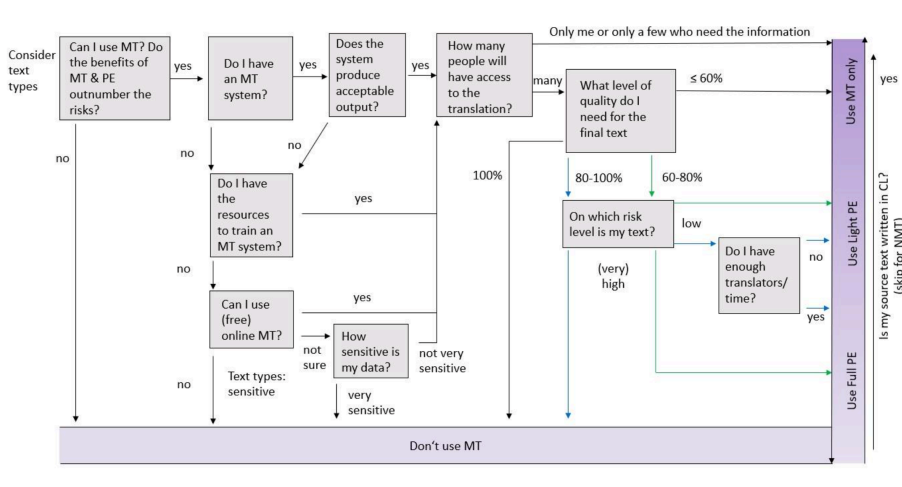


Figure 3.6: Decision tree guiding the adoption of MTPE (Nitzke 2019: 246)

First, some text types are more suitable for MT than others. Translations requiring high levels of creativity such as literature, marketing, and slogans should usually be translated by humans, whereas texts that are not very creative, contain redundancies, and have been created using specific rules or controlled languages can be machine-translated. Second, human translation should be preferred for high-risk texts such as warning messages since the risk of mistakes or misunderstandings when using MT could be too high. Another factor underlying the decision whether or not to use MT is data security. If the text for translation contains sensitive information, using a publicly available MT system could be too risky. The quality of the MT outputs is also an important criterion because the lower their quality, the higher the PE effort. Nitzke & Hansen-Schirra (2021: 60) advise pre-editing texts for MT if there are available resources to do so.

Other considerations involve the turnaround time, i.e. how urgent the translation is. For example, products such as software are released simultaneously on different markets, which is why translations for such products or product documentation must be produced very quickly and the adoption of MT is essential to meet tight deadlines. Furthermore, the translation of texts with a very short

shelf-life, e.g. internal communications in a global company, may be unfeasible or too expensive. Finally, the number of qualified translators available (or not) also has an impact on the decision whether to use MT. If there is a huge volume of content to be translated, but few language professionals available, MTPE might be a feasible solution to meet the deadlines.

3.2.4 **Research on PE**

A wealth of literature has concentrated on PE. One of the seminal texts in this field was written by Krings (2001), who investigated the PE effort in comparison with translation, using rudimentary methods such as a camera to record the process. Krings also contributed to the measurement of PE effort by dividing it into temporal, i.e. the time needed to PE a text, technical, i.e. keyboard and mouse actions, and cognitive effort.

In research on PE effort, different methods are applied. Keylogging software such as Translog (Jakobsen 2017) is used to analyse the keyboard activity. However, edit distance measures have also been introduced, such as TER (Snover et al. 2006). These metrics count the minimum number of operations needed to post-edit a text. Cognitive effort is more complex to measure and different methodologies have been applied ranging from verbalisations such as think-aloud protocols and retrospective interviews (Krings 1986, Ericsson & Simon 1984) to eye-tracking (Jakobsen 2017, O'Brien 2009).

While research on PE effort has garnered great attention, other aspects of PE have also been investigated. These include:

- Comparison of PE effort between statistical and neural MT (e.g. Jia et al. 2019b).
- PE of literary texts (e.g. Moorkens et al. 2018).
- Development of a competence framework for PE training and education (e.g. Nitzke et al. 2019).
- Distinction between editing, PE, and revision as well the number and types of corrections in revision and PE and instances of over-editing (e.g. Koponen et al. 2020).
- User attitudes and experience with PE interfaces (e.g. Moorkens & O'Brien 2013).
- User experiences with interactive and/or adaptive MT (e.g. Daems & Macken 2019).

As already explained, the present work is the first to investigate both the translation and PE process of gender-fair translation. PE is still currently required to

remove bias from MT outputs despite different debiasing techniques being proposed by computational linguists (Savoldi et al. 2021).

3.3 Relevance for the empirical study

The scope of the present section is to summarise key aspects presented in this third chapter and highlight their relevance for the empirical study described in Chapter 5, thereby illustrating how the literature on gender bias in MT shaped the methodology of the present work.

The chapter outlined the persistence of gender issues in MT systems, which often require PE to eliminate bias from their outputs. Though different debiasing techniques have been proposed (Vanmassenhove et al. 2018, Saunders & Byrne 2020, Tomalin et al. 2021), gender bias remains an issue in MT and also in Large Language Models (LLMs) that have recently seen an exponential increase in interest both from the general public and researchers (Kotek et al. 2023).

It is important to underline that gender issues in both human and MT arise for various reasons. For instance, gender may need to be explicitly marked in the target text while being ambiguous or neutral in the source text. In such cases, human translators frequently rely on social gender, i.e. stereotypical associations between a word, usually a noun, and a specific gender (Nissen 2002). Similarly, MT systems rely on patterns learned from training data that reflect societal biases (Forcada 2017). Two aspects are particularly pertinent to the present work.

First, bias in translation technology raises societal concerns due to the widespread use of MT, hence individuals might be unaware they are reading machine-translated content (Martindale & Carpuat 2018). This can exacerbate gender bias, resulting in the invalidation and discrimination of marginalised groups (Dev et al. 2021). The theoretical framework of this monograph posits that language, and consequently translation, can also be used to counter discrimination and promote societal change.

Second, the nature of gender bias in MT is interdisciplinary. It encompasses linguistics (e.g. how gender is expressed in languages and its societal connotations), computer science and computational linguistics (e.g. how machine learning systems function), and Translation Studies (TS) (e.g. how gender issues arise in cross-linguistic scenarios). However, research in this area often overlooks literature and methodologies from TS.

Gender bias is therefore best conceptualised as a complex sociotechnical issue (Savoldi et al. 2021). Essentially, machine learning systems are trained on extensive datasets that reflect societal biases and are optimised to maximise prediction

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accuracy for the training data, thus exacerbating these biases (Zou & Schiebinger 2018). Combining theories from feminist and queer TS with a process-oriented investigation was identified as a promising approach to examine a specific form of bias, namely the lack of non-binary representation, in (machine) translation. This approach considers translation as a powerful socially embedded practice (Wolf 2007).

Consequently, methodologies from translation process research were employed in the study of gender-fair language to understand how translation professionals actively engage with it to enhance the linguistic visibility of non-binary individuals. In this context, the differences between translation and PE proved particularly enlightening for understanding the impact of MT outputs on the work of translation professionals. Moreover, the professional insights of translators might inform debiasing techniques that recognise the multifaceted nature of bias in cross-linguistic contexts. Different translations may be produced for the same target text based on a variety of factors, including linguistic structures and divergent societal perspectives.

The following chapter concludes the theoretical framework of the present work. It presents strategies for GFL across different languages, illustrating that various approaches are feasible within the same language. The selection of a particular strategy carries significant implications, underscoring the importance of investigating the GFL translation and PE process from the standpoint of language professionals.

4 Gender-fair language in (machine) translation

The visibility of non-binary individuals has increased in the last few years. Celebrities such as Sam Smith and Demi Lovato came out as non-binary and television series such as *Sex Education* or *One Day at a Time* are more frequently featuring non-binary characters. In some languages, this represents no challenge: for example, in English where singular *they* can be used as a gender-neutral pronoun (APA Style 2019b) and most nouns are not gender-specific. In others, gender must be marked and new GFL strategies have been proposed, such as the inclusive *e* in Spanish (López 2019). The chapter delivers an introduction to the topic and presents GFL strategies in three different languages, namely English, German, and Italian. It then moves on to present research on GFL in TS and MT.

4.1 Introducing gender-fair language

Sczesny et al. (2016) use the term gender-fair and highlight how other expressions such as gender-neutral and gender-inclusive are also used, whereby the two have actually different meanings. Gender-neutral means that linguistic forms are not marked for gender, whereas gender-inclusive refers to linguistic forms that represent all gender identities (Hornscheidt & Sammla 2021: 28). This also reflects the process some languages are undergoing. In English, for example, as a notional gender language that is moderately inflected, speakers are headed towards a de-gendering, i.e. gender neutrality, to counteract sexism (Weatherall 2002: 17). For this reason, gender suffixes such as *-ess* in *actress* are becoming obsolete (Eckert & McConnell-Ginet 2013: 220). In grammatical gender languages, such as German and Italian, which are morphologically richer, new suffixes to include all gender identities have been introduced (see for example Hornscheidt & Sammla 2021 and Vitiello 2021 for some overviews), which corresponds to a process of engendering. This means that strategies also differ based on the gender system of a language. As a consequence, gender-fair is used in this work as a generic term to include every strategy to foster linguistic equality between the sexes/genders.

Note also that gender-inclusive and gender-neutral are often erroneously used interchangeably even though they refer to two distinct processes.

Similarly, in their work López (2019) distinguishes between Direct Non-Binary Language (DNL) and Indirect Non-Binary Language (INL). The former strategy describes those approaches that aim to address non-binary individuals through new gendered suffixes (e.g. the use of *e* and *es* in Spanish), typographical characters (e.g. @), neopronouns (e.g. *elle* and *elles*) and/or neologisms. The latter refers to the use of sentences structured in order to avoid gender markers, for instance through passive constructions and indefinite pronouns.

An interesting reflection on the term inclusion was offered by Acanfora (2021: 121). Its definition as “the act of including someone or something as part of a group, list, etc., or a person or thing that is included” (Cambridge Dictionary 2021) reveals that there is a power imbalance between those who include and those who are included. It is the former who decide whether to include the latter or not. Acanfora (2021: 125f) thus proposes a re-branding of this concept by introducing the expression *convivenza delle differenze* (coexistence of differences): *coexistence* suggests that relationships between people are based on mutual respect and equality, while *differences* is used to move away from the concept of norm which is often used as a standard to define what is considered as not normal and therefore inferior. In Italian, for example, other terms to refer to inclusive language, in general, are *linguaggio aperto* and *linguaggio ampio* (lit. open and broad language to illustrate the inclusion of all oppressed identities) and describe a language that is free from stereotypes and respectful of all differences, including gender, race, age, class, and so on (e.g. Gheno 2022).

Finally, Subsection 2.4.4 described how sexist language use covers a different range of phenomena such as the use of masculine generics, asymmetrical forms, or gender stereotypes. The present work focuses on the representation of non-binary individuals in language, who are often excluded from public discourse or misgendered. Expressions like gender-fair and gender-inclusive are consequently to be understood as making them visible and delivering the appropriate linguistic means to refer to them. Though discrimination can be perpetrated on other, subtler levels, e.g. through stereotypes, this is not the focus of the present work.

4.1.1 Gender-fair language in English

In Subsection 2.4.2, the gender system of the English language was briefly introduced, highlighting how gender is mainly marked in third-person singular

pronouns (he/she/it) and in some specific noun pairs (boy/girl) or job titles (actor/actress). In order to represent non-binary individuals, the process English is undergoing is primarily degendering or neutralisation (Weatherall 2002: 17, Sczesny et al. 2016), i.e. markers for gender are being removed and/or gendered words are being replaced by gender-neutral alternatives. This tendency, however, started with second-wave feminism movements and, especially, in the United States where feminists focused on two issues in language use, i.e. the use of masculine generics and of gender-specific titles in contexts where gender is not relevant (Kramer 2016: 73).

According to APA Style (2019a) and European Parliament (2018),¹ binary terms and gendered occupational titles must be avoided and replaced by gender-neutral forms, e.g. *housewife* by *homemaker*. More examples can be found in Table 4.1 as well as APA Style (2019a). Masculine generics should be avoided through gender-neutral words and/or expressions when talking about humans in general, e.g. *human rights* instead of *men's rights* (APA Style 2019a) as in Table 4.2. Singular *they* should be applied when talking about a person whose gender is unknown or a hypothetical person whose gender is irrelevant, e.g. “The client is usually the best judge of the value of their counseling” (APA Style 2019a).

Table 4.1: Gender-fair alternatives to binary terms and gendered professions in English (APA Style 2019a, European Parliament 2018)

Problematic terms	Preferred terms
mother, father	parent
sister, brother	sibling
steward, stewardess	flight attendant
waiter, waitress	servant
actor, actress	actor
chairman	chairperson, chair

More generally, terms and expressions that imply binaries, such as *opposite sex* or *opposite gender* should be avoided. For instance, expressions such as *mixed sex/gender* should be preferred to describe people of different sex/genders in a relationship. Gendered expressions should also be avoided since they convey stereotypes. For instance, *mothering* can be replaced by *parenting* since the first

¹Note that the list provided by European Parliament (2018) is still based on a binary conception of gender. For a more detailed list of gendered terms with alternative neutral versions, see Nonbinary Wiki (2021).

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Table 4.2: Gender-fair alternatives to masculine generics in English (APA Style 2019a, European Parliament 2018)

Problematic terms	Preferred terms
men’s rights	human rights
men, mankind	people, individuals, humanity
manpower	workforce
statesmen	political leaders

term implies that only mothers take care of their children and that there are only mothers and fathers. For the same reason, *mother tongue* is changed to *first language*. Furthermore, gender should not be specified unless it is relevant to the context of the conversation (see APA Style 2019a). Likewise, it is better to avoid stereotypical expressions suggesting how sexes/genders are and/or should be.

An alternative strategy to this process of neutralisation is the introduction of the sometimes so-called neopronouns, a term that is not actually correct as they date back centuries (see Baron 2020 for a detailed historical account). Since the 1780s, there have been many attempts to introduce gender-neutral pronouns in the English language with a uniquely intense ongoing debate (see Baron 2020, who gathered a wide collection of sources to account for this debate and the creation of more than 200 pronouns). These attempts have their origin in the distinction among genders in the third-person singular pronoun only, while other pronouns and most nouns are gender-neutral. When talking about people in general or a hypothetical person in sentences like “everyone forgets his passwords”, the gender of the indefinite pronoun is specified through the third-person singular masculine pronoun. The use of the supposedly generic *he* has been deemed as inclusive of women in a time where non-binary people were practically ignored and yet used to exclude women from access to basic rights, such as to vote (Baron 2020). As a consequence, different groups of people started to work without particular success on gender-neutral pronouns to overcome this issue and nowadays some are used specifically by non-binary people (see an overview in Table 4.3).

Grammarians such as Fisher (1753) have historically recommended generic *he* based on Lily (1549). The scholar, building on Latin grammar, believed the masculine gender to be worthier than the feminine and the neuter. Note that singular *they* has nevertheless been used since 1375 as a gender-neutral pronoun to refer to a hypothetical person whose gender is irrelevant or unknown (Baron 2020).

Table 4.3: Neopronouns in English (LGBT Resource Center 2021)

1	2	3	4	5
(f)ae	(f)aer	(f)aer	(f)aers	(f)aersel
e/ey	em	eir	eirs	eirself
per	per	pers	pers	perself
they	them	their	theirs	themself
ve	ver	vis	vis	verself
xe	xem	xyr	xyrs	xemself
ze/zie	hir	hir	hirs	hirself

It is only recently, also thanks to the increasing visibility of non-binary people, that the use of this latter pronoun has been endorsed by language authorities (see for example APA Style 2019b and Merriam-Webster 2019b) and also became Merriam-Webster’s word of the year in 2019 (Merriam-Webster 2019a). According to a survey conducted by Gender Census (2021) on language use by people whose gender is not binary, nearly 80% of respondents used the pronoun *they* (see Figure 4.1)

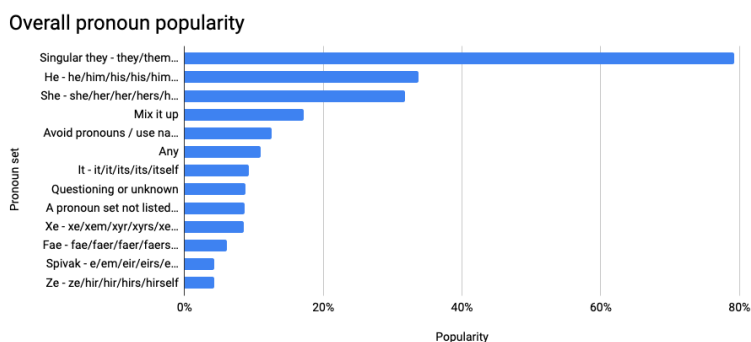


Figure 4.1: Use of gender-neutral pronouns by trans or non-binary people (Gender Census 2021). Note that respondents could choose more than one pronoun, so percentages add up to more than 100.

The use of forms of address has changed as well. First, *Ms* has been introduced as the feminine equivalent to *Mr* to overcome the sexist distinction between *Mrs* and *Miss* (see Subsection 2.4.4) (Weatherall 2002: 21). Second, a gender-neutral version, i.e. *Mx* has not only been introduced, but also officially recognised by public institutions in the UK (Henry 2015). This shows how the English language

is undergoing a process of neutralisation to remove gender markers from every word class except pronouns. In this last case, the use of singular *they* is becoming the norm when referring to people of unknown gender, mixed-gender groups and non-binary individuals.

4.1.2 Gender-fair language in German

As a grammatical gender language and hence described as morphologically rich, German is currently undergoing the process of including and representing non-binary people, i.e. multigendering or engendering (Weatherall 2002: 17, Sczesny et al. 2016). Nevertheless, some strategies aim to achieve gender neutrality. Importantly, due to its structure, gender markers are not being removed but rather substituted with typographical characters, such as asterisk, or new suffixes that are meant to be neutral (see Hornscheidt 2012: 294, Hornscheidt & Sammla 2021: 53). This shows, as previously mentioned, the lack of consistency in GFL terminology. For the German language, four main approaches can be identified:²

1. Gender-neutral rewording: these strategies aim to use the available inventory of language without introducing new pronouns or suffixes, by rephrasing/rewording in gender-neutral ways.
2. Typographical gender-inclusive strategies: typographical characters such as gender star (*) are introduced to separate masculine from feminine forms of words and thus represent all genders.
3. Gender-neutral strategies: typographical characters or new suffixes that are meant to be gender-neutral and used in contexts where gender is unknown or irrelevant.
4. Gender-fair neosystems: introduction of a fourth gender in addition to masculine, feminine, and neuter, and a new inventory of suffixes for (almost) each word class to make non-binary individuals visible and/or to refer to mixed groups of people.

Other aspects of language change are: (i) pronouns; (ii) forms of address; (iii) binary terms; and (iv) gender stereotypes. They are covered in more detail below. Due to the richness of the debate on gender-inclusivity, it is nevertheless impossible to cover all proposed strategies. The author decided to cover the most used and those found in the literature. However, in social media and more generally on the internet, one can find many more approaches to GFL, see for example Nichtbinär-Wiki (2021a).

²This classification is based on the commonalities the strategies share across all approaches.

The simplest possible approach to GFL is gender-neutral rewording. Sentences can be built such that gender is not marked (Universität Leipzig 2020: 6, En et al. 2021: 26), for instance:

- Collective and singular nouns may be built with specific suffixes and compounds such as *-kraft*, *-ung*, *-hilfe*, *-person*, *-berechtigte* as in *Lehrkraft* (teacher) or *Fachperson* (expert).
- Gender-neutral nouns and nouns referring to functions and institutions may be used, for example *Person*, *Mitglied* (member), *Mensch* (person), *Ministerium* (ministry), *Leitung* (direction), and *Team*.
- Plural nouns may be built by nominalising of the present participle, for example *Studierende* and *Lehrende* – note that such forms have become quite widespread but might not be considered as neutral or inclusive anymore as they are associated with maleness (Hornscheidt & Sammla 2021: 49).
- Reference to gender may also be avoided by using infinitive and passive constructions.

The second approach is the use of typographical characters (see Table 4.4) to separate masculine from feminine forms of words and make all genders visible. These typographical characters are pronounced as a brief pause in the spoken language. Since there are no official rules on how to build these forms, some creativity is required (Universität Leipzig 2020: 4, En et al. 2021: 27). These strategies have now become quite common in written language use (see En et al. 2021).

Table 4.4: Strategies for gender-inclusive German (adapted from AG Feministisch Sprachhandeln 2015: 16)

Strategy	Nouns	Personal Pronouns	Possessives	Definite Articles	Interrogatives
*	Student*in, Student*innen	sie*er, si*er	ihre*seine	die*der, di*er	welche*r?
³ ₋	Student_in, Student_innen	si_e, er_sie	ihre_seine	die_der, di_er	welche_r?
:	Student:in, Student:innen	si:er, er:sie	ihre:seine	die:der, di:er	welche:r?
/	Student/in, Student/innen	si/er, er/sie	ihre/seine	die/der, di/er	welche/r?
'	Student'in, Student'innen	si'er, er'sie	ihre'seine	die'der, di'er	welche'r?

In contexts where sex/gender is or should be irrelevant, usual masculine and feminine suffixes can be replaced by typographical characters (Hornscheidt 2012: 294) or new gender-neutral suffixes (Hornscheidt & Sammla 2021: 53; see Table 4.5 for an overview of some). The former solutions are attached to the verb stem,

³The underscore is also used as *dynamischer Unterstrich* (German for dynamic underscore), i.e. it is placed anywhere within the word to allow a more critical questioning of the binary gender system (Hornscheidt 2012: 304).

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and the latter to the noun stem to build nouns. Moreover, they can stand alone to replace some elements, such as pronouns. These strategies are normally used to question the gender binary as well as to irritate and provoke readers (see Hornscheidt 2012 for a detailed explanation of how and when to use them). In particular, Hornscheidt & Sammla (2021: 53) propose the use of gender-neutral Ens-forms which derive from the German word Mensch (person, human). Ens-forms are easy to pronounce and provide a complete inventory of suffixes for each word class, hence they might also be classified as a neosystem (see Paolucci et al. 2023).

Table 4.5: Selected strategies for gender-neutral German (Hornscheidt 2012: 294, AG Feministisch Sprachhandeln 2015, Hornscheidt & Sammla 2021: 53)

Strategy	Nouns	Personal Pronouns	Possessives	Articles	Indefinite Pronouns
x	Studierx Studierxs Lesx Lesxs	x xs (genitive)	xse	dix einx	x
*	Studier* Studier** Les*, Les**	* *s (genitive)	**s	d* ein*	*
-ex	Studentex Lesex	ex	ex	-	-
-ens ⁴	Studentens Lesens	ens	ens	dens einens	ens

Some examples formulated with these strategies are:

- “x sollte zukünftig besser besser auf xes Sprachhandlungen achten” (they should in future be more careful with their choice of words).
- “einx schlaux Studierx liebt xs Bücher” (a smart student loves their books) (Hornscheidt 2012: 294).
- “Lann liebt es, mit anderen zu diskutieren. Ex lädt häufig dazu ein, einen Roman zu besprechen” (Lann loves to discuss things with others. They frequently invite people to discuss a novel).
- “Lann und ex Freundex haben ex Rad bunt angestrichen” (Lann and their friend have colourfully painted a bike) (Hornscheidt 2021).
- “dens singend Radfahrens” (the singing biker).
- “dens solidarisch Rechtsberatens hat eine Kollektivausbildung absolviert” (the solidary counsellor has completed a specialised education).
- “einens hat das Recht zu gehorchen” (they have the right to comply) (Hornscheidt & Sammla 2021: 55).

The last approach to GFL in German is the development of neosystems that introduce a fourth gender in addition to masculine, feminine, and neuter, to make non-binary people visible in language and/or to refer to mixed groups of people.

⁴Other possible forms are: *keinens, niemensch, jemensch, jedens*.

This approach consists of a new inventory of suffixes for (almost) each word class. A brief overview of some systems is given below.

The *NoNa System* was (and is being) developed by two non-binary activists (Geschlechtsneutrales Deutsch 2021). As a personal pronoun, they propose the use of *hen/hem*, derived from the Swedish gender-neutral pronoun *hen* (see for example Bäck et al. 2015). Two sets of suffixes are introduced, i.e. *-ai/-ais/-am/-ai* for definite articles, relative pronouns and demonstratives and *-t* plus a mix of usual masculine and feminine suffixes for indefinite articles and the negation word *kein*, making it *keint* (see Table 4.6). With respect to nouns, the activists suggest that the gender star is used to build them and the adjective declension mixes masculine and feminine suffixes (see Table 4.7 for an overview of the system with the phrase “a good friend”).

Table 4.6: Pronouns and articles according to the NoNa-System (Geschlechtsneutrales Deutsch 2021)

Case	Definite Articles	Indef. Articles	Personal Pronouns	Posses.	Demon.	Relatives
Nom.	dai	eint	hen	meint	diesai	welchai
Gen.	dais	einter	hens	meinter	diesais	-
Dat.	dam	eintem	hem	meintem	diesam	welcham
Acc.	dai	eint	hen	meint	diesai	welchai

Table 4.7: Declension system according to the NoNa System (Geschlechtsneutrales Deutsch 2021)

Case	Declension with indefinite articles	Declension with definite articles
Nom.	eint gute Freund*in	dai gute Freund*in
Gen.	einter guten Freund*in	dais guten Freund*in
Dat.	eintem guten Freund*in	dam guten Freund*in
Acc.	eint gute Freund*in	dai gute Freund*in

Another system is the Sylvain system. During the Workshop *Geschlechtergerechte Sprache* that took place as part of the *Trans*Tagung* in Berlin in 2007, *nin* was proposed as a gender-inclusive pronoun (see Baumgartinger 2008). This delivered the input for author De Sylvain to create a whole new grammar system to make German more inclusive and they also decided to write a novel using it (De Sylvain & Balzer 2008: 40). This system introduced a new grammatical gender, the Indefinitivum (also liminales Geschlecht). Therefore, in addition to the

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terms *der Mann* (the man) and *die Frau* (the woman), *din Lim* is to be used for people who reject the gender binary. An overview of the system can be found in Table 4.8.

Table 4.8: Pronouns and articles according to the Sylvain system (De Sylvain & Balzer 2008)

Case	Def. Art.	Ind. Art.	Pers. Pron.	Poss. Pron.	Dem.	Rel. Pron.
Nom.	din	einin	nin	meinin	diesin	welchin/din
Gen.	dins	einins	nims/nimser	meinins	dienins	-/derin
Dat.	dim	einim	nim	meinim	diesim	welchim/dim
Acc.	din	einir	nin	meinin	diesin	welchin/din

Other forms that are proposed include *mensch*, *jemensch*, *niemensch*, and *jed-mensch* to replace the indefinite pronoun *man*,⁵ *jemand* (someone), *niemand* (no one) and *jedermann* (everybody). Finally, nouns are created by adding the suffix *-nin* to the word stem in the singular (e.g. *Studentnin*) and *-ninnen* for non-binary people only or *-Ninnen* for mixed-gender groups in the plural (e.g. *StudentNinnen*). New adjectival suffixes for the indefinite article are proposed as well, as shown in Table 4.9, whereas feminine suffixes are used for the definite article. In the table, four phrases in German are reported, i.e. “a young man”, “a young woman”, “a young non-binary person”, and “a young animal”.

Table 4.9: Declension system according to the Sylvain system (De Sylvain & Balzer 2008: 45)

Case	Singular			
	Masculine	Feminine	Liminal	Neutral
Nom.	ein junger Mann	eine junge Frau	einin jungin Lim	ein junger Tier
Gen.	eines jungen Mannes	einer jungen Frau	einins jungen Lims	eines jungen Tieres
Dat.	einem jungen Mann	einer jungen Frau	einim jungen Lim	einem jungen Tier
Acc.	einen jungen Mann	eine junge Frau	einir jungin Lim	ein junges Tier

Heger (2020) introduced and has updated throughout the years the personal pronoun *xier*. In 2009, the non-binary artist and activist started to engage with language to develop gender-inclusive alternatives to make German more inclusive for non-binary people (see Heger 2021). They proposed new neopronouns,

⁵The impersonal pronoun *man* because of its similarity to *Mann* (i.e. man in English) is often regarded as a sexist element of the German language and some propose therefore to replace it.

constantly updating their forms. Currently, Heger uses *xier*, derived from a fusion of the masculine and feminine pronouns *er* and *sie*. As shown in Table 4.10, definite articles, relative pronouns and possessives (as in “their friend”, see the last column in the table) can also be built accordingly.

Table 4.10: Overview of the pronoun *xier* (Heger 2020)

Case	Personal Pronouns	Def. Articles/ Relative Pron.	Possessives
Nom.	xier	dier	xiesa Freund_in
Gen.	xieser	dies	xiesas Freund_in
Dat.	xiem	diem	xiesam Freund_in
Acc.	xien	dien	xiesan Freund_in

The debate on GFL also takes place on social media. For example, there is a private Facebook group called Geschlechtsneutrales Deutsch,⁶ where about 500 members discuss language and try to find new solutions to overcome the binary structure of German.⁷ Recently, the group administrators founded the Verein für Geschlechtsneutrales Deutsch (Association for gender-neutral German) (Verein für geschlechtsneutrales Deutsch 2024). On its website, they propose three similar gender-fair systems, namely De-E-System, Dey-E-System and Dier-E-System. The first one is described to provide an overview of these more specific proposals.

In the De-E-System, it is suggested that the gender-fair suffixes *e* or *re* as well as *rne* should be used to build nouns respectively in the singular and plural. The personal pronoun should be *en* (Verein für geschlechtsneutrales Deutsch 2021). Table 4.11 shows an overview of the declension of adjectives when they are preceded by no, definite and indefinite articles (within the phrase “a good teacher”), and indefinite pronouns.

As concerns general pronoun usage, in German pronouns *er* and *sie* are used to refer to people. There is no official equivalent to singular *they* to refer to a person whose sex/gender is unknown (or irrelevant to the context of the conversation). Nevertheless, there are various solutions to this issue:

- Impersonal, gender-neutral pronouns such as *alle*, *diejenigen*, *jene* can be used (Universität Leipzig 2020: 7).
- An individual’s name can be repeated instead of using pronouns.

⁶<https://www.facebook.com/groups/geschlechtsneutral/>

⁷Even though there is a third grammatical gender – the neuter – this is not used for people as it is seen as degrading and offensive.

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Table 4.11: Overview of the *De-E-System* (Verein für geschlechtsneutrales Deutsch 2021)

Case	Def. Art.	Indef. Art.	No Art.	Indef. Pronouns	
Nom.	de gute Lehrere	ein gute Lehrere	gutey Lehrere	jedey	jemand
Gen.	ders guten Lehreres	einers guten Lehreres	guters Lehreres	jeders	jemanders
Dat.	derm guten Lehrere	einerm guten Lehrere	guterm Lehrere	jederm	jemanderm
Acc.	de gute Lehrere	ein gute Lehrere	gutey Lehrere	jedey	jemand

- Sentences can be reworded to avoid the use of pronouns.
- Gender-fair neopronouns can be used (see Table 4.12 for an overview⁸) – note that trans*, non-binary, and inter people may use different pronouns. For this reason, one should always ask each person for their pronouns (En et al. 2021: 38).

Table 4.12: Gender-fair pronouns in German (Hornscheidt & Sammla 2021: 36f, Heger 2020, Hornscheidt 2012: 294, ProNa 2015: 33, De Sylvain & Balzer 2008: 47)

Nom.	Gen.	Dat.	Acc.
xier	xieser	xiem	xien
er*sie	sein*ihr	ihm*ihr	ihn*sie
y	y	y	y
they	their	them	them
dey	deren	demm	demm
hen	hens	hem	hen
ens	ens	ens	ens
nin	nimser/nims	nim	nin
x	xs	x	x
ex	ex	ex	ex
per	pers	per	per

Another field requiring GFL concerns forms of address. When addressing people, for example in emails, salutations are often gendered in German, e.g. “Liebe Frau Smith” not only refers to the person explicitly as *Frau* or *Herr*, but it also

⁸In this case too, the overview is by no means complete, for other pronouns see Nichtbinär-Wiki (2021b) and de.pronouns.page (2021).

includes a gendered adjective since it is in the attributive position. Different strategies can be used to formulate a gender-fair form of address (Hornscheidt & Sammla 2021: 77):

- Salutation followed by name and surname, such as *hallo, Sehr geehrt*, Lieb*, Guten Morgen/Tag*.
- Salutation only, e.g. *Hallo*.
- *Frau* and *Herr* can be replaced by *Person* or *Mensch*.
- *Guten Morgen* followed by full name.
- Gender star or other strategies can be applied, for example for job titles, e.g. *Professor*in*.

The German language also comprises several terms which imply the gender binary. These include family terms, such as *Mutter/Vater* (mother/father) or *Tante/Onkel* (aunt/uncle) as well as job titles ending in *-mann/-frau*, e.g. *Feuerwehrmann* (firefighter) or *Obfrau* (chairwoman). A list of gender-inclusive alternatives can be found on Nichtbinär-Wiki (2021a). Finally, a GFL is also a stereotype-free language, i.e. free from expressions suggesting how sex/genders are or should be, such as *das starke Geschlecht* (the strong sex), that imply that men must be strong (En et al. 2021: 33).

4.1.3 Gender-fair language in Italian

In Italian, the debate on GFL is still very much focused on the inclusion of women in discourse; more specifically, on the use of the feminine forms of functions and high-level professions from which women were once excluded. For instance, Gheno (2019) provides an overview of the rules to feminise functions and job titles in Italian and summarises as well as rebuts the arguments against them. The debate on language sexism was started by Sabatini (1986) who noted the widespread use of masculine generics, the asymmetrical use of name and titles, e.g. men called simply by surname whereas women by surname preceded by definite articles, e.g. “la Thatcher”, and the lack of symmetry for offices and high-level professions. These are generally used in the masculine form even in reference to women.

Throughout the years, the debate on the inclusion of women in language has focused on this last point (see also the work of Robustelli 2012, 2014). Proposals to avoid the use of masculine generics have also been made, but they have been somewhat restricted to administrative texts; see for example the use of passive

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and impersonal⁹ forms, indefinite pronouns, double forms, and gender-neutral nouns such as *person* (Robustelli 2012: 22).

More specifically, strategies to avoid masculine generics in Italian and to reach gender fairness are (Vitiello 2021):

- Change of subject so as to avoid past participles – these are gendered in Italian.
- Use of periphrases.
- Use of synonyms for verbs, nouns, and adjectives.
- Change of perspective in the sentences.
- Use of neutral and collective nouns or impersonal pronouns, such as *persona*, *membro* (member), *personale* (staff), *chi* (who).
- Omission of nouns, pronouns and adjectives, and use of verbs as subjects.
- Use of passive and impersonal constructions (which are sometimes gendered, therefore one must be careful).
- Avoidance of the term *uomo* (man) to refer to people in general.

As mentioned in the section dedicated to the German language, such strategies can also be used to include non-binary people in discourse.

With regard to the main topic of this work, i.e. the visibility of non-binary people, the asterisk was the first strategy introduced so as not to mark gendered word suffixes (Asterisk Project: Pride in Palermo 2021). Curiously, it was introduced on the occasion of the Pride parade in Palermo in 2010 (Asterisk Project: Pride in Palermo 2021) and became quite popular on the internet (Marotta & Monaco 2016: 50), also gaining an entry in the grammar section of the Italian Encyclopedia *Treccani* (Treccani 2021). In expressions such as “car* tutt*” (dear all), the asterisk is used to avoid masculine generics and references to the gender binary (Treccani 2021). Though this may represent a simple but not definitive solution for the written language, it cannot be orally transposed as it is not pronounceable (Marotta & Monaco 2016: 50).

Other similar strategies have been proposed and are used (see Table 4.13 for an overview). These are mainly graphical solutions, such as @ or - and they are used to replace feminine and masculine suffixes in gendered elements like nouns, adjectives, and past participles as well as sometimes to build pronouns (see Vitiello 2021).

Among all of the summarised strategies to represent non-binary individuals, the use of the International Phonetic Alphabet (IPA) symbol for the mid-central

⁹Nevertheless, gendering cannot always be avoided in these cases and some of the examples provided are illustrative: “si entra uno alla volta” – i.e. in English “enter one at a time” – one is a masculine form in Italian.

Table 4.13: Strategies for gender-inclusive Italian based on Vitiello (2021)

Ending	Example	Pronoun
*	buongiorno a tutt*	l*i
@	buongiorno a tutt@	l@i
-	buongiorno a tutt- buongiorno a tutt	
x	buongiorno a tuttx	lxi
y	buongiorno a tutty	
u	buongiorno a tuttu	
ə/3	buongiorno a tutt3	ləi

vowel schwa is worth considering. Proposed in 2015 by Boschetto (Italiano Inclusivo 2021a), an enthusiast on linguistics and inclusivity, it was made popular by Gheno (2019: 167), who wondered whether such innovation would become the norm to refer to a multitude of people whose gender is unknown or irrelevant to the context of the conversation.¹⁰ Schwa has been adopted by publishing house effequ (Costantino 2020) and has also been used in articles in the magazine *L'Espresso* (see for example Murgia 2021a) and in some books (e.g. Meloni & Mibelli 2021¹¹). Finally, it has been introduced among symbols in smartphone keyboards by Android and Apple (Italiano Inclusivo 2021d), but the latter adopted ə only (see below).

The use of schwa is very interesting for different reasons: (i) except for -u, it is the only pronounceable strategy proposed; (ii) there is the possibility that singular and plural be maintained, as 3 has been proposed for plural; (iii) it is a mid-vowel that is not generally present in the Italian language except for some dialects, hence it should not evoke maleness (for example the use of u, which is a masculine ending in the Sardinian language) (Italiano Inclusivo 2021c, Gheno 2021). In Table 4.14, an overview of how to use schwa is proposed following suggestions by Italiano Inclusivo (2021b).

¹⁰Note that Gheno (2019: 167) uses schwa (ə) to refer both to a single person and a multitude of people, e.g. “carə collegə” (dear colleagues) whereas Italiano Inclusivo (the site founded by Boschetto) suggests using another symbol (3) to maintain two distinct suffixes for singular and for plural (Italiano Inclusivo 2021b).

¹¹These authors do not follow suggestions by Italiano Inclusivo (2021b); they rather use singular schwa as in Gheno (2019: 167).

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Table 4.14: Overview of schwa in Italian as proposed by Italiano Inclusivo (2021b)

Case	Feminine	Masculine	Inclusive
Regular Nouns , e.g. teacher	maestra maestre	maestro maestri	maestrə maestrɜ
Irregular Nouns , e.g. hero and painter	eroina pittrici	eroe pittori	eroə pittorɜ
Epicene Nouns , e.g. artist and poet	un’artista la poeta	un artista il poeta	un*artista lə poeta
Articles	la le	il/lo i/gli	lə lɜ
	una/un’	un/uno	unə/un*
3rd-Person Singular Pronoun	lei	lui	ləi

Basically, schwa is used to replace feminine and masculine suffixes respectively in the singular and plural (as mentioned before, in Italian, adjectives, articles, pronouns, and past participles are also gendered). In the case of irregular nouns, that are hence not symmetrical, schwa replaces the last vowel of the masculine form. As regards articles, since the grammatical gender of the indeterminate *un’* and *un* can be distinguished by the apostrophe, the use of an asterisk (*) is suggested (Italiano Inclusivo 2021b). Nevertheless, a solution for gendered pairs such as brother/sister (in Italian there is no such term as sibling) or mother/father has yet to be proposed.

The use of schwa is also controversial (De Benedetti 2022, Arcangeli 2022, for example, strongly criticise it) and it is impossible to predict whether it will become the norm for GFL in Italian (see for example Gheno 2021). In particular, two main points are worth considering: schwa is considered to be ableist and it is used inconsistently across authors. Specifically, it is supposed to represent an obstacle for neurodivergent people reading it (Gheno 2021), though there are no sources or studies on the topic, and some screen readers cannot pronounce it (Orrù 2021). Orrù, a copywriter and technical translator, explains other problems concerning the accessibility of schwa, its implementation in assistive technologies being the most prominent. Additionally, schwa has been appearing in books and articles (see the aforementioned Costantino 2020, Meloni & Mibelli 2021, Murgia 2021a) but only the ə has been used as in Table 4.15. However, having two symbols, one for plural and one for singular, is part of what makes such a strategy appealing whereas the use of only one symbol may complicate reading.

Table 4.15: Alternative use of schwa as in effequ (2022)

	Singular	Plural
Nouns with articles	lə maestrə	ə maestrə
Nouns with prepositions	dellə maestrə	deə maestrə

The present and the other two preceding subsections detailed specific strategies for GFL in English, German, and Italian. Note that GFL is language-dependent because grammatical structures differ among languages. Some good practices, however, may be applied regardless of the language and are briefly discussed in the following section.

4.1.4 Good practices in gender-fair language

It is often thought that the sex/gender of a person can be deducted from external elements like physical appearance, voice, behaviour, clothing, and so forth. Based on that, people are referred to with masculine or feminine (pro)nouns (ProNa 2015: 32). This may lead to misunderstandings that can be hurtful for people; research shows that being misgendered may cause feelings of stigma and psychological distress (McLemore 2018). One should never assume to know people’s sex/gender and should ask for identified pronouns instead.

Another good practice consists in disclosing one’s own identified pronouns, for example in the email signature or when introducing oneself to others (En et al. 2021: 23). Recently, social media such as LinkedIn and Instagram have introduced the option to add pronouns in the profile precisely in order to foster inclusivity (McLaren 2021, Olson 2021). This helps normalise the habit of asking for identified pronouns and acknowledges that many people’s gender identity is not coincident with their sex assigned at birth.

Finally, many expressions in language imply that a particular gender follows from one’s sex assigned at birth. Expressions such as “women with menstruation” or “male condoms” imply that a person with menstruation identifies as a woman, whereas a person who uses external condoms is a man (En et al. 2021: 33). Such expressions can be avoided and replaced by neutral alternatives like “people with menstruation” or “external condoms”. This shows how gender fairness encompasses knowledge of terminology concerning gender and sexuality as well.

4.1.5 The societal impact of gender-fair language

Subsection 2.4.5 addressed how the use of GFL can actually foster the visibility of women, especially in the workplace. The above-mentioned studies are fundamental in supporting the call for GFL. Nevertheless, such studies still concentrate either on binary linguistic strategies or only consider discrimination against women, thus ignoring non-binary individuals. Since the introduction and proliferation of the Swedish gender-neutral pronoun *hen* (see Gustafsson Sendén et al. 2015), some researchers focused on new, non-binary strategies in comparison to old neutral alternatives, finding that the latter still contain cognitive male bias.

Bäck et al. (2015) carried out two studies on the perception of *hen*. In the first experiment, the researchers asked travellers at Stockholm Central Station to read the description of a person applying for a job as an insurance sales manager and to remember their gender. This profession was selected as it is equally occupied by men and women in Sweden. The description referred to the applicant as *he*, *she*, *the applicant*, and finally *hen*. Even though the Swedish equivalent for *applicant* is also gender-neutral, it was found to evoke male imagery, whereas *hen* was found to be far less biased. In the second experiment, other participants were asked to perform the same task but this time the job descriptions were presented using either typical masculine or feminine characteristics. The results were similar to those of the first experiment, and *applicant* was found to be male-biased even when the job description was phrased with stereotypically feminine terms.

Lindqvist et al. (2019) conducted two other experiments, one in Swedish and the second in English, asking participants to read the description of a candidate for a job position. The candidate was referred to using (i) paired forms such as *he/she* in English and in Swedish with the corresponding forms, (ii) traditional neutral words and singular *they*, (iii) newly created gender-neutral forms such as gender-neutral pronoun *hen* in Swedish and *ze* in English. They found that in scenario (ii), people actually visualised men to a greater extent, hence traditional neutral wordings being cognitively biased towards men. Strategies (i) and (iii), instead, eliminated it. The scholars, however, call for the use of newly, actively created gender-neutral words instead of paired forms as the former are also inclusive of non-binary people.

Similarly, Renström et al. (2024) tested whether paired and gender-neutral pronouns evoke a normative gender bias in Swedish and English. Normative bias is defined in the study as “when words lead to stronger associations with individuals with normative gender expressions than with individuals with non-normative gender expressions” (Renström et al. 2024: 2). The first experiment was conducted in Swedish and participants were asked to read single sentences

with the pronoun *hen* or the paired pronoun *he/she* used in reference to a person. The setup for the second experiment was similar and participants read about a candidate for a job position. Finally, a third experiment was conducted to replicate experiment two, but in English. The pronouns used were *ze*, singular *they*, and *he/she*. The results suggest that neopronouns evoke fewer normative associations when they are newly created and recognized by language users as non-binary pronouns, since they are not linked to masculine or feminine genders..

Some initial studies on the effects of German gender star on mental representation have appeared, e.g. Kurz & De Mulder (2023) who replicated the study by Stahlberg & Sczesny (2001) presented in Subsection 2.4.5. The researchers focused on the use of masculine generics and gender star and found that this last strategy leads to more inclusive mental representations.

To the best of my knowledge, these are some of the few studies in which gender-neutral, non-binary strategies are taken into account, although they do not always focus on the effects of such strategies on the visibility of non-binary individuals. It can only be presumed that their usage will help dismantle the gender binary in the long run, creating a more respectful (not only) linguistic environment for non-binary people.

4.2 Research on gender-fair language in (machine) translation

GFL poses a challenge for human and MT alike. Since non-binary people are featured in different typologies of popular texts and formats, ranging from magazine articles to television series and films, translators must be able to opt for appropriate strategies that reflect the language used in the source text (Misiek 2020) while translating and/or post-editing. In addition, MT systems still suffer from gender bias, thus gender identities beyond the binary are underrepresented or erased in MT outputs. Issues of the representation of non-binary people in language are neglected in both TS and MT.

This section summarises selected work focusing on non-binary GFL either from a TS, an MT, or both perspectives. For a more comprehensive literature review, see Lardelli & Gromann (2023a). Note that GFL research is still in its infancy, both in TS and MT, and numerous studies are appearing at the time being, hence the overview provided in this section is by no means exhaustive.

4.2.1 Gender-fair language in translation studies

Misiek (2020) analyses the Polish translation of three different English-language TV series that feature non-binary characters. The scholar finds that translators often misgendered the characters or even completely erased their gender identity. Polish is a grammatical gender language which has the feminine, masculine, and neuter. It has three third-person singular pronouns, i.e. *on*, *ona*, and *ono*, which respectively correspond to English *he*, *she*, and *it*. It has also two third-person plural pronouns, i.e. *oni* for the masculine and one for the feminine and neuter. There is hence no direct equivalent to singular *they* to refer to non-binary individuals. Misiek also provides an overview of the strategies for gender neutrality in Polish. They include distortion and/or omission of verb suffixes, use of passive constructions, present tense because the past is gendered, use of the vowel *u* as a new neuter suffix, or use of neuter forms (Misiek 2020: 167). As regards translation, the scholar proposes to: (i) always ask the actor interpreting the character how they are to be referred to in Polish; (ii) consult non-binary or genderqueer communities; (iii) use neuter forms or the aforementioned vowel *u*; (iv) use neologisms to describe professions; and (v) use general, non-gendered words like *person*.

Šincek (2020) focuses on Croatian and analyses a film translation as well as journal articles on Sam Smith's coming out as non-binary. The researcher also finds that non-binary gender identities are misrepresented or even erased in both the film and the news reports. Similar to Polish, Croatian is a grammatical gender language with three third-person singular pronouns, i.e. *on*, *ona*, and *ono* which correspond to English *he*, *she*, and *it*. In contrast to Polish, there are three plural forms: *oni* is masculine, *one* feminine, and *ona* neuter, hence there being no equivalent to singular *they* in Croatian either. When analysing Sam Smith's coming out as non-binary, Šincek found that they were referred to as *on*, *ono* or *oni* – this last being a literal translation of singular *they*. The researcher also conducts interviews with three non-binary people asking which pronouns they use in Croatian. One participant uses a pronoun that reflects their sex assigned at birth but thinks that *ono* could be a good gender-neutral pronoun whereas the other two consider it derogatory because it is used to describe inanimate objects. Another participant tried to use *oni* for some time but noted how this led to confusion. In general, participants agree to use archaic verb forms that are not gendered in Croatian as well as gender-neutral words such as *person*. To date, there are no dedicated strategies to achieve gender fairness in Croatian, and English gender-neutral pronoun singular *they* is often rendered as *oni*, which is nevertheless third-person plural and masculine. Šincek proposes (i) to use simple

tenses since these are not gendered; (ii) as well as the word person; and (iii) to try to add a new meaning to oni.

López's (2019, 2022) work focuses on the translation of media products as well. These include the Spanish dubbed and the subtitled versions of English language TV shows, such as *One Day at a Time*, which feature non-binary characters. López distinguishes between INL and DNL. In the former, sentences are structured in a way that gender can be avoided, e.g. using passive constructions and gender-neutral terms such as person. INL may be used to address a broad audience that includes different genders since it conceals them. In the latter, new gendered suffixes, typographical characters, and neopronouns may be used to specifically address non-binary individuals. The researcher finally notes that (i) the identity of non-binary characters is often erased in the TV shows analysed; (ii) gendered forms are used inconsistently; and (iii) non-binary language is often translated with unusual forms. Similarly, Attig (2023) analyses the Spanish and French dubbed and subtitled versions of *One Day At a Time* and calls for a community-informed translation practice of queer-oriented texts.

Lardelli & Gromann (2023b) provide an overview of GFL strategies beyond the binary in German and Italian. To investigate their actual use, they create a corpus with online reports about Demi Lovato's coming out as non-binary in both languages. They find that misgendering is still very common, especially in Italian language texts. Translation solutions for singular *they* greatly differ and show that journalists often lack knowledge about non-binary genders and language. While a dominant GFL strategy was not found in the analysed corpus, a mix of gender-neutral and gender-inclusive forms was used.

The present work employs a process-oriented framework to study the use of GFL in translation. Specifically, it analyses the impact of different strategies on cognitive processing, focusing on times, screen activity tracking data, and retrospective accounts. Such study is important because it contributes towards bringing translation professionals to the forefront of the discourse around gender fairness in language and translation, also showcasing their role as mediators and agents for social change. The participants in the study were not regarded as passive agents, but they were asked to experiment with different strategies to gain insights into their professional and personal opinions regarding GFL.

4.2.2 Gender-fair language in machine translation

Saunders et al. (2020) propose a first approach to address gender-fair MT beyond the binary. The researchers extended a gender-balanced dataset (Saunders & Byrne 2020) with gender-neutral sentences containing placeholders for gender

inflexions in German and French. They trained an NMT system through transfer learning and tested the effectiveness of their methods with a gender-neutral version of WinoMT (Stanovsky et al. 2019) that they produced. The NMT model scored a low overall accuracy, struggling with the generation of gender-neutral language.

Lauscher et al. (2023) analyse how three commercial MT systems translate third-person pronouns. They compare the translations of gendered, gender-neutral, and neopronouns from English to Danish, Farsi, French, German, and Italian as well as from Danish to English. They find that gender-neutral pronouns often lead to grammatical and semantic translation errors, e.g. pronouns are often wrongly translated, omitted, or even copied in the MT outputs. In addition, gender neutrality is often not preserved leading to misgendering.

In the literature, there is a clear trend toward developing diverse benchmarks, spanning multiple languages and focusing on either gender-neutral, gender-inclusive, or non-binary language (Piergentili, Savoldi, et al. 2023, Piergentili et al. 2024, Lardelli et al. 2024). Moreover, the field is increasingly focusing on LLMs, for instance testing their capabilities to produce GFL while bias evaluation methods are evolving from traditional reference-based scoring to LLM-driven, reference-free approaches, often leveraging chain-of-thought prompting for improved accuracy (Piergentili et al. 2025). A brief overview of such research directions is provided in the following paragraphs.

Piergentili, Fucci, et al. (2023) tackle the issue of gender bias in MT from a theoretical perspective. While they do not specifically address non-binary language issues, they focus on gender-neutral language strategies such as indefinite pronouns, omissions, repetitions, and gender-neutral alternatives to gendered terms. These strategies should be used, for example, to avoid assigning a gender in the target text when the source is ambiguous, and to avoid the generic masculine. The scholars reviewed gender-neutral strategies in English and Italian, outlined a definition of gender-neutral translation and discussed the technical challenges involved in implementing it in MT such as the dynamicity of the gender-neutral strategies, the lack of training data and methods as well as evaluation procedures. Based on this, Piergentili, Savoldi, et al. (2023) introduce GeNTE, a benchmark for evaluating gender-neutral translation from English to Italian. They propose both reference-based and reference-free evaluation methods and show that current MT systems still struggle in this area, whereas Piergentili et al. (2024) explore gender-inclusive translation into Italian using neomorphemes (e.g. *tutta*) and LLMs. They release NEO-GATE, a benchmark for evaluating gender-inclusive translation, and demonstrate that LLM prompting can improve inclusive language generation. Finally, Piergentili et al. (2025) propose an LLM-as-a-judge

4.2 *Research on gender-fair language in (machine) translation*

method for evaluating gender-neutral translations in a scalable, reference-free manner. Using phrase-level prompting improves LLM judgment reliability across multiple languages and enhances automation of gender bias evaluation in MT.

Lardelli et al. (2024) present a benchmark for gender-fair translation from English to German using real-world texts. Their analysis reveals a strong tendency toward masculine defaults and limited use of inclusive language, even among state-of-the-art systems. Robinson et al. (2024) introduce MiTTenS, a multilingual benchmark covering 26 languages to evaluate gender mistranslation. The dataset includes both naturalistic and synthetic data targeting known failure modes, revealing persistent gender bias across MT systems and language pairs.

Finally, Chen et al. (2024) and Jourdan et al. (2025) strongly focus on gender bias against non-binary people. The former propose AmbGIMT, a multilingual benchmark measuring gender bias in translation using ambiguous attitude expressions. They introduce the Emotional Attitude Score (EAS) and find that translations of non-binary contexts often carry more negative sentiment and reduced quality. The latter release FairTranslate, a gender bias evaluation dataset for English–French MT with a focus on occupational terms and inclusive forms. Through LLM prompting experiments, they show that non-binary forms receive lower translation quality and exhibit persistent stereotypes.

Amrhein et al. (2023) propose a gender-fair rewriting model in line with other research in the field of debiasing MT (see, e.g. Vanmassenhove, Emmery, et al. 2021). While in previous studies training data for these models were created through linguistic rules, Amrhein et al. (2023) use MT models to create gender-biased text from real gender-fair text via round-trip translation. They train a rewriting model for German which was assessed in a human evaluation study and showed an increase in gender fairness in comparison with biased MT outputs.

The present study differs from the research presented in this subsection for various reasons. It neither analyses the outputs of various commercial MT systems nor proposes debiasing approaches. Instead, it examines the PE process with the aim, among others, of determining whether PE is a viable instrument for producing gender-fair datasets that could further research on gender-fair MT. Additionally, it collects the opinions of language professionals to call for more user-centred and interdisciplinary-informed debiasing approaches, potentially prompting a shift in how gender bias research is conducted within computational linguistics.

4.2.3 Interdisciplinary research on gender-fair language

Burtscher et al. (2022) and Gromann et al. (2023) propose a method building on participatory action research to include experiential experts – i.e. queer and non-binary people, translators, and MT experts – in the discussion around gender fairness in human translation and its implementation in the MT design process. The researchers organised a three-day participatory workshop focusing on (machine) translation from English into German. Some of the major findings include the importance of context dependency to avoid identity invalidation, the need for materials and resources on GFL, and a desire for customisable MT solutions.

The present work contributes to discussion among TS and MT researchers by investigating both the translation and PE process of non-binary GFL. Similar to existing research, the focus is on the translation from English into languages for which the debate on GFL is much more developed. Although to a different extent and with varying degrees of success, in German-speaking countries GFL and GFL policies have been discussed since the 1970s (see Sczesny et al. 2016), and several strategies to represent and/or include non-binary individuals in discourse have been developed, as explained in Section 4.1.

While related research in TS generally provides overviews of common strategies for GFL and analyses their use in the translation of different media products, the present study combines process and product data to provide more comprehensive insights into the translation and PE process which can support translation professionals in their work and also provide inspiration for the debiasing of MT.

4.3 Relevance for the empirical study

The scope of this final section is to summarise key theoretical aspects, highlighting their relevance for the empirical study and analysis proposed in this book.

An overview of various concurrent GFL strategies was provided, illustrating different approaches to gender fairness both within and across languages. Languages have different grammatical structures, thus complicating the translation of gender, as also presented in Subsection 2.2.3. A source language may express gender neutrally or ambiguously, whereas the target language may not. In such cases, GFL may be needed, but the choice of strategy can meaningfully affect the translation's interpretation.

A fundamental principle in the translation practice and TS is the existence of multiple translations for a single source text. Different versions may be generated based on the translation's intended purpose (Reiß & Vermeer 1984). Furthermore,

translation can be seen as a process of intercultural communication, resulting in a text that functions appropriately in specific situations and contexts (Holz-Mänttari 1984).

Consequently, participants were instructed to experiment with three GFL approaches to facilitate comparative analysis and discussion on the implications of their usage. During the retrospective interviews, the concept of the context-dependent selection of strategies was discussed, with participants elucidating the impact of GFL on the target texts (see Chapter 7).

Additionally, the literature review presented in this chapter revealed a lack of interdisciplinarity, with researchers from TS and computational linguistics often overlooking studies outside their respective fields. To date, the only interdisciplinary research efforts have been collaborative projects (Burtscher et al. 2022, Gromann et al. 2023). This study aimed to investigate the perspectives of translation professionals and integrate viewpoints from TS, a discipline that has historically overlooked research on translation technology (Rozmysłowicz 2024).

Finally, excerpts from this chapter were used to create a handout on GFL, which was distributed to study participants to prepare them for the GFL translation and PE tasks. These contents also served as a basis for analysing the strategies employed and the errors made by participants, as discussed in Chapter 5.

5 Methodology

The present work consists of a translation and PE experiment from English into German with 12 professional translators. The selected method combines non-participant observation, screen recordings, retrospective interviews, and target text annotation. It was inspired by the field of TPR and Albl-Mikasa et al. (2017), who combine screen recordings and retrospective interviews in research on the translation of non-native English in the context of the European Parliament. TPR focuses on the translation process and typical research questions include problem-solving, decision-making, revision, time pressure, and motivation (Jääskeläinen 2012: 123f). It borrows methodologies from other disciplines, such as psychology, linguistics, and the cognitive sciences (Alves 2015: 18). Usually, data on the translation process are collected through keylogging, eye tracking and/or screen recordings, and triangulated with verbal reports, such as think-aloud protocols (TAP) or retrospective interviews. In the last few years, new technologies such as electroencephalograms (EEGs) have also been applied in research settings (Jakobsen 2017: 21).

The experiment was conducted between July and October 2022. Twelve professional translators were recruited (see more details on the study participants in Subsection 5.1.3). Before the experiment, participants received instructions on the tasks, PE guidelines by the Translation Automation User Society (TAUS),¹ and a handout on various strategies for GFL to prepare for their participation. Participants also completed an online questionnaire to collect demographic data as well as data on their use of GFL and its perceived difficulty.

Participants were divided into two groups. The first group was asked to translate three brief texts of approximately 150 words each on three English-language TV series that feature non-binary actors and characters (see Subsection 5.1.2). The second group was asked to manually post-edit the same texts that had been previously machine-translated, i.e. participants received a text document divided into two columns, one containing the English language source text and the other containing the MT. The three texts were retrieved online, and in the case of PE, translated with the freely available MT system DeepL² in June 2022. DeepL was

¹<https://info.taus.net/mt-PE-guidelines>

²<https://www.deepl.com/translator>

selected for this study because it is a widely known and used commercial system. For each text, participants were instructed to adopt a different approach to GFL, that is, (i) gender-neutral rewording, (ii) gender-inclusive characters, and (iii) neosystems. For each approach, they could freely select specific strategies from the provided handout, e.g. gender star (*) or underscore (_) amongst others for (ii). The handout as well as other electronic resources such as online dictionaries were allowed during the experiment.

The study was conducted online for several reasons. It aimed for an authentic and nonintrusive experimental setting in which translators could work from a familiar environment, notably their office at home, with familiar computers and, if desired, known CAT tools. The only restriction was to use one screen for the whole process to be recorded, and the translating participants could not use MT. The target country for participant recruitment was Austria because UNIVERSITAS, the Austrian Interpreters and Translators Association, financially supported the study. Due to difficulties in finding a large number of participants, especially non-binary people, who are still a marginalised group, recruitment was not restricted to one specific city. These difficulties are common to research on non-binary GFL (see Burtscher et al. 2022, Gromann et al. 2023), probably because it is still a controversial, political, and ideologically loaded topic.

Participants joined a video conference and were asked to share their screens as in Figure 5.1. During the translation and PE process, the video conference was open in the background and the shared screen was recorded. Participants could decide whether and when to make pauses and did not have to work under time constraints. After the translation and PE tasks, they were interviewed on their impressions, strategies, and challenging aspects of the study, amongst other things. They also discussed the possibility of using MT to translate GFL.

The analysis of the translation and PE process was inspired by Krings' (2001) division into temporal, technical, and cognitive effort. In order to shed light on the cognitive processes involved in integrating GFL in translation and PE, it focused on the translation and PE times, the screen activity tracking data, and participants' experiences and impressions. These different aspects of the translation and PE process allowed investigation of the effort as well as strengths and limitations of the three GFL strategies tested in the present work.

Screen recordings were used to measure translation and PE times. Though they are usually combined with other methods, such as keylogging (e.g. Rosa et al. 2020) and/or eye tracking (e.g. Cui & Zheng 2021), in studies conducted in natural settings they tend to be used individually (e.g. Kujamäki 2010) or in combination with retrospective interviews (e.g. Albl-Mikasa et al. 2017).

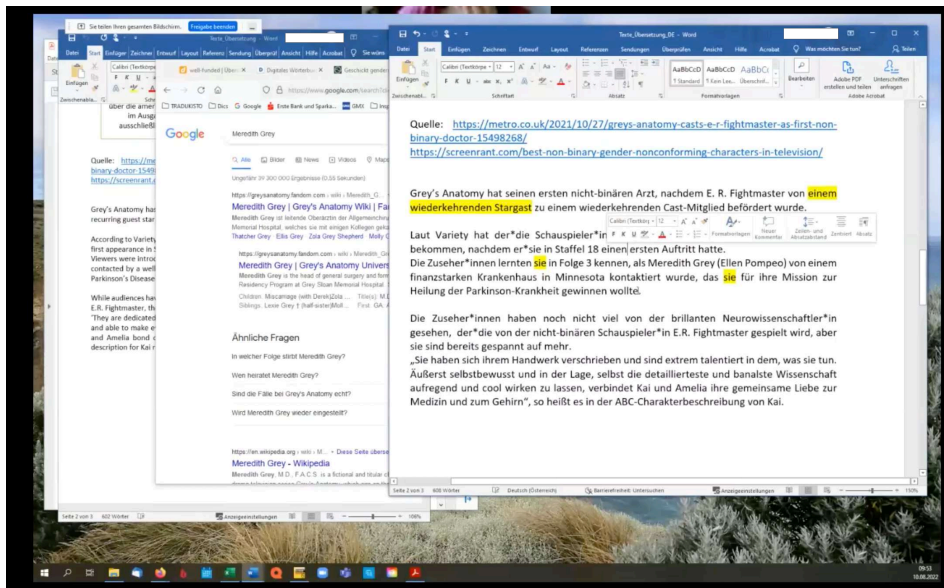


Figure 5.1: Study participant's screen during the translation process

Observation protocols were produced by combining non-participant observation and screen recordings. This is a research method that allows the researcher to gather data on a phenomenon while observing activities in their natural context without taking part in the observed activities (Liu, Maitlis, et al. 2010: 609). Protocols were used to track and reproduce the screen activity of the participants to shed light on the translation and PE process of GFL. Muñoz Martín & Cardona Guerra (2019: 536) maintain that broken words, pauses, changes, and searches may generally be indicators of translation problems. In the present work, they were used as types of screen activity to study the translation and PE process of participants. Broken words refer to translators leaving words unfinished. Changes are text modifications which correspond to a change in the original planning of the translation, which does not include typos.

Retrospective interviews were used to gather data on the perceived difficulty as well as the personal experience of participants. The use of TAP and/or retrospective interviews originally comes from psychology (Ericsson & Simon 1984) and was first applied to TS by Krings (1986). Some concerns arose about the validity of the method. These include the completeness and reliability of verbal reports and their possible effect on cognitive processes. They also cannot yield information about unconscious processes (see Jakobsen 2017). Nevertheless, TAP

and retrospective interviews are used in TPR to gain further insights into translation tasks (Dimitrova & Tiseliuss 2009). They are of great value, especially if used to triangulate data (Jakobsen 2017), as in the current experiment with screen recordings and observation protocols.

In addition, the produced target texts were annotated. Since the focus of this study is not on translation quality but rather on the ease of integrating GFL into the translation and PE process, only the selected GFL strategies and the success of their use were annotated. Further details on the study preparation, data collection and analysis are described in the next section.

5.1 Study preparation

In the following subsections, different aspects of the study preparation are described in detail. These include criteria for language, text, assignment, and participant selection.

5.1.1 Languages and GFL strategy

English was selected as the source language because it is a notional gender language that carries inflection only in third-person singular pronouns and few nouns. Reference to all genders is hence relatively straightforward and GFL strategies are more common. Notably, singular *they* has nearly become the standard for referring to non-binary individuals (Gender Census 2021) and has historically been used to refer to people of unknown gender (Baron 2020). Also, in English-speaking countries, especially the USA, UK, Canada, and Australia, awareness of non-binary genders seems to be more widespread with, for instance, celebrities coming out as non-binary and TV series portraying non-binary people.

German was selected as the target language since it has grammatical gender and requires extensive gender marking in articles, nouns, and adjectives. Furthermore, a multitude of GFL strategies exist in German and no standard has yet been widely adopted. Due to this multiplicity of GFL strategies as well as differences in language structures, achieving linguistic inclusion or neutralisation of all genders is challenging for both human and MT from English into German. The language pair English-German is also a good selection for the MT-based PE component of this study because MT is known to work better for high-resource languages due to the amount of training materials available (Forcada 2017).

As shown in Figure 5.2, GFL strategies to be used in the translation and PE experiment included (i) gender-neutral rewording, (ii) gender-inclusive characters, and (iii) neosystems. In the figure, different word classes are highlighted in different colours: violet for adjectives, yellow for singular non-binary nouns, green for plural nouns, light blue for articles, and red for pronouns. Each GFL approach differs in its application and was therefore expected to present different challenges. While (i) and (ii) are already used to a certain extent, especially to avoid masculine generics, (iii) represents a new approach to gender fairness and is devised by and for non-binary people. Three texts were then retrieved based on several selection criteria, i.e. text typology, variety of gendered elements, length, and readability.

Die Serie „Sex Education“ auf Netflix ist für ihre hemmungslose Annäherung an Sex und Sexualität bekannt. Einen ganzähnlichen Ansatz verfolgt die dritte Staffel in puncto Geschlechtsidentität: Hier stehen vor allem Menschen im Vordergrund, die sich selbst als nichtbinär bezeichnen. Hierbei nimmt vor allem die nichtbinäre Person Dua Saleh eine Vorreiterrolle ein: Im Bereich Schauspiel und Musik aktiv schlüpft Dua Saleh in die Rolle von Cal, einem neuen Mitglied in Moordale, das sich unverblümt mit der Schulleitung Hope (Jemima Kirke) anlegt.

Während das Publikum die*n von der*m nicht-binären Schauspieler*in E.R. Fightmaster gespielte*n brillante*n Neurowissenschaftler*in bislang noch nicht viel zu Gesicht bekam, wartet es schon ungeduldig auf weitere Begegnungen. Laut der Charakterbeschreibung von ABC „macht Kai ihre *seine Arbeit mit Leidenschaft und ist höchst talentiert. Sie*r ist extrem selbstbewusst und schafft es, selbst kleinste, eigentlich langweilige wissenschaftliche Details spannend und cool erscheinen zu lassen. Kai und Amelia finden in ihrer Liebe zu Medizin und Gehirnforschung eine gemeinsame Leidenschaft.“

Baig, spielt bei dieser Serie außerdem die Hauptrolle. Baig ist einens südasiatisch, queer, muslimisch Schauspielens, dens sich zuvor mit ens Arbeit im Theater einen Namen gemacht hat und dens in Sort Of an den Tag gelegter trockener Humor ens größtes Markenzeichen ist. Ens spielt die Rolle des Kinderbetreuens Saba Mehboob, dens die Möglichkeit hat, ens Traum eines besseren als ens bisher so unbefriedigenden Lebens in Berlin zu verwirklichen. Nachdem die Mutter der Kinder allerdings einem schweren Fahrradunfall zum Opfer fällt, entscheidet sich Saba dafür, doch in Toronto zu bleiben.

COLOUR-CODING

Adjectives

Non-binary nouns (singular)

Plural nouns

Articles

Pronouns

Figure 5.2: Extracts from the translations according to the three different study conditions

5.1.2 Texts and assignments

Since non-binary individuals are increasingly featured in TV series and films, they also appear in popular texts, such as magazine articles (Misiek 2020). General language texts on non-binary actors starring in English-language TV series were retrieved online. Another reason for choosing this type of text was that they present and/or describe non-binary individuals in positive terms. The first text is about the Netflix series *Sex Education*, where non-binary actor and singer Dua Saleh played the non-binary student Cal. The source of the text is the online website of the American magazine *Newsweek*.³ The second text is about the promotion of non-binary actor E.R. Fightmaster as a recurring cast member in *Grey's Anatomy*. It has two sources, namely the online British news site *Metro*⁴ and the movie and TV news website *Screen Rant*.⁵ The third and final text presents the Canadian TV series *Sort Of*, both created and interpreted by non-binary artists. Also in this case, two sources were used to create the text, i.e. the Canadian pop culture website *Complex*⁶ and the Canadian news site *CBC*.⁷

Texts were also selected based on how many and which kind of personal references to non-binary people and mixed-gender groups they contained. Each text contains singular *they* and at least one inflected form, two nouns in the singular form as well as articles or relative pronouns in reference to a non-binary individual. In addition, each text also contains a noun in the plural form referring to a mixed-gender group or non-binary individuals. Texts were also slightly modified to comply with these requirements. Study materials are included in the Appendix. Here is an extract:

Leading the way is *a non-binary actor and musician* (non-binary article, adjective, and noun) Dua Saleh as Cal, *a new student* (non-binary article, adjective, and noun) at Moordale (...) Hope is determined to turn Moordale's reputation as a "sex school" around and in doing so, she sets out to squash *students'* sexual freedoms and *their* right to express *their* (mixed-gender group) identity. Cal stands up against Hope (...) *They* (non-binary pronoun)

³<https://www.newsweek.com/finally-sex-education-season-3-praised-groundbreaking-non-binary-representation-1629326>

⁴<https://metro.co.uk/2021/10/27/greys-anatomy-casts-e-r-fightmaster-as-first-non-binary-doctor-15498268/>

⁵<https://screenrant.com/best-non-binary-gender-nonconforming-characters-in-television/>

⁶<https://www.complex.com/pop-culture/bilal-baig-cbc-sort-of-interview>

⁷<https://www.cbc.ca/arts/new-show-sort-of-doesn-t-just-break-representational-barriers-its-also-more-than-sort-of-great-1.6186771>

Table 5.1: Details on translation and PE materials

Text No.	TV Series	Instructed Gender-Fair Strategy	Word Count	Gender Phrases
1	Sex Education	Gender-Neutral Rewording	152	9
2	Grey's Anatomy	Gender-Inclusive Characters	151	12
3	Sort Of	Neosystems	163	10

refuse to wear the girls uniform and is reluctant to wear the tight-fighting boys uniform.⁸

In order to annotate the target texts, the gender phrases they contained (nine, twelve, and ten respectively) and their translations were identified and analysed.

In TPR involving the use of eye tracking, short texts of around 200–300 words are usually selected, for example, because the researcher must analyse a significant amount of data and participants' motivation may decrease because of tiredness or boredom (O'Brien 2009: 261). This also applies to the present study as participants were required to translate or post-edit three different texts, one after the other. As a consequence, only the first paragraphs of the retrieved texts were used in the study. They consist respectively of 154, 149, and 152 words. Details on the texts are summarised in Table 5.1.

To ensure comparability of estimated translation and PE times, readability scores were computed using the Flesch-Kincaid readability test (Kincaid et al. 1975) as in Rojo López & Caro (2016). All texts were scored as "fairly difficult to read". This test accounts for the number of words and their length but ignores semantic difficulty. For this reason, the perceived difficulty of the source texts was discussed in the retrospective interviews.

Participants received a translation brief together with instructions on tasks, PE guidelines and a handout on GFL. This was designed to resemble a realistic order to aim for an authentic experimental setting (see materials linked in the Appendix). Therefore, the assignment description included client specifications, target audience, instructions concerning resources that could be used, and for the PE group also guidelines on full PE.

Austrian TV magazine *TV-Media* was selected as the client for whom the translations were to be prepared. The translation brief specified that the source texts were general-language texts, thus prepared terminological resources would not be provided. The target audience was said to be general. Translators received the texts and had to deliver them as text documents in Word format. They were

⁸Grammar and spelling errors in the original are reproduced here.

allowed to use their usual translation environment and tools as well as online resources. The texts were to be delivered as completed translations that would not need proofreading.

Finally, people receiving the translation assignment were asked not to use MT. People who received the PE assignment were also given rules for full PE. The translation brief also included instructions on which approach to GFL was to be used. Five minutes before the video conference, participants received the texts with a shortened version of the translation brief and the texts to translate and post-edit.

5.1.3 Participant selection

The number of participants, namely twelve, was established based on existing literature in the field of TPR (see for example O'Brien 2009: 255). O'Brien reviews previous studies in the field and notes how these feature a number of participants ranging from five to twenty. This is generally due to funding constraints. She also underlines how results from studies with twelve participants cannot be used to make generalisations, but have nevertheless significant value. Criteria for participant selection in the translation experiment were then established and aimed at ideal conditions in the translation market.

Translators needed (i) to have at least two years of work experience in the translation industry; (ii) to be highly proficient in English (C1+ level of the Common European Frame of Reference for Languages (CEFR)), i.e. the source language; (iii) to be German native speakers, as the translation into the L1 is most common; and (iv) to hold a relevant degree, e.g. translation/interpreting, linguistics, languages. An additional criterion was limited knowledge of GFL.

Previous joint studies on this topic (Burtscher et al. 2022, Gromann et al. 2023) suggest that few professional translators in Austria have extensive experience with GFL beyond the binary. As this study strives for representativity, translators with little or no knowledge of GFL were sought. This criterion could only be met in part. First, it does not apply to non-binary individuals who, for obvious reasons, were expected to be more sensitive to the topic and have higher expertise. Second, the topic is particularly sensitive and still controversial: only people with positive attitudes towards or a general interest in GFL, i.e. with reasonable expertise in the topic, participated in the study.

Participants were selected by means of opportunity sampling. A call for participation in the translation experiment (see Appendix) was disseminated on the main social media platforms, i.e. Facebook, Twitter, Instagram, and LinkedIn, used by the author and by UNIVERSITAS. UNIVERSITAS also disseminated the

request in their mailing lists and newsletters. The association was involved since it supported the study financially. Candidates were paid and worked from their familiar environment to increase the experimental realism of the study. The call for participation was disseminated from 31 May 2022 onward. Originally, a deadline for the application had been set for 6 June 2022. Since few translators had responded, the deadline was extended to 21 June 2022. After that, the author started to search for participants in his and his co-supervisor's personal network.

An additional selection criterion was gender. Since translators are mostly women, eight out of twelve participants in this sample were women. Three men and one non-binary participant were selected in order to create a sample that was as representative as possible. However, no statistics concerning translation professionals in Austria were available, thus limiting the representativity of the study. To collect information concerning the participants' profiles, a pre-study questionnaire was created with Google Forms (see Figure 5.3), consisting of twenty open and closed questions on demographics, work, GFL and PE experience as well as GFL use. The profile of the candidates is further discussed in Chapter 6.

5.2 Data collection and analysis

In the following subsections, the data collection and analysis process is described in more detail, delving into interview preparation, transcription and analysis, translation times measurements, observation protocols preparation and analysis, and segment-based target text annotation.

5.2.1 Interviews

Semi-structured interviews were conducted immediately after the study. This format was selected as it allows for the author's flexibility to study specific aspects of the participants' experience in depth. The interview guide consisted of twelve questions for the group of translators and fourteen questions for the group of post-editors, respectively. These included impressions of the tasks, such as the challenges faced, the participants' preferred GFL approach, the impact of GFL use on their productivity during the translation and PE process, as well as their use of MT in general for translators, and within this study for post-editors (see Appendix for the specific questions). Specifically, post-editors were asked how their experience with PE was, whether the MT draft made their work easier and/or faster, and whether they would consider using MT again in the future for a similar assignment.

Questions Responses 12 Settings

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Genderfaire Übersetzung

Liebe*r Teilnehmer*in,

Danke, dass Sie sich bereit erklärt haben, an meiner Übersetzungsstudie zu genderfairer Sprache teilzunehmen!

Bitte beantworten Sie kurz die vorliegende Umfrage vor Ihrer Teilnahme an der Studie. Alle Daten werden anonymisiert und streng vertraulich behandelt. Somit werden die Daten Ihrer Person nicht zuordenbar sein.

Falls Sie Fragen haben, können Sie mich jederzeit unter manuel.lardelli@uni-graz.at kontaktieren.

Email *

Valid email address

This form is collecting email addresses. [Change settings](#)

...

Zu welcher der nachfolgenden Alterskategorien gehören Sie?

Multiple choice

- ☐ 17 oder jünger
- ☐ 18-24
- ☐ 25-31
- ☐ 32-38
- ☐ 39-45
- ☐ 46-52

Figure 5.3: Screenshot of the questionnaire for participants

In studies involving the use of eye tracking and keylogging, the log file generated during the translation experiment is at times replayed and the writing process is disclosed to the participants (Dimitrova & Tiselius 2009: 112). In this study, the target and source texts were used as cues, therefore participants were allowed to read them during the interviews. Interviews lasted approximately 45–50 minutes, with deviations among participants.

Data were collected from July to October 2022. At the same time, the interviews were transcribed following the rule system proposed by Kuckartz et al. (2008), which was later updated by Dresing & Pehl (2018), who created a content-semantic transcription system that prefers readability over details concerning pronunciation, colloquialisms, and tone. This decision was taken since elements such as pronunciation and the use of colloquialisms are not relevant to the scope of the present study. The transcripts of the interviews were also sent to participants to comply with good scientific practice.⁹

⁹In compliance with regulations at the University of Graz, ethical approval was sought and received for the present study (Ethikantrag Nr. 87 - 2021/22).

The interviews were then analysed in MAXQDA,¹⁰ a software for computer-assisted qualitative and mixed-method data analysis, and following the basic process of thematic content analysis by Kuckartz (2014) as depicted in Figure 5.4. During the analysis, suggestions and good practices for work in MAXQDA proposed by Kuckartz & Rädiker (2020) were also followed to ensure a systematic analysis of qualitative data. For instance, a project memo with the research topic, research questions, and information related to the study participants was created. Also, the project logbook was used to track the progress of the data analysis. During the first reading of the interviews, memos were added and text passages were highlighted with colours or with emojis, as shown in Figure 5.5. The transcripts were then summarised.

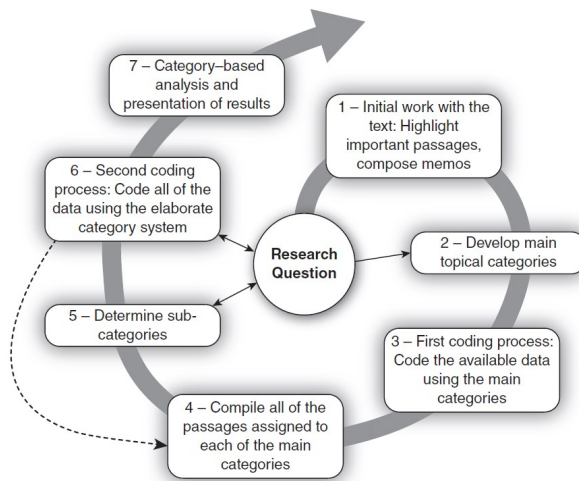


Figure 5.4: Steps for thematic qualitative text analysis as in Kuckartz (2014: 70)

After this step, a first category system was inductively developed based on the interview guide. For almost every question, a category was created and defined in the code memos. Additionally, three other categories were developed following Kuckartz & Rädiker (2020), i.e. *other*, *interesting*, and *quotable passages*. The *other* category was used for aspects that are relevant to the research questions but did not belong to a category yet. The *interesting* category was used for passages that are not fully relevant to the research questions but still cover potentially

¹⁰<https://www.maxqda.com>

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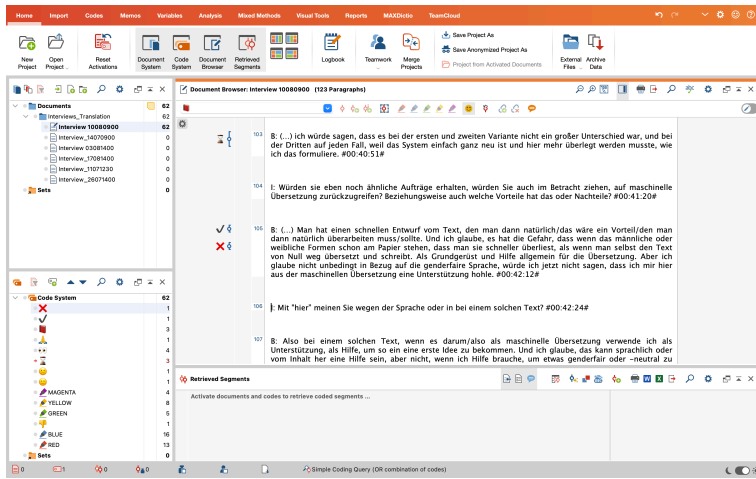


Figure 5.5: First text reading in MAXQDA

interesting aspects for the present work. The *quotable passages* category was used to select statements to be quoted in the result section of the work. Basic coding was then performed on all interviews.

In the third step, segments assigned to each category were read to expand the category system with subcategories. During the fine coding phase, an order of category relevance was followed to create subcategories and then further develop and/or adapt them to better fit the data. The coded segments were consequently read numerous times until a saturation point was reached, i.e. no further subcategories were added anymore. The final category system comprises eleven main categories:

1. **ST – Challenges in Source Texts:** used when participants talk about the source texts in general and not about translation-related problems.
2. **GNL – Gender-Neutral Language:** used when the translation process of the first assignment, i.e. the translation with gender-neutral rewording, was discussed.
3. **GIC – Gender-Inclusive Characters:** used when participants elaborated on the translation process of the second assignment, i.e. the translation with gender-inclusive characters.
4. **N – Neosystems:** used for impressions concerning the translation process during the third assignment, i.e. the translation with neosystems.
5. **GI – General Impressions:** used when participants discussed personal impressions of the study such as what they liked doing or not.

6. **MT – Use of MT and Dis/advantages:** used when participants elaborated on the use of MT in general as well as for texts featuring GFL.
7. **PEMT – Impressions of the PE Task:** this category was applied to interviews with translators who were assigned the PE task and therefore discussed their impressions regarding their speed, the quality of the MT outputs and similar matters.
8. **SP – Speed and Productivity:** used for the participants’ impressions of the impact that the use of GFL exerted on their work speed and productivity.
9. **O – Impact of Observation:** used when participants reflected on how they behaved during the study because of the researcher’s observation.
10. **RA – Real Assignments:** used for discussions on how the participants would approach the translation or PE of GFL in a real-world scenario.
11. **D – Desires:** used when participants expressed their desires for optimal work with GFL.

Finally, MAXQDA was used to compare the coded data and to create tables that will be presented in Chapter 6.

5.2.2 Segment-based target text annotation

After data collection, target texts produced by the participants were annotated for GFL strategies used and the success of their application. The devised method was inspired by approaches to both TQA (Castilho et al. 2018) and qualitative content analysis (Kuckartz 2014). The process is explained in more detail below.

First, two Excel documents were created, one for translation and the other for PE analysis. For each source text, nine, twelve, and ten gender phrases were identified with references to groups, e.g. “audiences”, and non-binary actors/characters, e.g. “non-binary neuroscientist”, that required gender marking in German. These were pasted into the Excel sheets along with all corresponding translations. Following typical approaches to qualitative content analysis (Kuckartz 2014), the texts were read several times and a basic category system was developed. The category system differed for each text since participants were instructed to use a different gender-fair strategy, namely gender-neutral rewording, gender-inclusive characters, and neosystems.

The system included different GFL strategies, instances of misgendering, use of masculine generics, and mistakes in the application of GFL. Each category was assigned a number, as shown in Table 5.2. Texts were accordingly annotated, as in Figure 5.6 and the category system was refined. During this step, some categories were merged to provide a clearer overview of the results.

Changes included merging the categories *rewording*, *abstraction*, and *change of perspective* into the general category *rewording* for the first text, and the use of the broad category *other strategy* for gender references in the second and third text for which participants did not use gender-inclusive characters and neosystems. For this reason, in Table 5.2 some numbers and categories were removed. Finally, a count function in Excel was used to calculate the frequency of each category both in general and for each analysed gender phrase.

Table 5.2: Numbers assigned to categories for the translation annotation of the first text

Code	Strategy
0	No strategy
1	Indefinite Pronoun
2	Gender-Neutral Words
4	Rewording
5	Collective Noun
6	Participial Form
7	Name Repetition
10	Pronoun Omission
11	Misgendering/Mistranslation of <i>they</i>
12	False Term
13	Gender-Inclusive Characters
14	Relative Pronoun
15	Masculine Generics
16	Omission

Similarly, the machine translations produced with DeepL were annotated for bias. For each text, the nine, ten, and twelve phrases containing gender references to be post-edited were selected and pasted into the previously mentioned Excel sheet. The MT outputs were annotated for three types of bias: (i) misgendering (i.e. a binary gender was assigned to a non-binary person mentioned in the text), (ii) mistranslations of singular *they* (i.e. the pronoun use was misinterpreted and translated in the third-person plural into German), and (iii) masculine generics.

5.2.3 Translation times and observational protocols

Screen recordings are considered an appropriate method to elicit data on the translation process, even though they are not as precise as keylogging. They al-

SOURCE TEXT	P1		P2	
Audiences	dens Zuschauens	1	Das Publikum	6
They were also like enough	Ens hatten auch das Glück	1	sondern durfte sich auch so glücklich schätzen	0
This is a comedy about a nanny navigating their complicated existence	geht es um ein genderfluides Kinderbetreuens bei der Bewältigung von ens komplizierter Existenz.	1 5	von einem genderfluiden Nanny handelt, dens sich in ens komplizierten Dasein zurechtfinden muss	1 5
by co-creators	einer Co-Produktion von Bilal Baig und Fab Filippo	6	Erfunden wurde die Serie von	6
a South Asians, queer Muslim actor	einens südasiatisch, queer, muslimisch Schauspielens	1	einens südasiatisch, queerens, muslimisch Schauspielens	1 5
who was previously known for their work	dens sich zuvor mit ens Arbeit im Theater einen Namen gemacht hat	1	dens vorher vor Allem für deren Arbeit im Theater bekannt war	1 5
and whose deadpan delivery	dens in Sort Of an den Tag gelegter trockener Humor ens größtes Markenzeichen ist.	1	durch Baigs todernteste Art	6
They star as nanny Sabi Mehboob	Ens spielt die Rolle des Kinderbetreuens Sabi Mehboob	1	Ens spielt dens Nanny Sabi Mehboob	1 4
who is given a chance to move to Berlin in pursuit of something more than their clearly unsatisfying life	dens die Möglichkeit hat, ens Traum eines besseren als ens bisher so unbefriedigenden Lebens in Berlin zu verwirklichen	1	dens die Chance bekommt, nach Berlin zu ziehen, um dort mehr zu finden als deren bisheriges so unbefriedigendes Leben	1 5
Sort of represented their first major experience with television	ist Sort Of ens erste große Erfahrung mit dem Fernsehen.	1	ist die Serie Sort Of deren erstes richtiges Fernsehprojekt	5

Figure 5.6: Example of target text annotation

low the researcher to collect additional data on the type of consultations made by translators and to see what happens on-screen. For this reason, they are regarded as a valid method for data triangulation (Angelone 2013). In the present study, they were used to measure translation times as well as to complete observation protocols produced with the researcher's notes.

Translation times were measured and imported into an Excel sheet, together with the demographics and data on the use and perception of GFL by participants as collected in the pre-study questionnaire. The quantitative analyses in this study aimed to examine whether different tasks (translation vs. PE) and GFL strategies (gender-neutral rewording, gender-inclusive characters, and neosystems) had an impact on participants' performance. To achieve this, both descriptive and inferential statistical methods were employed, using Microsoft Excel for basic analysis and the statistical analysis software Jamovi¹¹ for more advanced statistical testing.

Descriptive statistics, including the mean (average), median (middle value), and standard deviation (a measure of how spread out the values are), were calculated to summarise the translation and PE times. These metrics help describe the general tendencies in the data and provide a foundation for further statistical testing. This approach is in line with standard practice in TPR (O'Brien 2009).

¹¹<https://www.jamovi.org>

Inferential statistics are used to determine whether observed differences in the data are statistically significant, that is, unlikely to have occurred by chance alone. Before selecting appropriate tests, assumptions such as normal distribution (whether the data follow a bell-shaped curve) and homogeneity of variance (whether different groups have similar variability) were checked. Since these assumptions were not met, alternative robust tests were applied.

Inferential statistics were performed importing the Excel sheet with the experiment data into Jamovi. The different types of variables, e.g. age, gender, and translation times, were assigned to different categories, i.e. nominal, ordinal, and continuous.

A *Welch's t-test* was conducted to establish whether the task, i.e. translation and PE, affected the participants' speed. Unlike the standard t-test, Welch's test does not assume equal variances between groups, making it more appropriate for small or unequal samples (Ruxton 2006). Because the data were also not normally distributed, a *rank transformation* was applied. This means that the actual data values were replaced by their rank positions (e.g. 1st, 2nd, 3rd), which reduces the impact of outliers and skewed data while allowing parametric test structures to be retained (Zimmerman & Zumbo 1993). This method is recommended in TS when working with small and heterogeneous samples (Mellinger & Hanson 2016).

To examine whether the different GFL strategies had an effect on processing times, a *repeated measures ANOVA (rANOVA)* was used. This test is suitable when the same participants complete multiple tasks, allowing within-subject comparisons. However, rANOVA assumes sphericity (that the variances of the differences between all pairs of conditions are equal). Since this assumption was violated, the *Greenhouse-Geisser correction* was applied to adjust the degrees of freedom and reduce the risk of false statistical significance (Girden 1992). This procedure is common in psycholinguistic and TPR experiments involving multiple within-subject measurements.

Following a significant rANOVA result, *post-hoc tests* were conducted using the *Bonferroni correction* to adjust for multiple comparisons. When several comparisons are made, the chance of finding a significant result by chance increases. The Bonferroni method addresses this by lowering the threshold for significance, thus preserving the overall error rate (Bland & Altman 1995). Although conservative, this method ensures the reliability of the results and is recommended for studies with multiple conditions.

Screen recordings along with observation notes were used to produce observation protocols. The source texts were split into segments based on their punctuation and entered in an Excel sheet. The aim of the protocol analysis was to

summarise the translation and PE process, more specifically the activities needed to integrate GFL in the target texts. Muñoz Martín & Cardona Guerra (2019: 536f) consider broken words, changes, searches, and pauses as indicators for possible translation problems and hence cognitive processes. Such activity categories were used in the analysis of the observation protocols.

While in Muñoz Martín & Cardona Guerra (2019) the first category was *broken words*, for the present work, noun phrases, e.g. “non-binary actor”, were regarded as larger units that are translated together and hence the category *broken units* was preferred. The annotation included frequencies of each phenomenon as well as explanations of what happened on the screen, e.g. which sources were searched for and when pauses were taken during the translation and PE process, as shown in Figure 5.7. A column was also added for general comments and observations that could potentially provide more insights into the participants’ workflows.

SOURCE SEGMENT	Broken Words		Changes		Searches	
	n.	type	n.	type	n.	type
Sex Education on Netflix is known for its no-holds-barred approach to sex and sexuality. In season three, the show takes a similar approach to gender identity, shining a bright light on those who identify as non-binary.						
Leading the way is a non-binary actor and musician Dua Saleh as Cal, a new student at Moordale who clashes directly with Headmistress Hope (Jemima Kirke).	2	Schaus - pause / Musik - pause	Schauspiel deleted and then wrote "Schauspielperson" / Musik deleted / "Mitglied der Studierenden" -> "der Schülerschaft"	3	4	Handout
Hope is determined to turn Moordale's reputation as a "sex school" around and in doing so, she sets out to squash students' sexual freedoms and their right to express their identity.	1	Schul -	1	Schul -		
Cal stands up against Hope, refusing to wear the new Moordale school uniform that is deemed acceptable in Hope's eyes.						
They refuse to wear the girls uniform and is reluctant to wear the tight-fighting boys uniform.			1	"Cal" deleted	1	Handout (looking for pronouns)
Cal likes to keep their uniform loose and baggy, as it makes them feel more comfortable in who they are, but Hope just won't listen.			"sich wohler mit der eigenen Person" -> "sich wohler in Bezug auf die eigene"	1		
RESULTS	3		6		5	

Figure 5.7: Example of observational protocols

In TPR, pauses have been operationalised differently (see Kumpulainen 2015) based on the methodology used in studies as well as on their objectives. In screen recordings, it is almost impossible to measure pauses of less than three seconds. Hence, only pauses of three seconds or more (3+) were counted for the analysis. Finally, absolute and relative frequencies of the analysed activities were computed and compared among participants, assignments, and text segments. The absolute frequencies of activities per participant were also pasted into the aforementioned Excel sheet containing the data of the translation and PE times.

In this case as well, inferential statistics were performed. As in the analysis of the translation times, two tests were conducted. The first was a Welch’s test

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on rank-transformed data to test whether the task type affected the number of activities observed in the screen recordings. The second was a rANOVA with Greenhouse-Geisser correction and a post-hoc test with Bonferroni correction to test whether the assignments affected the number of the activities tracked.

This chapter outlined the methodology of the present study, which integrates both quantitative and qualitative methods from process- and product-oriented research in TS. Twelve participants were recruited and divided into two groups: one was tasked with translating, and the other with post-editing three texts related to TV series featuring non-binary characters. The translation and PE process was recorded, and the author conducted observations while taking detailed notes. Processing times were measured, and participants were interviewed after each study session. Finally, the target texts were analysed with particular emphasis on the strategies employed and the errors made.

6 Results

This chapter presents the main findings from the study described in Chapter 5. The results are presented as follows: first, the participants' profile is described. Translation and PE times are then presented and compared, followed by the analysis of the screen activity tracking data. Process data are complemented with the interviews concerning the participants' experience and impressions of the tasks, and their most relevant comments are summarised. Finally, the analysis of the specific GFL strategies used and the success of their application is presented and gender bias in the MT outputs is briefly discussed.

6.1 Participants' profile

As initially intended, 12 participants were recruited for the translation experiment. Six received the translation assignment (Task: TR in Table 6.1) and six the PE one (Task: PE in Table 6.1). Participants were aged from 25 to 59 years old. Eight identified as women, three as men, and one as non-binary. Their work experience spanned from 3–5 to more than 20 years, and all except for one had some experience with PE. For the PE assignment, participants with extensive ($n=4$) or little ($n=2$) experience in this task were selected. In addition, all participants indicated that they used GFL in their daily work except for a patent translator who stated that this is not desired in the field. The majority of the participants ($n=7$) indicated that they normally used gender-inclusive characters. Three indicated an alternation with gender-neutral rewording based on the client's requirements and/or the assignment whereas one used all strategies. An overview of the participants' profiles is given in Table 6.1. Their reasons for using GFL include (i) the desire to include everyone in language, (ii) to keep up with linguistic change, and (iii) feminist beliefs.

Participants, with the exception of P8 who did not use GFL in their daily work, were also asked to rate the difficulty of GFL on a Likert scale from 1 to 5 where one stands for "very difficult" and five for "very easy". The majority ($n=4$) indicated "neutral" and the rest were divided between "easy" ($n=3$) and "difficult" ($n=4$), as shown in Figure 6.1, where the percentages relate to the number of participants giving this answer. Only one participant indicated "very easy". Finally,

Table 6.1: Participants' profiles

Participant	Task	Age	Gender	Work Experience	PE Experience	GFL Experience	GFL Use
P1	TR	39-45	Woman	11-15	Little	Little	Depends on client/assignment
P2	TR	25-31	Woman	3-5	Very Little	Yes	Gender-inclusive characters
P3	TR	25-31	Man	3-5	No	Little	Gender-inclusive characters
P4	TR	32-38	Non-Binary	6-10	Little	Yes	All
P5	TR	32-38	Woman	6-10	Very Little	Yes	Gender-inclusive characters
P6	TR	32-38	Woman	6-10	Very Little	Little	Gender-inclusive characters
P7	PE	32-38	Man	6-10	Extensive	Yes	Depends on client/assignment
P8	PE	39-45	Woman	16-20	Extensive	Little	No
P9	PE	53-59	Woman	20+	Little	Little	Gender-inclusive characters
P10	PE	32-38	Woman	6-10	Extensive	Yes	Depends on client/assignment
P11	PE	25-31	Woman	3-5	Little	Little	Gender-inclusive characters
P12	PE	39-45	Man	11-15	Extensive	Yes	Gender-inclusive characters

participants were also asked to comment on challenging and simple aspects of GFL use. Challenges included: (i) the use of neopronouns, especially because they are not yet established in the German language; (ii) the need to adapt and select strategies based on context; (iii) the selection of gender-neutral alternatives to gendered terms; (iv) consistent use of the strategies; and (v) negative impact on the readability of target texts. The use of gender-inclusive characters is simple according to two participants whereas two indicated that GFL is simple regardless of the strategy.

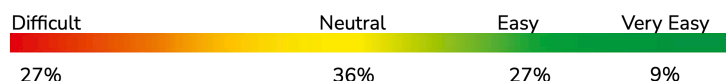


Figure 6.1: Ratings of GFL difficulty

6.2 Translation and PE times

Figure 6.2 shows the translation and PE times for each participant in each assignment. It must be noted here that the assignment order was not changed among participants and the most challenging approach to GFL, i.e. neosystems, was applied in the third text. Longer translation and PE times could hence be caused not only by the specific strategy applied but also by participants' tiredness.

First, a difference was found between tasks. PE was generally faster than translation in the first two assignments, i.e. gender-neutral rewording and gender-inclusive characters. Second, differences in speed among participants were considerable. For instance, Figure 6.2 clearly shows that P2, and, in part, P10, spent

longer times on the third assignment as opposed to the other participants. As observed in 6.3, they did not split their screen so as to have the translation open alongside the GFL handout, a common strategy among their study colleagues. Finally, neosystems required the longest and gender-inclusive characters the shortest processing time. These differences, however, were not statistically significant. The main findings from the analysis of the translation and PE times are presented in more detail below.



Figure 6.2: Translation and PE times for each assignment in minutes

Table 6.2 reports the median translation and PE times for each assignment along with the standard deviation (SD). SD values are reported to show the variability in translation and PE speed among participants; this prompted the use of the median in this chapter as a better indicator of the data’s central tendency. The translation and PE of the second text with gender-inclusive characters required the least time, i.e. a median of 00:21:20 minutes. The median value was higher in the first assignment, i.e. 00:24:56 minutes. Finally, neosystems took the longest median time, i.e. 00:27:02 minutes.

To test whether the task affected the participants’ speed, a Welch’s test was performed on the rank-transformed data. This test was selected since no equal variance and normal distribution of the data could be assumed. A statistically significant difference between translation and PE was found in the first ($t[9.98]=1.98$, p -value=0.038, $d=1.14$) and in the second assignment ($t[9.37]=3.07$, p -value=0.006,

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Table 6.2: Overview of translation and PE times among assignments

Assignment	Median	SD
1	00:24:56	00:06:36
2	00:21:20	00:07:33
3	00:27:02	00:09:06

$d=1.77$). The difference in the third assignment was not statistically significant ($t[10.0]=0.307$, $p\text{-value}=0.383$, $d=0.177$). The result suggests that PE was faster than translation in the first and second assignment, but not in the third. These differences can also be seen in Figure 6.3, where the box plots compare temporal differences between the translation (TR) and PE (PE) task among assignments.

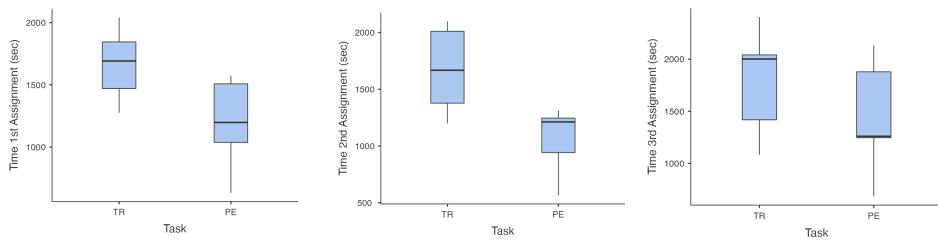


Figure 6.3: Box plots showing differences in translation and PE times (seconds) among assignments

As shown in Table 6.3, the translation task took longer than PE in each assignment. Participants in the translation group completed the first assignment in a median of 00:28:11 minutes – about eight minutes longer than participants in the PE group with 00:19:58 minutes. A similar difference was found when using gender-inclusive characters: translation required a median of 00:27:48 minutes and PE, 00:20:12. In the case of neosystems, the difference among tasks increased: translation took a median of 00:33:21 minutes and PE, 00:21:00.

Differences in translation times among assignments were considerable. Gender-neutral rewording and gender-inclusive characters took a median of 00:28:11 minutes and 00:27:48 minutes respectively, increasing by five minutes when using neosystems, i.e. 00:33:21. Median values for PE times were similar among assignments: the first required 00:19:58 minutes, the second 00:20:12, and the third 00:21:00 minutes.

Table 6.3: Comparison of translation and PE times among assignments

Assignment	Descriptive statistics	TR	PE
1	Mean	00:27:45	00:19:59
	Median	00:28:11	00:19:58
	Shortest Time	00:21:16	00:10:33
	Longest Time	00:34:00	00:26:12
	SD	00:04:49	00:06:03
2	Mean	00:27:53	00:17:49
	Median	00:27:48	00:20:12
	Shortest Time	00:19:59	00:09:26
	Longest Time	00:34:59	00:21:53
	SD	00:06:23	00:04:54
3	Mean	00:29:54	00:24:04
	Median	00:33:21	00:21:00
	Shortest Time	00:18:02	00:12:32
	Longest Time	00:40:04	00:35:32
	SD	00:08:39	00:09:59

In order to produce a more precise comparison of translation and PE times across assignments, measurements were normalised. A standard approach in TPR is to divide the translation times for each assignment by the number of words in the source texts (Läubli et al. 2013), as presented in Table 6.4. The results are separated for translation and PE because the translation times were divided by the number of words in the English-language source texts, but PE times by the number of words in the MT outputs. Again, the median was used, but the SD is also reported. To test whether the assignments had an impact on translation and PE times, a rANOVA with Greenhouse-Geisser correction was computed. The test did not show a statistically significant difference among assignments ($F[1.59,15.86] = 3.046$, $p\text{-value} = 0.085$). SD was generally high and was higher in the third assignment than in the first two and in PE than in translation. This indicates that there were considerable speed differences among participants.

Table 6.5 provides an overview of times per participant for each assignment, with the fastest and slowest assignment for each participant indicated in yellow and in red respectively. The second text with gender-inclusive characters was the fastest assignment for half of the participants, namely P2, P6, P7, P8, P11, and

Table 6.4: Median value of translation and PE times, SD (seconds per word), and relative SD by assignment

Assignment	TR & PE Times	TR	PE
1	Time (s/word)	11 ± 1.9	7.9 ± 2.4
	Relative SD	17%	30%
2	Time (s/word)	11.2 ± 2.6	7.9 ± 1.9
	Relative SD	23%	24%
3	Time (s/word)	13.2 ± 3.4	7.8 ± 3.5
	Relative SD	26%	44%

P12. Three participants only, namely P1, P3, and P4, took the longest for this second assignment. Gender-neutral rewording was the fastest for four participants, P1, P5, P9, and P10, and the slowest for three, P6, P7, and P11. Translation and PE times for the third assignment were generally higher and neosystems were the fastest assignment for two participants only, namely P3 and P4. Conversely, P2, P5, P8, P9, P10, and P12 took the longest for this. One important difference between translation and PE emerges: while participants in the translation group were generally faster in the third assignment than in the second, post-editors were generally slower in the third. Perhaps, post-editors felt less under time pressure because of the MT and therefore focused more intensively on the third assignment because neosystems were judged to be more complex.

As concerns translation speed in general, P4 was generally the fastest participant in the translation group. They were the only non-binary participant in the study and rated the difficulty of using GFL as easy in the pre-study questionnaire. Also, as described in Section 6.5, they selected a neosystem that they already used in their daily life with friends. P4 was the fastest in completing the first and the third assignments. In the second, P6 was the fastest. P6 admittedly had less experience with GFL than P4 and found it difficult to use. Among translators, P3 was generally the slowest participant in the first and second assignments. When using neosystems, P2 needed the longest time. Both rated GFL as neither difficult nor easy, but P2 had more experience with it. P12 was the fastest participant overall, post-editing each text in about ten minutes even though they rated GFL use as difficult. P8 was the slowest participant in the PE group and, in fact, the only participant who did not integrate GFL in their everyday work.

When asked to comment on the impact of GFL on their speed and productivity, half of the participants stated that they felt GFL generally slowed them down.

Table 6.5: Participants' translation and PE times per assignment in minutes

Participant	Task	1st Assignment	2nd Assignment	3rd Assignment
P1	TR	00:30:38	00:34:34	00:34:06
P2	TR	00:30:47	00:30:20	00:40:04
P3	TR	00:34:00	00:34:59	00:33:44
P4	TR	00:21:16	00:22:12	00:18:02
P5	TR	00:24:06	00:25:15	00:32:58
P6	TR	00:25:45	00:19:59	00:20:32
P7	PE	00:22:12	00:20:47	00:20:55
P8	PE	00:26:12	00:21:53	00:34:43
P9	PE	00:17:09	00:20:43	00:21:05
P10	PE	00:17:44	00:19:41	00:35:52
P11	PE	00:26:07	00:14:23	00:20:44
P12	PE	00:10:33	00:09:26	00:11:26

This perception was more common in the translation ($n=4$) than in the PE group ($n=2$). Also, five participants (P5, P7, P8, P10, P12) felt the least productive when using neosystems. This impression was confirmed when analysing translation and PE times except for P7. Finally, P4 and P6 felt they were the slowest in the first assignment, which was only confirmed for P6. No participant commented on gender-inclusive characters as the most time-consuming approach to GFL.

To summarise, differences in times could be partially observed between assignments. In translation, gender-neutral rewording and gender-inclusive characters each required a median of 28 minutes, while neosystems took a median of 33 minutes. Neosystems required the longest for half of the participants. PE was significantly faster than translation in the first two assignments and the median for each assignment was about 20 minutes. Furthermore, there were considerable differences in translation and PE among participants, but longer or shorter times did not always correlate with the perceived difficulty of GFL. Finally, the perception of the negative impact of GFL on speed and productivity was more common among translators than post-editors. This might be because PE is usually faster than translation and, as explained in Section 6.4, post-editors felt that the high-quality MT outputs they received improved their productivity.

6.3 Screen activity tracking data

In order to analyse the screen activity and gain insights into the gender-fair translation and PE process, observation protocols were produced with the observation notes and the screen recordings. The observation notes included comments on general approaches to and/or interesting aspects of the process. The screen recordings analysis was focused on the number of activities, i.e. (i) broken units, (ii) changes, (iii) searches, and (iv) pauses, performed by the participants.

First, as regards general approaches to the translation and PE, the majority of the participants ($n=8$) looked online for more information concerning the TV series described in the texts and/or the characters and actors mentioned. Only P2, P5, P6, and P12 did not. P3 and P9 also looked online for images of the non-binary actors. P4 even read the biography of Dua Saleh, a celebrity mentioned in the first text, in order to reformulate one text passage that contained the nouns *actor* and *musician*. P4 replaced the former by citing a series in which the actor played (“die auch in Generation mitgespielt hat”, i.e. who also acted in Generation).

Participants also highlighted gender references for which GFL was to be used. P3 and P5 did so in every assignment, i.e. they often translated gender references in the masculine or feminine form, highlighted them as shown in Figure 6.4, then used GFL in a subsequent step. P8 and P11 adopted the same strategy in the first assignment only, whereas P9 did so for a sentence in the third assignment which they found particularly difficult. P4 used a gender-inclusive character and a neopronoun in the first assignment, putting them into brackets in order to replace them with a gender-neutral strategy in a subsequent step.

A common strategy in the third assignment was to split the screen so as to have the translation open alongside the GFL handout or a similar resource. This allowed P3, P5, P7, P9, P11, and P12 to avoid constantly opening and closing windows with resources on how to use the selected neosystem as P2 and P10 did. Another strategy was to focus on GFL as a separate task from translation and PE. For instance, P9, P11, and P12 corrected gender references first and then post-edited the MT outputs.

Second, the analysis of specific screen activities, i.e. broken units, changes, searches, and pauses, is presented below. In Figure 6.5, the median values and sum for each activity are reported for each assignment.

The median number of the observed activities was similar in the first two assignments, i.e. 24.5 and 24 respectively, but considerably increased to a median value of 50 when using neosystems. Activities occurred with different frequencies. Broken units were found sporadically in the observation protocols. Changes

6.3 Screen activity tracking data

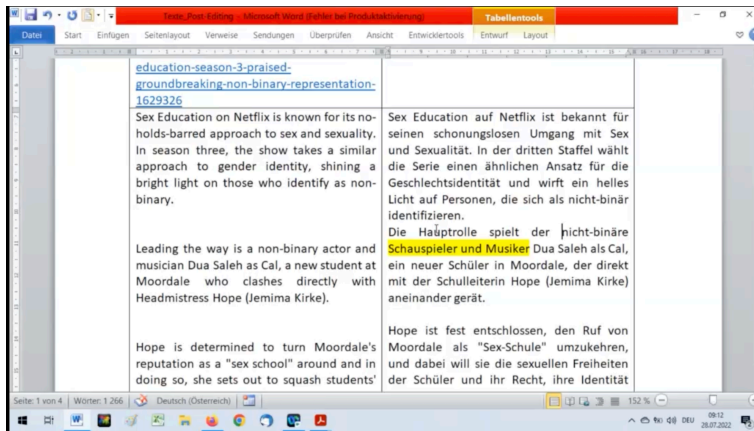


Figure 6.4: Use of the highlight function to later revise GFL

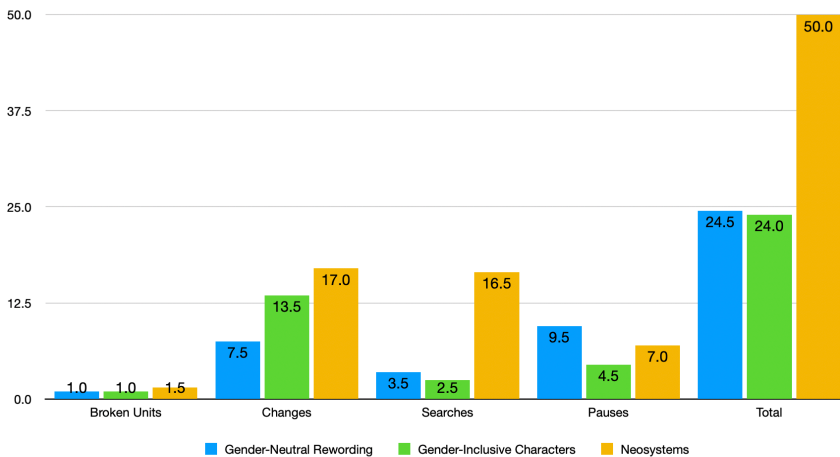


Figure 6.5: Overview of the keyboard activity for each assignment with median values

and searches were most frequent when participants used neosystems, with a median of 17 and 16.5 occurrences respectively, as against a respective median of 7.5 and 3.5 when gender-neutral rewording, and 13.5 and 2.5 when applying gender-inclusive characters. Pauses were the most frequent when participants reworded in gender-neutral ways, with a median 9.5 times, as against 4.5 in the second and 7 in the third assignment. To test whether the different strategies had an impact on the screen activity tracking, a rANOVA with Greenhouse-Geisser correction was performed. It showed a statistically significant difference in the observed ac-

tivities among assignments ($F[1.45,14.46]=10.54$, $p\text{-value}=0.003$). Post-hoc analysis was conducted with Bonferroni correction to determine the statistical significance of relationships among the specific strategies. There was a statistically significant difference between the first and the third ($p\text{-value}=0.017$) as well as the second and third assignment ($p\text{-value}=0.021$). This means that activity increased in the third assignment and no statistically significant difference was found between the first and the second assignment.

Differences can also be found in the comparison between translation and PE, as shown in Table 6.6. The median of screen activities was higher in the PE task in gender-inclusive characters (TR: 20.5 vs PE: 32) and neosystems (TR: 41 vs PE: 52). This difference is largely influenced by the number of searches in the third assignment, which were conducted more frequently in the PE (25.5) than in the translation task (14). In gender-neutral rewording, the analysed screen activities were higher in translation (27) than in PE (16.5). To test whether the task had an influence on activity frequency, a Welch's test was performed on ranked data. The test was selected because the data violated the assumption of equal variance and were not normally distributed. It showed no statistically significant difference between translation and PE in each assignment (first assignment: $t[7.89]=-1.53$, $p\text{-value}=0.166$, $d=-0.881$, second assignment: $t[9.80]=1.06$, $p\text{-value}=0.317$, $d=-0.609$, third assignment: $t[9.68]=-0.788$, $p\text{-value}=0.450$, $d=-0.455$).

Post-editors made more changes, whereas translators made more pauses. A higher number of changes in the PE task was expected since participants had to correct almost every machine-translated gender reference, while participants in the translation task could produce an appropriate solution without modifying it. Nevertheless, there were differences among assignments in the number of changes. While translating, the median for changes was 6.5, 8, and 12 respectively. Hence, translators made more changes when using neosystems. While in PE, the median for changes was 10, 18, and 18.5 respectively, hence there were no difference between gender-inclusive characters and neosystems. Pauses were more frequent in the translation task in the first (TR: 10 vs PE: 4.5) and third (TR: 7.5 vs PE: 2.5) assignments.

Since the number of changes cannot be directly compared between tasks (i.e. translation and PE), the analysis of the screen recordings was repeated and the count of changes was omitted as shown in Table 6.7. In this case too, the highest screen activity was found in the third assignment both in translation and in PE. PE in this assignment required a higher number of activities than translation. Nevertheless, the difference seems to be caused by two participants in particular, as shown in Table 6.8. P8 and P10 did 71 and 72 searches respectively. When using gender-neutral rewording, the frequency of the observed activities was higher in

Table 6.6: Overview of screen activity tracking with median values by assignment and task (normalised values based on the number of gendered phrases per assignment in brackets)

Assignment	Screen activities	TR	PE
1	Total	27 (3)	16.5 (1.85)
	Broken units	2 (0.2)	0.5 (0.05)
	Changes	6.5 (0.7)	10 (1.1)
	Searches	4.5 (0.5)	0.5 (0.05)
	Pauses	10 (1.1)	4.5 (0.5)
2	Total	20.5 (1.7)	32 (2.65)
	Broken units	0.5 (0.4)	1 (0.08)
	Changes	8 (0.65)	18.5 (1.55)
	Searches	4 (0.35)	2 (0.15)
	Pauses	5 (0.4)	4.5 (0.38)
3	Total	41 (4.1)	52 (5.2)
	Broken units	1.5 (0.15)	1.5 (0.15)
	Changes	12 (1.2)	18 (1.8)
	Searches	14 (1.4)	25.5 (2.55)
	Pauses	7.5 (0.75)	2.5 (0.25)

the translation task, whereas no differences were found in the second text with gender-inclusive characters.

Table 6.7: Total number of screen activities with their median in brackets after omitting the count of changes

Assignment	TR	PE	Total
1	103(18)	66(10)	169(13.5)
2	77(9.5)	74(9.5)	151(10)
3	147(27)	246(35)	393(30.5)

Table 6.8 shows the total number of observed activities per participant for each assignment, where B indicates broken units, C changes, S searches, P pauses, and T the total of their sum. For each participant, the assignment with the lowest and highest screen activity is highlighted in yellow and red respectively. The

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table reflects the intuitive notion that screen activity was the highest in the third assignment (using neosystems) for ten participants, i.e. P1, P2, P3, P5, P7, P8, P9, P10, P11, and P12. For instance, an increased number of changes and searches was observed. The lowest activity was observed in the first assignment for six participants, namely P1, P5, P7, P9, P10 and P12, and in the second for five, i.e. P2, P3, P6, P8, P11.

Table 6.8: Comparison of screen activity tracking per participant in each assignment

P	1st Ass.					2nd Ass.					3rd Ass.				
	B	C	S	P	T	B	C	S	P	T	B	C	S	P	T
P1	1	6	12	10	29	0	6	11	13	30	0	6	17	7	30
P2	3	6	5	10	24	0	11	7	3	21	6	16	16	14	52
P3	0	17	11	2	30	0	6	2	1	9	2	30	21	8	61
P4	6	14	1	12	33	1	10	6	3	20	1	8	1	5	15
P5	0	6	3	9	18	3	17	1	16	37	2	22	12	16	52
P6	4	7	4	10	25	2	6	1	7	16	1	5	11	7	24
P7	0	8	0	1	9	0	12	0	0	12	1	5	11	7	18
P8	4	19	7	23	53	1	21	21	8	51	4	27	72	9	112
P9	0	7	0	3	10	2	25	3	7	37	0	16	20	12	48
P10	0	7	4	6	17	1	15	16	5	37	3	20	71	1	95
P11	2	14	1	11	28	2	20	1	4	27	4	18	31	3	56
P12	1	12	0	3	16	1	17	0	2	20	0	19	7	2	28

P7 was the participant for whom the lowest number of activities was observed, with a total of 9, 12, and 18 activities respectively for the three assignments. They made no searches and almost no pauses in the first two assignments. They were also the participant who performed the lowest number of activities in the first assignment, followed by P9. The lowest screen activity in the second assignment was registered by P3, followed by P7. When using neosystems, P4 was the participant who produced the least screen activity tracking data, with e.g. only one search.

P8 was the participant whose screen activity was the highest in each assignment with a considerable number of changes, i.e. 19, 21, and 27 respectively. This indicates that they were neither satisfied with nor sure about the use of GFL. They also made several pauses (23) in the first assignment while thinking of possible gender-neutral solutions and did a very high number of searches (72) in the third.

The difference in the number of searches by P8 and P10 (72 and 71 respectively) versus the other participants is substantial, with P11 conducting 31 searches and the others even fewer.

In Table 6.9, translation and PE times are compared with the total number of analysed screen activities. The lowest values are highlighted in yellow and the highest in red. For half of the participants (P2, P5, P6, P8, P9, and P10), there was a correspondence between longer translation times and higher activity. For P3, conversely, longer translation times corresponded to lower activity. A higher number of screen activities was observed for P7 and P11 when using neosystems, while for the first assignment with gender-neutral rewording, they needed more time. A Kendall's Tau was performed to test for a relationship between translation/PE times and screen activity; this showed no correlation ($\tau = 0.074$, p -value = 0.530).

Higher screen activity does not correlate with the perceived difficulty of GFL either. P10 indicated that GFL use is very easy but their screen activity was one of the highest during the study. P12 indicated that GFL use is difficult, but they were the participant who produced the least screen activity tracking data. Use, or actually lack thereof, in everyday work-life seems to impact the number of screen activities. P8 was the sole participant who did not use GFL in their work and their screen activity was the highest in the study.

In order to shed light on which gender references require more effort in terms of screen activity in translation and PE, the three texts translated during the experiment were divided into segments based on punctuation rules. For each, the number of activities performed was annotated. Also, notes were taken to reconstruct the translation and PE process of GFL. An overview of the total activities performed in each text segment is presented below, along with examples of gender-specific terms that were particularly challenging in the source and target texts.

The first text contained the following seven segments (analysed gender references are shown in bold):

1. Sex Education on Netflix is known for its no-holds-barred approach to sex and sexuality.
2. In season three, the show takes a similar approach to **gender identity**, shining a bright light on **those who identify as non-binary**.
3. Leading the way is **a non-binary actor and musician** Dua Saleh as Cal, **a new student** at Moordale who clashes directly with Headmistress Hope (Jemima Kirke).

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Table 6.9: Triangulation of translation and PE times with screen activity tracking data

Participant	1st Assignment	2nd Assignment	3rd Assignment
P1	00:30:38 29	00:34:34 30	00:34:06 30
P2	00:30:47 24	00:30:20 21	00:40:04 52
P3	00:34:00 30	00:34:59 9	00:33:44 61
P4	00:21:16 33	00:22:12 20	00:18:02 15
P5	00:24:06 18	00:25:15 37	00:32:58 52
P6	00:25:45 25	00:19:59 16	00:20:32 24
P7	00:22:12 9	00:20:47 12	00:20:55 18
P8	00:26:12 53	00:21:53 51	00:34:43 112
P9	00:17:09 10	00:20:43 37	00:21:05 48
P10	00:17:44 17	00:19:41 37	00:35:52 95
P11	00:26:07 28	00:14:23 27	00:20:44 56
P12	00:10:33 16	00:09:26 20	00:11:26 28

- Hope is determined to turn Moordale's reputation as a "sex school" around and in doing so, she sets out to squash **students'** sexual freedoms and **their** right to express **their** identity.
- Cal stands up against Hope, refusing to wear the new Moordale school uniform that is deemed acceptable in Hope's eyes.
- They** refuse to wear the girls [sic] uniform and is [sic] reluctant to wear the tight-fighting [sic] boys [sic] uniform.
- Cal likes to keep **their** uniform loose and baggy, as it makes **them** feel more comfortable in **who they** are, but Hope just won't listen.

As shown in Table 6.10, no activity was performed in the first and fifth segments. The third showed the highest tracking activity, e.g. 50 changes and 47 pauses (values highlighted in red). In the last segment as well, participants faced great difficulties. The greatest problems thus relate to the translation of “actor and musician”, “student”, and the singular *they* pronoun in different cases.

Table 6.10: Total for screen events per segment in the first assignment (Translation|PE)

Segment	Broken units	Changes	Searches	Pauses	Total
1	0 0	0 0	0 0	0 0	0 0
2	1 0	4 4	8 2	0 0	13 6
3	3 2	21 29	18 4	25 22	67 57
4	2 1	5 8	2 1	6 3	15 13
5	0 0	2 0	0 0	0 0	2 0
6	1 2	6 12	5 2	6 10	18 24
7	7 2	18 14	3 3	16 12	44 31

For instance, before translating “student” P1 made a ten-second pause. They used the word “Neuzugang” (new entry) and made another pause, this time of nine seconds. They then looked on the internet for the frequency of use of “Neuzugang” and “Neuankömmling” (newcomer) and made another ten-second pause. Then, they deleted the former term and replaced it with the latter. They made a 15-second pause, deleted the term, made a 6-second pause and finally opted for “neuen Mitglied” (new member).

While translating the third segment, P2 looked four times in the provided handout for inspiration. Also, before finding a solution for “actor and musician”, they made two long pauses of 38 and ten seconds respectively. Then, they started writing “Schau” (act), made another pause of nine seconds, wrote “Schauspie” (acting, incomplete word), deleted it, and wrote “Schauspielperson und Musik” (acting person and music). They did not complete the translation but made a 21-second pause. Finally, they opted for “Musikmensch” (music person).

P8 had difficulties in translating singular *they* and made a long pause of 90 seconds in the seventh segment, before PE the MT for “it makes them feel more comfortable in who they are”. They also looked three times online for synonyms for “comfortable” in order to restructure the sentence and made six changes to the segment. Finally, during the revision phase, they made another seven pauses lasting from 15 to 40 seconds.

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The second text contained six segments (analysed gender references are shown in bold):

1. Grey's Anatomy has its **first non-binary doctor** after E. R. Fightmaster was promoted from a **recurring guest star** to a **recurring cast member**.
2. According to Variety, **the actor** has landed a recurring role as **Dr. Kai Bartley** after making **their** first appearance in Season 18.
3. **Viewers** were introduced to **them** in episode three when Meredith Grey (Ellen Pompeo) was contacted by a well-funded Minnesota hospital that wants her to join their mission to cure Parkinson's Disease.
4. While **audiences** have not seen much of **the brilliant neuroscientist**, played by **nonbinary actor** E.R. Fightmaster, **they** are already eager for more.
5. **They** are dedicated to **their** craft and extremely talented at what **they** do.
6. Confident as hell and able to make even the most detailed and mundane science seem exciting and cool, Kai and Amelia bond over **their** shared love of medicine and the brain,' the ABC character description for Kai reads.

During the translation process, one participant looked at the handout before starting their revision. Other participants generally used resources when they encountered a term for which the translation or the gender-fair alternative was difficult. As shown in Table 6.11, the first and the fourth segments caused difficulties for both translators and post-editors (values in red). The screen activity was high in the second and third segments as well. Terms that caused difficulties were "doctor", "recurring guest star", "recurring cast member", and singular *they*. In addition, the translation of the fourth segment required an abundance of gender-inclusive characters, thus prompting a high number of changes and searches in order to reword the text in an attempt to avoid overusing characters that would impact readability.

P5, for example, was unsure how to translate the term "doctor" with gender star because the masculine and feminine forms of the noun differ for the umlaut on the first letter ("Arzt", "Ärztin"; male doctor, female doctor). Hence, they first translated the English term using the masculine form, and then highlighted it. Subsequently, they tried to write an alternative, which they immediately deleted. Then they rewrote the sentence, this time using the feminine form. In brackets, they added an alternative translation using the synonym "Mediziner*in". During the revision phase, they also searched online how to use gender-inclusive characters in the word "Arzt". Similarly, P8 searched online five times how to gender the term.

Table 6.11: Total of screen activities per segment in the second assignment (Translation|PE)

Segment	Broken units	Changes	Searches	Pauses	Total
1	5 6	29 26	18 12	20 11	72 55
2	1 0	9 21	2 6	7 2	19 29
3	0 1	5 14	1 6	4 4	10 25
4	0 0	9 25	5 13	10 3	24 41
5	0 0	3 17	1 0	3 4	7 21
6	0 0	1 7	0 4	1 2	2 13
pre-revision	0 0	0 0	1 0	0 0	1 0

P10 faced difficulties while post-editing the MT for “recurring guest star”. While post-editing, they stopped writing “wiederkehrende...” (recurring) and looked online five times for a translation, using different electronic vocabularies. They also made three pauses while PE the segment and opted for rewording the term as “Star mit wiederkehrenden Gastauftritten” (star with recurring guest appearances). In this case, however, the use of GFL was not the sole cause of difficulties. Participants struggled to find a German equivalent for the source term.

Finally, participants made changes because they were unsure regarding the order of masculine and feminine forms when using pronouns and articles. For instance, P11 changed the MT of “actor” from “der Schauspieler” (masculine article and noun) to “der*die Schauspieler*in” (masculine*feminine article + masculine*feminine noun) and then changed it to “die*der Schauspieler*in” (feminine*masculine article + feminine*masculine noun). Similarly, P1 translated using masculine forms first, then read the provided handout and decided to change the order of gendered forms accordingly.

The third text contained seven segments (analysed gender references are shown in bold):

1. **Audiences** at last week’s Toronto International Film Festival didn’t just watch movies.
2. **They** were also lucky enough to get a sneak peak at what will surely become one of the most celebrated TV series of this fall: Sort Of.
3. This is a comedy about a **gender-fluid nanny** navigating **their** complicated existence by **co-creators** Bilal Baig and Fab Filippo.

4. The series also features Baig. Baig is a **South Asian, queer Muslim actor who** was previously known best for **their** work in theatre, and **whose** deadpan delivery in *Sort Of* is one of its many great assets.
5. **They** star as **nanny** Sabi Mehboob, **who** is given a chance to move to Berlin in pursuit of something more than **their** clearly unsatisfying life.
6. But after the mother of the children gets in a serious bike accident, Sabi decides to stay put in Toronto.
7. As Bilal mentions, *Sort Of* represented **their** first major experience with television.

In this case, some participants also read the handout or used other resources before and after their translation and/or PE in order to learn more about how the selected neosystems are used. Table 6.12 shows an overview of the screen activity tracking data produced in each segment. The greatest difficulties were faced in the third, fourth, and fifth segments for which the number of changes and searches is particularly high. Difficult gender constituents were “nanny”, “co-creators”, adjective declensions, and pronouns.

Table 6.12: Total screen activities per segment in the third assignment (Translation|PE)

Segment	Broken units	Changes	Searches	Pauses	Total
pre-translation	0 0	0 0	3 4	0 0	3 4
1	2 0	4 1	3 1	0 0	9 2
2	0 0	1 1	3 1	2 1	6 3
3	3 4	33 36	39 84	20 13	95 137
4	5 6	29 35	14 62	18 7	66 110
5	2 0	14 28	11 39	15 4	42 71
6	0 0	3 3	2 3	1 2	6 8
7	0 1	3 7	1 12	1 2	5 22
pre-revision	0 0	0 0	2 0	0 0	2 0

For instance, P6 looked for a possible translation of “nanny” in an online dictionary. Then they searched for “Kinderfrau” (synonym for “nanny”) in the Duden,¹ a German language dictionary. They finally translated the term as “Kinderpflegens” (childcare worker with the Ens-form) and then changed it to “Kinderbetreuens” (caregiver). P3 struggled with the use of possessives in the Sylvain

¹<https://www.duden.de>

system. While post-editing the MT for “whose deadpan delivery”, the participant first opted for omitting the pronoun, then they changed it to “seiner/ihrer” (his/her), “ihrin” (a tentative use of possessives with the Sylvain system), “ihrins”, “seiner/ihrer”, and finally “ihrin”. They also looked at the handout six times as well as at a paper explaining the system.

From the observation, it also emerges that most of the participants used the handout or other resources concerning neosystems for the whole duration of the study. During the whole PE of the third assignment, e.g. P8 and P10 conducted 72 and 71 searches respectively. A common strategy was to open the document with source and target text along with the handout or another resource describing the use of the selected neosystem. Almost every time a gender reference was encountered, the resource was consulted. This also resulted in a continually interrupted writing flow, for instance in segment four, where adjectives, nouns, and pronouns occurred with a certain frequency.

To summarise, differences in screen activity tracking data were found among strategies as well as participants. Activity frequency was similar in the first and second assignment, but significantly increased when neosystems were used. When using gender-neutral rewording, participants made more pauses, probably to reflect on gender-neutral alternatives to gender-specific terms. In the third assignment, the number of searches and – to a lesser extent – changes increased. There were also differences between translation and PE, but these are less clear-cut. In this case, the number of changes cannot be directly compared. Since machine-translated gender references were often wrong or contained masculine generics, changes in the PE tasks occurred more frequently. However, if this indicator is excluded from the analysis, post-editors performed more activities in the third assignment. Results are probably skewed since P8 and P10 did considerably more searches than the other participants. No difference between translation and PE was found when using gender-inclusive characters, and gender-neutral rewording was the only case where translation required higher activity than PE. This outcome may be coincidental, as inferential statistical analysis indicates that the observed difference lacks statistical significance. Finally, translators made more searches than post-editors in the first two assignments and more pauses in the first and third. While the perceived difficulty of GFL use does not correlate with a higher number of activities, the frequency of its use seems to play an important role. The only participant who does not use GFL in their work, i.e. P8, is the person who made more pauses, changes, and searches.

6.4 Participants' experience

In order to investigate the participants' personal experience with GFL during the study, retrospective interviews with the translators were conducted at the end of each study session. Participants were asked to comment on difficulties in the source text; on the translation process, with a particular focus on easy and challenging aspects of GFL use; and their impressions of the advantages and disadvantages of using MT for GFL (see the interview guide and the transcripts in the Appendix).

While gender-neutral rewording was the easiest assignment for five participants, a recurring comment was that this strategy also requires great effort, mostly because one has to extensively reword simple nouns which could be gendered with inclusive characters. For example, the word "high school student" is gender-specific in German and neutral alternatives are rather uncommon. Gender-inclusive characters were perceived as the easiest assignment by most of the participants ($n=7$) and relatively straightforward to use. Neosystems were never selected as the easiest strategy. Rather, they were the most difficult assignment ($n=6$) and considered particularly challenging. Nevertheless, the resulting cognitive overload did not always increase translation or PE times. No major differences emerged between the translation and PE tasks in the first and second assignments. When using neosystems, post-editors selected them more often as the most difficult strategy and also reported a more negative experience with them. Finally, half of the participants ($TR=4$, $PE=2$) commented that using only one strategy per text is limiting and they would rather use a mix of gender-neutral rewording and gender-inclusive characters ($TR=3$, $PE=5$).

Table 6.13 provides an overview of the most interesting statements that participants made concerning gender-neutral rewording, while the interview transcripts in German are linked in the Appendix. While only P1 found it the most difficult strategy, half of the participants ($TR=3$, $PE=3$) commented that it requires great effort because, for example, it is sometimes difficult to find alternatives to gendered terms ($TR=3$, $PE=3$), which "are otherwise totally natural for me, for example, 'students' " (P4). In general, participants commented that they had to restructure the texts and hence constantly reflect on possible alternatives. Nevertheless, this was the easiest assignment for five participants ($TR=3$, $PE=2$). There was some consensus regarding the fact that rewording is generally easy ($TR=1$, $PE=4$) because one can use one's own language repertoire without introducing new characters and/or forms. While some recognised that rewording requires creativity ($TR=3$, $PE=1$), this was considered both positive and negative. For example, P7 commented that they "could think in every possible direction [...]" I

could remove personal references, and use proper names [...] I simply had great margin to reformulate". Conversely, P2 felt that the margin for creativity was "really unnatural" as well as "time-consuming and difficult".

Table 6.13: Statements from the interviews regarding the first assignment

Statement	Participant	TR	PE	Total
It was the most difficult assignment	P1	1	0	1
It requires great effort	P1, P4, P6, P9, P10, P11	3	3	6
It was the easiest assignment	P3, P5, P6, P7, P12	3	2	5
It is generally easy to apply	P3, P8, P9, P10, P12	1	4	5
Creativity is allowed and/or required	P2, P5, P6, P7	3	1	4
It is difficult to find gender-neutral alternatives	P2, P4, P6, P9, P10, P11	3	3	6
I would prefer to use gender-inclusive characters	P4, P6, P9, P10	2	2	4
The translation reads differently than the source text	P1, P5, P6, P11	3	1	4
It is difficult in reference to a single person	P2, P5, P11	2	1	3
I am not satisfied with my solutions	P2, P6, P8, P11	2	2	4
Difficulties with "actor and musician"	P1, P2, P4, P6, P8, P10, P11	4	3	7
Difficulties with "student"	P4, P7, P8, P9, P10, P12	1	5	6
Difficulties with pronouns	P1, P2, P5, P10, P11	3	2	5
Difficulties with "students"	P6, P8, P9	1	2	3

Despite the generally positive attitudes towards this strategy, four participants (TR=2, PE=2) expressed a preference for gender-inclusive characters. In general, gender-neutral rewording is believed to shift the meaning of the text because the translation reads differently than the source text (TR=3, PE=1), for example, P11 stated "I do not know whether there is some meaning loss [...] It is not something

that Cal [the non-binary character in the text] would say, but rather something general". This strategy is also considered difficult when referring to a single person (TR=2, PE=1) because it is normally used to neutralise plural references. As a consequence, some participants (TR=2, PE=2) stated that they were not satisfied with the solutions they found for the translation of gender-specific terms. The most common challenges in the first assignment were the translation of "actor and musician" (TR=4, PE=3), "student" (TR=1, PE=5), pronouns (TR=3, PE=2), and "students" (TR=1, PE=3). For instance, P8 stated that they had "great difficulties with the terms 'actor' and 'musician'. I had to reflect for quite a long time on how I could reword them" whereas P5 felt that "(in the case of singular *they*) I repeated the proper noun of the character too often [...] The general challenge is to avoid many repetitions so that the text is fluent but also [...] That the text is not too abstract" (see, for example, P5's translation in the materials linked in the Appendix). These difficulties are also confirmed by the analysis of the screen activity tracking data, which was higher in the third segment containing the aforementioned nouns and in the last containing several pronouns.

As shown in Table 6.14, even though two participants (TR=1, PR=2) commented that gender-inclusive characters were the most difficult assignment for them, there was quite a consensus that it is the easiest assignment (TR=3, PE=4) and generally easy to use (TR=1, PE=5). For example, P12 commented that "[the non-binary person] was constantly mentioned in different inflections, dative, genitive (see the source texts and the MT outputs linked in the Appendix). And one could not actually avoid the pronouns" while P10 thought that "it is something we are already used to". Most of the participants also agreed that adjective declension is straightforward (TR=3, PE=5), even though they have to combine masculine and feminine suffixes with a character if and when they differ. Only two participants (P1, P8) felt that the adjective declension is at times difficult. Also in this case, some participants (TR=1, PE=2) commented that characters are more difficult to use in reference to a single person, because this strategy is quite common in written language to avoid masculine generics.

In general, most of the translators (n=4) and post-editors (n=5) agreed that the frequent use of characters negatively impacts readability which could possibly transform the text into a "storm of characters" (P5). The greatest challenge concerning gender-specific terms was "doctor" (TR=2, PE=5) because, as already explained, the masculine and feminine form of the noun differ with an umlaut on the first letter, i.e. "Arzt" (male doctor)/"Ärztin" (female doctor). This is uncommon; usually the word stem is not altered and the suffix *-in* is added to the masculine form. Hence, participants were not sure how to use gender-inclusive characters. Another challenge that was often mentioned was the translation of

Table 6.14: Statements from the interviews regarding the second assignment

Statement	Participant	TR	PE	Total
It was the most difficult assignment	P5, P12	1	1	2
It was the easiest assignment	P1, P2, P4, P8, P9, P10, P11	3	4	7
It is generally easy to apply	P1, P7, P8, P9, P10, P11	1	5	6
The adjective declension is straightforward	P4, P5, P6, P8, P9, P10, P11, P12	3	5	8
It is difficult to use in reference to a single person only	P5, P10, P12	1	2	3
It affects readability	P3, P4, P5, P6, P7, P8, P9, P11, P12	4	5	9
Effort to ensure consistent use	P1, P3, P4, P5, P6, P7, P11	5	2	7
Difficulties with “doctor”	P5, P6, P7, P8, P9, 11, P12	2	5	7
Uncertainties with “recurring guest star”	P4, P7, P8, P10, P11	1	4	5
Uncertainties how to combine forms	P2, P6, P8, P10, P11	2	3	5
Thought of masculine and feminine forms to use it	P1, P2, P3, P4, P5, P7, P8, P10, P12	5	4	9

terms like “recurring guest star” (TR=1, PE=5). The interviews made clear that the problem did not only concern finding an inclusive form. Sometimes it was difficult to find a viable translation in German or to understand whether gender-inclusive characters were to be used at all costs, as explained by P7: “my question was ‘do I have to reformulate ‘member’? [...] This is neutral in German as well”. Other issues include deciding how to combine masculine and feminine forms of articles and pronouns (e.g. “der*die” (masculine*feminine definite article) or “die*der” (feminine*masculine definite article)) (TR=2, PE=3) and ensure their consistent use as well as not to overlook gendered forms (TR=2, PE=5). For example, P1 looked at their translation and commented “So here, too, the feminine comes first. Here too. Well, with the interrogative pronouns it is inevitably the

same and with the nouns it is the other way round [...] It is not a very consistent strategy”.

These uncertainties and difficulties are also confirmed by the observation protocols (see materials linked in the Appendix). First, the highest number of screen activities occurred in the first segment containing the noun “doctor”, followed by the fourth segment with “recurring guest star”, and “recurring cast member”. Second, participants generally made more changes than in the first assignment and part of these included changing the order of masculine and feminine forms. Furthermore, this strategy does not seem to challenge the gender binary and may in fact reinforce it since the majority of the participants stated they needed to think of feminine and masculine forms of gender-specific terms in order to be able to combine them with characters (TR=5, PE=4). Finally, P5 made a very interesting remark: they translated the text first and subsequently adjusted gender references.

Table 6.15 provides an overview of statements made concerning the use of neosystems. This strategy was generally considered the most difficult (TR=2, PE=4), and participants described it as challenging (TR=1, PE=3), e.g. “it took up a lot of my brain capacity so that I often had the feeling there might be less capacity left for finding other nice formulations” (P6). Nevertheless, for P2 and P11 the experience in the third assignment was better than they expected, e.g. “yesterday I read the handout and thought that it sounded artificial [...] Today I found it actually pleasant” (P2), and another three participants (P1, P10, P12) found it positive.

There was a general consensus that further training would be required to effectively apply the strategy (TR=4, PE=5) and that neosystems are unknown, thus being perceived as a foreign and/or artificial language (TR=2, PE=6) and affecting the target text readability (PE=5). P3, for instance, commented that “everything is completely new and it does not sound how one would speak or write”. Participants were thus unsure about how the neosystems are used even though they had access to a handout describing them (TR=2, PE=2). They were also unsure about whether they applied the selected system correctly (TR=2, PE=4). Specific challenges concerned the translation of “nanny” (TR=5, PE=5), because the profession is stereotypically feminine and the German equivalents are of feminine gender or have a different connotation. The term “co-creators” caused some difficulties as well. In this case, as in the previous assignment with “recurring cast member”, this was sometimes due to terminology rather than GFL, e.g. “it gave me a big headache. I then thought about what I would say in the non-gendered version” (P6). Finally, participants struggled with declensions, especially adjectives (TR=6, PE=2) and stated that they used the handout provided or other re-

Table 6.15: Statements from the interviews regarding the third assignment

Statement	Participant	TR	PE	Total
The experience was better than expected	P2, P11	1	1	2
It was a positive experience	P1, P10, P12	1	2	3
It was the most difficult assignment	P1, P3, P7, P9, P10, P11	2	4	6
It is challenging	P6, P8, P9	1	2	3
Training is required	P1, P3, P5, P6, P7, P8, P10, P11, P12	4	5	9
It feels like a foreign/artificial language	P3, P6, P8, P10, P11, P12	2	4	6
It affects readability	P8, P9, P10, P11, P12	0	5	5
Uncertainties about use	P3, P5, P9, P12	2	2	4
Uncertainties about correctness	P5, P6, P8, P9, P10, P11, P12	2	4	6
Difficulties with “nanny”	P2, P3, P4, P5, P6, P7, P8, P9, P11, P12	5	5	10
Difficulties with “co-creators”	P2, P5, P6, P7, P12	3	2	5
Difficulties with declensions	P1, P2, P3, P4, P5, P6, P8, P9	6	2	8
Need for support	P5, P8, P9, P10	1	3	4
Difficulties in selecting word class	P5, P10, P11	1	2	3
Resources used for the whole experiment	P1, P2, P7, P8, P9, P10, P11, P12	2	6	8
First content, then neosystem	P8, P11	0	2	2

sources during the whole task (TR=2, PE=6). They also commented that they would prefer to work with and/or ask another person to proofread their target text in order to ensure the correct use of the systems. Interestingly, P8 and P11 commented that their strategy was to first concentrate on the content and then post-edit all gender references with the selected neosystem.

While no clear-cut differences emerged between the translation and the PE group when commenting on the first and the second assignment, post-editors seemed to have more negative attitudes toward neosystems. For example, P7, P8, and P9 commented that they would refuse assignments with neosystems because they would require longer PE times and it would not be lucrative enough, e.g. “I would have to put so much preparatory work into it that it probably would not be financially profitable” (P8). Actually, throughout the interview with P8 (see transcript linked in the Appendix), a quite negative attitude towards more disruptive GFL strategies emerged. P9 also mentioned that they would otherwise increase their rates. Nevertheless, P7 needed less time to complete the third assignment than the first two, and P9 needed around 21 minutes for both the second and the third. While cognitively demanding, neosystems did not necessarily require more time and might therefore not be less lucrative. There were greater differences between the translation and the PE group when discussing the general use of MT as well as its (dis-)advantages, as shown in Table 6.16.

Half of the participants (TR=3, PE=6) stated that they do not generally use MT. However, some (TR=1, PE=5), especially those in the PE group, use it as a source of inspiration, P11 said: “So I like to work with machine translation, I use it mostly as a tool, I just have DeepL or context Reverso open most of the time”. In addition, most of the participants (TR=4, PE=2) find that MT is not a valuable tool for texts similar to those they translated or post-edited during the experiment, mostly because the MT outputs would require extensive PE of GFL since “it is simply also a text or an area where you have to think a lot, and where so many people have to have a lot of background knowledge, a lot of thinking, simply reading between the lines” (P1). In comparison, two participants commented that MT can nevertheless be helpful. The main advantages of MT concern the fact that PE is faster and/or cheaper than translation (TR=1, PE=5), especially because one has already a relatively good translation draft that can be adapted (TR=2, PE=3). Two participants even stated that the MT draft allows them to concentrate more on GFL since the output quality is rather good (PE=2). From the analysis of the (dis)advantages, it emerges that post-editors have a more positive attitude towards this technology than translators, who primarily remarked that MT outputs are biased and lack GFL (TR=5, PE=1) and extensive PE is required (TR=5, PE=5). Another disadvantage of this technology is that PE constrains creativity because the idea is to use as much as possible of the MT output and hence make fewer changes (TR=2, PE=3). Finally, one participant also commented that gendered elements can be overlooked. For instance, P10 stated that “when I translate, I am in the text and I know which person this is about and whether they’re binary or non-binary or whatever. And with machine

Table 6.16: Statements from the interviews regarding MT

Statement	Participant	TR	PE	Total
I don't generally use MT	P1, P3, P6, P7, P10, P12	3	3	6
I use MT sometimes as a source of inspiration	P4, P7, P8, P9, P11	1	4	5
I would not use MT for similar texts	P1, P2, P3, P5, P9, P12	4	2	6
I would use MT for similar texts	P4, P11	1	1	2
MT is faster and/or cheaper	P6, P7, P8, P10, P11, P12	1	5	6
MT provides a translation draft	P4, P5, P7, P8, P10	2	3	5
It allows to concentrate on GFL	P10, P11	0	2	2
Extensive PE is required	P1, P2, P3, P5, P6, P7, P8, P9, P10, P12	5	5	10
MT lacks GFL	P1, P3, P4, P5, P6, P9	5	1	6
One is constrained by the MT draft	P4, P5, P7, P8, P10	2	3	5
It is easy to overlook gendered elements	P10	0	1	1

translation, you really have to carefully check what elements refer to this person who is non-binary”.

Participants from the PE group were also asked about their specific impressions of the task. The majority (n=4) said it was difficult to decide to what extent the MT output should be adapted since the guidelines provided for full PE specified that one should use as much MT output as possible. Participants generally thought that, apart from lacking GFL, the biggest challenge was the stylistic component of the texts (n=3) – for which the MT outputs were not appropriate, even though they were good and comprehensible (n=5). Two participants also said that the high-quality MT allowed them to concentrate on GFL, even though they felt that they then overlooked other important aspects of the PE process, e.g. “what is in the source text? Does it correspond to the target text?” (P11). Other comments included that GFL obviously requires extensive PE (n=1), thus there is a need to constantly revise the MT outputs to ensure that gender references are not over-

looked ($n=1$). In general, participants ($n=5$) felt they achieved higher productivity through PE and they would have been slower if translating the study texts. Only one participant felt there would be no difference in productivity between the two tasks.

To summarise, strategies have different advantages and disadvantages. Gender-neutral rewording is considered a relatively easy approach to GFL but it requires creativity. While some participants felt free to experiment with the language, others felt it was unnatural, time-consuming, and challenging (see, e.g., P1's and P9's interview transcripts). Gender-inclusive characters were easier to use, mostly because people are already used to seeing them in newspapers or administrative texts. However, there is no standard concerning how to concatenate masculine and feminine forms with characters and some participants felt that this is arbitrary. In addition, the sole use of characters can negatively impact readability. Hence, participants expressed a preference for combining this strategy with gender-neutral rewording. Neosystems were the most challenging strategy mostly because they are unknown to the general public and require new grammatical rules, thus feeling like a foreign, unnatural language. Attitudes towards this strategy were more negative in the PE group: participants felt that this strategy negatively impacts their productivity, and so would at least require them to increase their rates. Participants still had a sceptical attitude towards MT; many of them do not use it, and find it unsuitable for creative and heavily gendered texts. However, post-editors found it increases their translation speeds and can be a valuable tool despite its biased outputs.

6.5 Strategy selection and application

This section presents an overview of the strategies used by participants in each text, followed by an error analysis. In the first assignment with gender-neutral rewording, participants had to find creative ways to avoid gendered elements. This most often involved the use of gender-neutral words or rewording whole passages. In the second assignment with gender-inclusive characters, the majority of the translators and post-editors opted for gender star. However, this was put into practice quite differently. Furthermore, translators opted more often for gender-neutral words and rewording than post-editors, even if this was against the assignment instructions. Finally, the most selected neosystems were the Sylvain system and the Ens-forms. Interestingly, the former was a more popular choice in the PE group, whereas the translation group preferred the latter. No other relevant differences were found between the two groups.

The error analysis also reveals that, apart from sporadic misgendering and use of masculine generics, participants rarely made mistakes in the first two assignments. Post-editors made fewer mistakes, but the difference from translators was minimal. While using neosystems, there was a significant increase in errors due to difficulties in applying the systems. Misgendering was more common due to difficulties in finding an adequate German equivalent for the term “nanny”.

In the first text, nine gender phrases were selected for analysis and included references to non-binary individuals and mixed-gender groups. A total of 54 phrases were analysed and each was annotated for specific strategies used. The annotations totalled 55 and 59 for the translations and post-edited texts respectively, as shown in Table 6.17. This difference was due to more strategies being used for the same phrase. Gender-neutral words were the most common strategy in both groups (TR=22% and PE=24%), followed by rewording (TR=20% and PE=22%), and pronoun omission (TR=16% and PE=12%). Personal and possessive pronouns could sometimes be omitted and, when this was not possible, participants usually repeated the actor’s or character’s name (TR=7% and PE=8%). Another relatively common strategy (although only in the translations) was the use of participial forms. However, articles for these must be gendered in the singular form, e.g. in “Schauspielende und Musizierende” (actor and musician). For this reason, this strategy was and should be generally applied to plural nouns. While in the target text analysis other strategies were found less frequently, i.e. collective nouns, segment omissions and, against the instruction, gender-inclusive characters, sometimes no strategy was needed, i.e. in the case of pronouns referring to mixed-gender groups since the third-person plural pronoun in German is already gender-neutral. Table 6.18 shows examples of the great variety of strategies applied by participants in the first assignment.

In the second text, 12 gender phrases were selected for analysis, and 81 and 77 annotations were made to the translations and post-edited texts respectively. In this case as well, more strategies were used for a specific phrase, showing that participants sometimes mixed gender-inclusive characters with rewording. Reasons for this include that some neutral words, e.g. “Publikum” for “audiences”, are relatively common and hence a natural translation choice while rewording should ensure better readability because it helps reduce the number of characters used to concatenate masculine and feminine forms. As in the previous assignment, sometimes no specific GFL strategy was needed, e.g. when “they” was used as a third-person singular pronoun in the source text. A general overview of the applied strategies is presented in Table 6.19.

Interestingly, participants in the PE group used gender-inclusive characters more often (59%) than those in the translation group (48%) who used gender-

Table 6.17: Strategies used in the first assignment

Strategy	TR		PE	
	N	%	N	%
Gender-Neutral Words	12	22	14	24
Rewording	11	20	13	22
Pronoun Omission	9	16	7	12
Participle	7	13	1	2
Name Repetition	4	7	5	8
Collective Noun	1	2	2	3
Relative Pronoun	1	2	1	2
Segment Omission	0	0	2	3
Gender-Inclusive Character	1	2	2	3
No Strategy	9	16	12	20
Total	55	100	59	100

neutral words and/or rewording more frequently (TR: 35%, PE: 23%). More interesting, however, is to analyse which gender-inclusive characters were selected by the participants and how they were applied.

Gender star (*) was the most selected character, namely by eight participants (TR=3, PE=5), followed by underscore (_) (TR=2), and colon (:) (TR=1, PE=1). Variations were found in their use. Here the focus is on gender star because it was applied in such a diverse way. While nouns are created with the word stem or the masculine form adding the gender-inclusive character and then the feminine suffix “-in”, there is more freedom when using articles and pronouns. The following variations were found:

- masculine forms were always used first, e.g. “der*die Schauspieler*in” (masculine*feminine article + masculine*feminine noun) (n=2).
- feminine forms were used first in articles and pronouns, e.g. “die*der Schauspieler*in” (feminine*masculine article + masculine*feminine noun) (n=4).
- Pronouns and articles were combined in a new form, e.g. “die*r Schauspieler*in” (n=1). This is, however, not possible for possessive pronouns.
- Slash (/) was used to combine masculine and feminine forms of pronouns and articles, e.g. “der/die Schauspieler*in” (n=1). This variation is rather uncommon.

Table 6.18: Examples of strategies from the first assignment

Strategy	Source Text	TR	PE
Gender-neutral words	actor and musician	Schauspielperson und Musikmensch	Schauspiel- und Musikprofi
Rewording	it (the uniform) makes them feel more comfortable in who they are	weil sich diese angenehmer anfühlt	da dies dem Identitätsgefühl der nicht-binären Person am meisten entspricht
Pronoun omission	their right	das Recht	das Recht
Name repetition	They refuse to wear	Cal weigert sich	Cal weigert sich
No strategy	shining a bright light on those who identify as non-binary	setzt den Fokus auf jene, die sich als nicht-binär identifizieren	stellt diejenigen, die sich als nicht-binär identifizieren, in den Mittelpunkt

Table 6.19: Strategies used in the second assignment

Strategy	TR		PE	
	N	%	N	%
Gender-Inclusive Characters	39	48	44	59
Gender-Neutral Rewording	28	35	17	23
No Strategy	14	17	13	18
Total	81	100	77	100

In the third text, ten gender phrases were selected for analysis. A total of 63 and 65 annotations were made to the translations and post-edited texts respectively. In this case as well, some phrases were annotated twice because participants used more than one strategy. An overview is provided in Table 6.20. In comparison to the second assignment, no differences emerge between translation and PE: all participants mostly used neosystems and less frequently resorted to gender-neutral rewording.

Table 6.20: Strategies used in the third assignment

Strategy	TR		PE	
	N	%	N	%
Neosystems	45	71	46	70
Gender-Neutral Rewording	14	22	13	20
No Strategy	4	6	6	9
Total	63	100	65	100

However, differences were found in the system selection:

- The Ens-forms, e.g. “Ens spielt dens Nanny Sabi Mehboob” (they star as nanny Sabi Mehboob), were selected by 5 participants (TR=4, PE=1).
- The Sylvain system, e.g. “Dabei handelt sich um eine Komödie über einir genderfluidin Nanny” (this is a comedy about a gender-fluid nanny) was selected by another 5 participants (TR=1, PE=4).
- The NoNa-System, e.g. “Baig ist eint südasiatische, queere, muslimische Schauspieler*in” (Baig is a South Asian, queer, Muslim actor) was selected by one participant (PE=1).
- One participant selected a system which they use in their everyday life and consists of neopronoun *dey/deren*, the gender-fair suffix *-i*, and the use of neuter for articles and adjective declension, e.g. “Eine Komödie über ein genderfluides Kindermenschi” (a comedy about a gender-fluid nanny).

While three participants stated that they selected the system arbitrarily, four used the Ens-forms because they considered it the easiest strategy and three chose the Sylvain system as it is the most developed, with the introduction of a fourth gender in the German grammar. Interestingly, translators opted for easier neosystems, while post-editors for more sophisticated ones. As explained in Chapter 7, PE perhaps requires less cognitive load, allowing for greater focus on neosystems.

Errors in each target text segment were also annotated. Errors included misgendering, misrecognition of singular *they*, i.e. translation in the third-person plural German pronoun, masculine generics, use of false terms, and mistakes in the application of the strategy, e.g. use of wrong pronouns or articles. Masculine generics are not an error *per se*, but they represent a sexist use of language. An overview of the results is presented in Table 6.21, where the number of errors for each analysed segment is reported. As already explained, the number of analysed segments refers to the gender phrases that required GFL in translation and in PE. For each assignment (1, 2, 3), results for translation and PE are both reported in terms of absolute (N) and relative (%) frequencies. In Table 6.22, the number of participants who made the errors is reported. This aims to provide an overview of how common certain error types were among the participants.

Table 6.21: Overview of the error analysis

Assignment	1				2				3			
	TR		PE		TR		PE		TR		PE	
Error Type	N/54	%	N/54	%	N/72	%	N/72	%	N/60	%	N/60	%
Misgendering	2	4	1	2	1	1	0	0	7	12	7	12
Misrecognition Sing. <i>They</i>	1	2	0	0	4	6	0	0	0	0	0	0
Masculine Generics	0	0	1	2	0	0	2	3	0	0	0	0
False Term	6	11	0	0	0	0	0	0	0	0	0	0
Mistake	0	0	0	0	5	7	1	1	17	28	17	28
Total	9	17	2	4	10	14	3	4	24	40	24	40

In general, the number of errors was quite low in the first (TR=17%, PE=4%) and in the second assignment (TR=14%, PE=3%); it increased in the third (TR=40%, PE=40%). The most common error in the first segment was the use of “Studierende” (university students) as the German equivalent for “students” which actually referred to high-school in the text. This mistake was not found in the post-edited texts, most probably because the MT selected the appropriate translation. Misgendering was found twice in the translations and once in the post-edited text. P3 used the participial form “Studierende” in the singular to refer to the non-binary character mentioned in the text. Participial forms are neutral only in the plural, otherwise a masculine or feminine article must be used. A similar mistake was made by P10 when post-editing “actor and musician”. P5 misgendered the actor mentioned in the text using masculine forms. During the interview, they commented that they thought the adjective “non-binary” made the actor’s gender clear and did not know how to translate “actor and musician” by rewording. Finally, P3 also misrecognised singular *they* in one passage: the

phrase “They refuse to wear” became “Die Studierenden weigern sich [...] zu tragen” (The students refuse to wear).

The most common mistake in the second text was the misrecognition of singular *they*. This, however, was due to the ambiguity of the text. In the final passage, ABC’s character description of the non-binary doctor portrayed in the TV series *Grey’s Anatomy* is reported but another character is also mentioned. Therefore, it was difficult to understand whether “they” referred to the non-binary doctor or both characters. This aspect was also mentioned by participants in the interviews: while the use of the said pronoun is generally clear (mentioned by five participants), certain text passages were ambiguous (mentioned by eight participants). Another common error type concerned the use of gender-inclusive characters, mostly in the case of “doctor” for which three participants (TR=2, PE=1) concatenated the masculine and the feminine form of the noun (e.g. “Arzt:Ärztin”) because these differ not only for the suffix but also for the umlaut on the first letter. Another participant in the translation group made a mistake in the adjective declension. Finally, misgendering was found in one text only: P1 used a masculine article in front of the title “Dr.” referring to the non-binary character mentioned in the text.

Table 6.22: Number of participants who made errors

Assignment	1		2		3	
Error Type	TR	PE	TR	PE	TR	PE
Misgendering	2	1	1	0	3	3
Misrecognition Sing. <i>They</i>	1	0	2	0	0	0
Masculine Generics	0	1	0	1	0	0
False Term	3	0	0	0	0	0
Mistake	0	0	3	1	5	6

In the third text, misgendering was quite common because participants faced difficulties in the translation of “nanny”. While some opted for gender-fair versions of the term “Kinderbetreuer” (caregiver), others did not because they found its connotations are different. “Nanny” has a more colloquial and familiar tone so that one can maybe picture a young person trying to earn some money while also studying. “Caregiver” would be more formal and might convey the idea of a professional specialised in childcare. As a consequence, half of the participants (TR=3, PE=3) kept the term in the target texts and applied GFL to articles and/or pronouns, as in “Nin spielt din Nanny” (they star as nanny). The number of errors

in the use of the neosystems was very high – 28% of the analysed segments both in translation and PE contained mistakes. This also means that every participant in the study made at least one mistake except for P4. In fact, they inadvertently used false pronouns throughout the text (e.g. “sein”, his/its) but these errors were counted as misgendering only. During the interview, the participant realised the false use of pronouns and thus commented on their translation adding the correct alternatives.

The increased number of mistakes in the third assignment confirms the intensified cognitive effort commented on by participants during the post-study interviews as well as the higher intensity of screen activity observed during the translation and PE process. Participants were not sure about the (correct) use of neosystems and constantly needed resources in order to apply them. Hence, they modified their target texts and looked for resources more often than in the first two assignments. Minimal differences were found between translation and PE in terms of the total number of errors (see Table 6.21). Since a large number of mistakes made by one participant could affect the values for absolute and relative occurrences among the analysed gender phrase, Table 6.22 reports only the number of participants who made mistakes. In the first two assignments, no post-editor misrecognised singular *they*, while one translator in the first and two translators in the second assignment did so. Besides, participants in the PE group never used the false term “Studierende” (university student), whereas half the translation group did.

To conclude, participants showed great levels of creativity in the first assignment since they had to avoid gender references at all costs. They usually opted for gender-neutral words and rewording whole passages, often also changing perspectives or abstracting information, e.g. a translation for “it (the uniform) makes them (the non-binary person) feel more comfortable” was “weil sich diese angenehmer fühlt” (because this feels better). In the second assignment, participants largely opted for gender star but its implementation varied. As a matter of fact, there is no established norm for the use of gender-inclusive characters and some participants (n=4) expressed the wish to have GFL dictionaries and/or more literature on the topic to make better-informed decisions. In the third assignment, participants from the translation group opted for easier systems while those from the PE group for more sophisticated strategies. While in the first two texts the number of mistakes was limited with post-editors performing slightly better than translators, it increased in the third text. The highest number of mistakes concerned the application of specific GFL strategies, whereas misgendering was found quite rarely except for the third text in which the translation of the term “nanny” proved particularly challenging.

6.6 Gender bias in the MT outputs

To contextualise the analysis of gender bias in DeepL’s MT output, this section starts with a brief overview of the system’s underlying architecture and functionality. DeepL Translator is an NMT system introduced in 2017 by the German company DeepL GmbH. It is based on proprietary developments derived from the Transformer architecture introduced by Vaswani et al. (2017). Although DeepL’s full model architecture has not been publicly released, available information and independent analyses suggest that it adheres to the encoder-decoder structure characteristic of Transformer-based models. These models rely on self-attention mechanisms, which enable dynamic contextual weighting of input tokens and are crucial for modelling long-distance dependencies in language (Bahdanau et al. 2015, Vaswani et al. 2017, Popel et al. 2020). This architecture is particularly advantageous for languages with rich morphology, flexible word order, or complex agreement systems – factors closely linked to the manifestation of grammatical gender.

Like most NMT systems, DeepL’s default mode of operation is sentence-based. Translation units are segmented primarily at punctuation or line breaks, with each sentence processed independently (Läubli et al. 2018, DeepL GmbH 2024b). This sentence-level granularity offers computational efficiency but imposes significant limitations when modeling features that depend on discourse-level information, such as coreference resolution and gender agreement across sentences (Hardmeier 2014, Voita et al. 2018). In the context of gender translation, this segmentation often leads to inconsistent gender assignments across contiguous discourse, especially when the referent’s gender is not lexically marked in the source language (e.g., English *doctor* vs. French *docteur/docteure*).

DeepL attempts to mitigate some of these limitations by offering contextual enhancement parameters through its API (DeepL GmbH 2024b). Specifically, the `split_sentences` parameter allows users to disable automatic sentence segmentation, thereby enabling longer text segments to be translated as a whole. However, this setting risks incorrect chunking if punctuation is non-standard or ambiguous, and excessively long input strings can trigger truncation or decreased translation quality.

Additionally, DeepL provides a context parameter that allows for the inclusion of auxiliary text intended to inform the translation of the primary input (DeepL GmbH 2024a). According to DeepL’s API documentation, this feature is particularly beneficial for “short, low-context source texts” and it enables the system to use adjacent information without directly translating it. Nonetheless, this parameter operates only within the scope of a single API call and does not

permit discourse continuity across multiple text parameters. As such, while the context parameter can improve local coherence, it does not solve the broader issue of document-level gender consistency unless the entire document is submitted as a single text unit.

In sum, while DeepL’s Transformer-based architecture offers notable advantages in handling syntactic complexity and producing fluent output, its sentence-level processing paradigm imposes inherent limitations for discourse-level phenomena such as gender coherence. These constraints are particularly pertinent for research into gender bias in NMT, as gender assignments often require information from broader discourse context, which may be inaccessible to sentence-level translation models.

In Section 6.3, it was highlighted that the main difference between the translation and PE process lies in the number of changes: post-editors were presented with MT outputs in which the gender references were generally wrong. MT phrases containing gender references to be post-edited were annotated for three typologies of gender bias, namely (i) misgendering, (ii) mistranslations of singular *they*, (iii) masculine generics. Table 6.23 provides an overview of the results of the gender bias analysis for each machine-translated text. Note that one phrase could contain different typologies of bias, thus the number of gender-biased phrases and the total number of annotations in Table 6.23 do not always match.

Table 6.23: Bias in the MT outputs

Text No.	MT errors per phrase	Types of MT errors
1	6/9	Misgendering (2), mistranslations of singular <i>they</i> (3), masculine generics (1)
2	10/12	Misgendering (5), mistranslations of singular <i>they</i> (4), masculine generics (2)
3	9/10	Misgendering (6), mistranslations of singular <i>they</i> (3), masculine generics (1)

Six out of the nine analysed gender references in the first text presented gender bias in the MT outputs. The other three phrases contained a plural indefinite pronoun, i.e. “those”, and plural *they* in the possessive case, i.e. “their right” and “their identity”. When plural nouns were translated, masculine generics were used, e.g. “students” as “Schüler”. Nouns in reference to the non-binary actor and character mentioned in the text were translated with masculine forms whereas

singular *they* was not recognised as such and translated with the third-person plural pronoun “sie”.

Similarly, ten out of the 12 analysed gender references in the second text were biased as well. The character depicted in the text was always misgendered except for two phrases. For the first, i.e. “Fightmaster was promoted from recurring guest star to a recurring cast member”, the German equivalent is gender-neutral as well, i.e. “Fightmaster wurde von einem wiederkehrenden Gaststar zu einem wiederkehrenden Cast-Mitglied befördert”. The second contained the English title “Dr.” which is also sometimes used in German as gender-neutral. Otherwise, the actor and the character mentioned in the text were systematically misgendered and singular *they* translated in the plural form. Interestingly, this time the term “neuroscientist” was translated in the feminine form, i.e. “Neurowissenschaftlerin”.

The same error typologies were found in the third text as well and only one phrase was translated correctly, i.e. “audiences” into the gender-neutral “Publikum”. The non-binary person mentioned in the text was generally misgendered with masculine, feminine and neuter forms. “Nanny” was translated as “Kindermädchen” which is grammatically neuter in German but socially female. During the annotation, two MT errors were found that deserve further attention. First, there was a coreference mistake. In the English language text, “audiences” are mentioned in the first sentence, and “they” is used in the second as an anaphoric pronoun. In the German MT, the plural noun was translated into the singular, i.e. “das Publikum” (the public), and the third-person singular pronoun “sie” was used as an anaphor instead of the grammatically correct “es” (it). This might happen because of DeepL operating at the sentence level. Second, the series is about a nanny who is genderfluid. This identity term was translated into “geschlechtsspezifisch” (gender-specific) which is wrong.

In line with Prates et al. (2020), Piazzolla et al. (2023), Lardelli et al. (2024), the analysis of the MT outputs seems to indicate that automated translation systems are still largely biased. A tendency to overuse masculine forms is noted while singular *they* is translated in the plural form. Nevertheless, as explained in Section 6.4, post-editors found that the MT output increased their productivity, which is also confirmed by comparing translation and PE times. Sometimes, MT even allowed more concentration on the gender aspect of the task.

7 Discussion

The results presented in Chapter 6 are discussed in the sections that follow. First, the focus is on the three GFL strategies tested in the present study. These are compared in terms of cognitive processes, i.e. translation and PE times, screen activity tracking, and personal impressions of the participants. Product data, i.e. strategy applications and the success of their integration in the translation and PE process, are also discussed. The strengths and limitations of each strategy are described. Second, translation and PE are compared. These sections build the basis for discussing the implications of the study for both language professionals and MT research. Based on the results, a tentative decision tree for integrating GFL into the translation and PE process is proposed. Finally, a section is dedicated to the research process, describing the difficulties faced during data collection and analysis as well as the limitations of the present work.

7.1 GFL approaches compared

Translation and PE times increased when using neosystems. While time is surely needed to understand how the selected system works and also to memorise new pronouns, suffixes, and declension systems, no statistically significant difference was found in translation and PE times when comparing neosystems with the other approaches. However, this assignment took the longest for half of the participants, and the median translation and PE times were higher than in the other assignments. It is unclear whether a larger sample could yield different results and it must be highlighted that great differences in translation and PE speed were found among participants.

Gender-neutral rewording and gender-inclusive characters are rather common in written language and almost every participant reported regularly using one or both. As language experts, participants can adapt language and texts according to different registers, contexts, client specifications, and so on. This might be why the increased effort in the use of neosystems, e.g. the high number of searches and changes as well as the perceived cognitive load, does not correlate with significantly longer translation and PE times.

The analysis of the screen activity tracking elicited similar results to the analysis of translation and PE times. No difference was found between the first two assignments and the screen activity increased in the third. The difference was statistically significant in this case. This may be because neosystems are not common, and translators needed to consult various sources to be able to use them. The number of changes was rather high in each assignment with almost no differences among gender-inclusive characters and neosystems. Currently, there is no standard for the use of gender-inclusive characters. As a matter of fact, participants used different characters and/or combined them differently. When in doubt, they also consulted resources, which sometimes led to changes, e.g. from “der*die” (masculine*feminine article) to “die*der” (feminine*masculine article). When applying neosystems, participants were for instance not sure how to use pronouns. These challenges led to a high number of text modifications both in the second and third assignments.

Screen recordings also provided interesting insights into the participants’ general approaches to translation and PE. The majority searched online for information concerning the TV series and the non-binary actors/characters mentioned in the study texts. Some even looked for photos. While more information may be useful to produce a better translation, searching for a person’s photo online is not likely to assist in the process. As explained in Section 2.3, there is a distinction between gender identity and gender expression, hence using the physical appearance in photos or in general to infer a person’s pronouns may lead to misgendering and identity invalidation. Participants also highlighted gender references to be adjusted. They often conceived the translation and PE process as a separate step from the use of GFL. Some participants in the PE task adjusted gender references first and then worked on the whole text. This seems to suggest that the use of GFL has a partially negative impact on cognitive processes at least at this stage of familiarity with GFL.

Increased translation times do not correlate with higher screen activity tracking. This may also be because the participants’ speed and productivity greatly differed as demonstrated by the high standard deviation in translation and PE times. It seems that some participants are faster than others, even though they make several changes or consult numerous resources. This may also be due to different levels of work experience (spanning from 3+ to 20+ years) and/or greater knowledge of GFL. In the pre-study questionnaire, participants were asked whether they had previous knowledge of GFL and participants could opt for “yes”, “little”, and “no”. While every participant in the study had at least a little experience, no more fine-grained assessment of their knowledge was carried out. During the interviews, it clearly emerged that some participants had greater awareness of

GFL use and issues. In future research, an assessment of GFL “skills” could reveal whether they have an impact on translation and PE times, effort, and target text quality.

Interviews suggest a preference for gender-inclusive characters, which is considered a relatively straightforward approach. Nevertheless, participants commented that the frequent use of characters may impact the target text readability. Opinions concerning gender-neutral rewording were more contrasting because finding gender-neutral alternatives to commonly used terms is seen both as a strength and a limitation of this approach. One can use the existing language inventory without introducing disruptive forms but this may also lead to the use of very uncommon terms or sentence structures. This may be the reason for the higher number of pauses and the lower number of searches in the first assignment. Participants commented that the use of neosystems was cognitively demanding, comparing them to a foreign language whose functioning is rather unclear. Their attitudes towards such systems differed: some were particularly negative, e.g. one described them as “raping the German language”, whereas others were generally positive even though participants argued that they would need further training to understand and apply them correctly.

Finally, the target text annotations could be used for triangulation with the process data. Many strategies are possible in the first assignment, e.g. gender-neutral words, passive constructions, and repetitions of proper names. For instance, participants used several neutral forms to translate the same gender-specific terms, e.g. “Schauspieltalent” (acting talent), “Schauspielstar” (acting star), “Schauspielprofi” (acting professional) to avoid using gender-specific “Schauspieler” (male actor) and “Schauspielerin” (female actor). In addition, they reworded whole text passages, often changing perspectives, abstracting, adding, or omitting information. This reflection and revision of one’s work is among the core competencies of translation professionals and differentiates them from MT systems, i.e. the creativity and the capacity to adapt texts to different cultures, requirements, and specifications.

Participants showed a clear preference for gender star (*) in the second text, although its use was inconsistent among them. During the study, several participants stated that they were unsure about the order of masculine and feminine forms, which may also have impacted the number of changes that were made, in part, to alter the order of gendered forms in articles and pronouns. The translation analysis also shows that sometimes gender star was combined with another character, i.e. slash. This could be a result of changes in language use. Once common gender-inclusive forms included Binnen-I, e.g. *LeserIn* (masculineFeminine noun), and slash, e.g. *Leser/in* (masculine/feminine noun). These forms are now

considered outdated because they only include the male and female genders and are being replaced by other characters. Furthermore, there is no norm concerning their use. This is also an aspect that was discussed in the interviews, with participants expressing the desire and need for rules to ensure systematic use of characters. Finally, the practical implementation of some forms may be difficult, mostly when the word stem changes after adding the feminine suffix to the masculine form, e.g. “Arzt/Ärztin” as in the translation experiment.

The most used neosystems were the Sylvain system and the Ens-forms. The former was selected due to its completeness, the second due to its ease of use. All participants made mistakes in their application, which seems to confirm the increased effort in comparison to the first two assignments. The consensus that this strategy is cognitively challenging was suggested by the target text analysis as well. The number of mistakes made in the application of neosystems was very high, whereas there were fewer, if any, application errors for the strategies in the first two assignments.

The analysis of the translation and PE times as well as screen activity tracking elicited interesting results, but interviews provided an opportunity to discuss the translation and PE process in more detail. They offered valuable insights that help describe the strengths and limitations of the strategies investigated in the present work. These are briefly summarised below.

Gender-neutral rewording requires rethinking the translation of very simple terms. This may be regarded as what translation practice is about: rewriting (Lefevere 1992) or “dire quasi la stessa cosa” (say almost the same thing, Eco 2016). However, constantly avoiding gender-specific terms leads to translation choices that are not always automatic and/or natural-sounding, as argued by the participants who were not satisfied with their translations. Participants stated that target texts may sometimes sound impersonal – e.g. because of the use of passive constructions – and/or may shift meanings. In terms of the translation and PE process, these may be the reasons why they made more pauses in the first text than in the other assignments during the experiment. A relatively high occurrence of changes was also observed, although to a lesser extent than in the other assignments. Furthermore, repetitions of proper names must be repeated to avoid pronouns, which led participants to rewrite whole text passages.

Gender-inclusive characters, especially gender star, are the most common GFL strategy in German-speaking countries. Their use is relatively straightforward since usual masculine and feminine forms are concatenated with a character. However, there is no established norm to do so. Even though participants showed a strong preference for gender-inclusive characters, they would definitely mix it with rewording to avoid a strong negative impact on readability. Indeed, these

two approaches were often combined in the second assignment. This is due to considerations of readability, but it also has to do with the nature of translation where a one-to-one correspondence between source and target language is not always possible and/or desirable. The relatively straightforward combination of masculine and feminine forms may be a limitation of gender-inclusive characters. Results from the interviews confirm that participants explicitly think of masculine and feminine forms of words to combine them with a character. This mindset reinforces the gender binary, while new gender-fair forms may be more inclusive of all genders. Using gender-inclusive characters could thus be just a first step towards more GFL because they are already common to some extent and may therefore be met with less resistance by a broad public.

While neosystems may be the best solution to foster the visibility of non-binary genders because they are new, unseen forms specifically devised for this, they are largely unknown. Only one participant in the experiment used a neosystem in their everyday life. Others had never or hardly heard of them before. In addition, neosystems also interfere with classical grammatical rules, so the number of searches was extremely high during the translation and PE of the third assignment. Participants had to repeatedly search for instructions regarding their use and then ensure the correctness of their translation solutions. Limited knowledge and difficulties/uncertainties also affected the number of changes, which was the highest for all assignments.

7.2 Translation and PE compared

PE was faster than translation in the first and second assignments. This is not surprising since a great body of research on PE already confirms this (e.g. Jia et al. 2019a). However, no statistically significant difference between the two tasks was found in the third assignment. This also seems to suggest that the use of neosystems involves higher effort, thus reducing the advantages in speed brought about by the automated translation. Participants in the PE group felt that MT increased their productivity. According to them, this was due to the good quality of the machine translations despite their biased nature.

Differences in screen activity tracking were less clear-cut. First, they were not fully comparable. As already explained in Chapter 6, the MT outputs contained masculine generics, misgendering, and literal translations of singular *they*. This meant that participants in the PE group had to revise nearly all gender references, something that the colleagues in the translation group did not have to do. Therefore, in a further analysis, the number of changes was excluded from the

total count of the activities performed. Median values seem to suggest higher activity for translation in the first assignment, no differences in the second, and higher activity for PE in the third. Such differences among tasks, however, were not statistically significant.

The retrospective interviews also showed few differences between the tasks. Participants in the PE group showed slightly more negative attitudes towards neosystems. The real difference concerned perceptions of the utility of MT in the assignments. Translators did not generally use MT and would not consider it for texts similar to those in the study. Post-editors felt that, despite the effort to correct biased outputs, they were more productive.

The most interesting differences between tasks emerge in the target text analysis. In the first and second assignment, translators made more mistakes. The most common mistake in the first was using the wrong term “Studierende” to refer to high-school students. In the second, some participants interpreted singular *they* as plural and also made mistakes when translating the challenging term “doctor”. Translators rewrote more often than post-editors in the second assignment, paying attention to readability and target text fluency. This may be due to the influence of the given MT, which constrains creativity and the ability to extensively adapt texts – a common PE issue mentioned by the study participants. Finally, participants from the translation group selected the Ens-forms since they were considered the easiest neosystem. Participants from the PE group selected the Sylvain system because it is the most developed and complete. This difference in the selection may be explained by the fact that post-editors have more capacity to concentrate on gender since they already have a good translation output that must simply be revised and stylistically adapted. Translators have to produce the target text without the support of MT, thus selecting the neosystem which presumably takes less time and effort.

7.3 Implications for translation practice and theory

The results of this study have several implications for both translation practice and theory. The first part of this section focuses on the implications for language professionals, including the context dependency of translation choices, the specialist knowledge required for producing gender-fair translations, the constant training needed to keep up with linguistic change, and the strengths and limitations of MT. The second part focuses on translation theory, exemplifying how this work expands the research object of feminist and queer translation and reinforces the conceptualisation of translation as a process of cross-linguistic, political, ideological, and ethical negotiation.

During the post-study interviews, participants discussed challenges concerning the use of GFL. They stated that translation choices, especially with regard to GFL, are extremely context-dependent as also observed in joint studies (Burtscher et al. 2022, Gromann et al. 2023). Therefore, several factors must be taken into account when translating. These include:

- **The source and target culture:** what are common views on gender? Do they differ from the source to the target culture?
- **The source and target language:** how is gender expressed in the source and target language? Are there differences in grammatical structures? Are there established gender-fair forms? How do these differ from source to target? What are the implications of the use of a specific strategy in contrast with the source language (and culture)?
- **The source and target text:** what is the purpose of the text? Does gender play an important role? Must it be visible? Is the purpose of the target text the same as that of the source?
- **The source and the target audience:** are all genders addressed? Are specific genders addressed? Does or should the text address a broad audience?
- **Accessibility** (connected to the audience): is the strategy comprehensible? To what extent does it affect readability? Can screen-readers pronounce it? Can it be pronounced at all or does it work in the written language only?
- **Referents:** which people are explicitly mentioned in the text? Are they real people or fictional characters? Is it possible to gather information on their pronouns? Is it possible to contact them and ask which pronouns to use in the translation?
- **Clients:** what are their attitudes toward GFL? Do they have preferences for a specific strategy? Are there any strategies that they absolutely reject?
- **Time constraints:** has the translator time to focus on the gender aspect? Are they paid for research, e.g. looking for pronouns used by people mentioned in texts and contacting them to discuss a translation strategy?
- **Resources:** which materials, e.g. instructions regarding GFL, guidelines, translation memories, etc., are available to translators? Do they have contact with non-binary and queer community members to make informed decisions?
- **GFL approach:** what are the advantages and disadvantages of each approach? Which strategies are more common in reference to a non-binary individual? Which strategies impact readability? Which strategies may be less comprehensible? And which are met with greater resistance?

These factors complicate the use of GFL in translation and highlight the impossibility of a one-size-fits-all approach. This is one of the main implications of this study for language professionals. Decisions must take account of all the above aspects. Hence, different approaches and/or strategies may be appropriate since they all have strengths and limitations as described in Section 7.1 and acknowledged by participants during the post-study interviews.

GFL may be regarded as a form of subject-matter expertise due to the large variety of available strategies and their constantly evolving nature as well as the terminology related to gender/sex identities. Such knowledge requires constant training, possibly establishing contacts with interested communities as put forward by Attig (2023). Language evolves rapidly and the acceptability of strategies changes. Those that were once common may now be outdated, e.g. the aforementioned *Binnen-I*, whereas new strategies have been and are being developed. Furthermore, the use of terminology is shifting. Terms such as “transsexual” are pathologising and nowadays considered unacceptable (López 2019). Hence, based on the results of this study and the knowledge gained by the author while working on the present monograph, a tentative best practice model is proposed in the next section. However, more research on this topic is needed and GFL use might change rapidly.

Finally, a major implication concerns the use of technology. While translators’ attitudes towards MT are still quite negative, they improve with the use of technology. PE can be a valuable tool in improving translators’ productivity and, in part, ensuring better translation quality even though the MT outputs are severely biased. Conversely, the value of human translation is demonstrated by bias in MT, and by the translators’ skills in creatively avoiding gendered elements as well as adopting different strategies based on clients’ specifications. Translators can rapidly adapt to linguistic change and even foster it through their work, possibly becoming agents for social change, not only through their translations but also their contacts with clients. For example, they may explain why GFL should be used on websites, in product specifications, and in other text types, showing how inclusion could be important for the client’s business. However, this should not be the only reason for adopting GFL. Furthermore, translators can understand identity terms, e.g. non-binary and transgender, and preserve them in their translations in comparison to MT. They can also produce many different versions of the same texts, adapting them to different requirements including different GFL approaches. To date, this is something MT systems can only accommodate to a very limited extent, if at all (whereas LLMs show promising results).

At a broader level within the discipline and translation theory, this monograph expands the scope of feminist and queer translation research for two rea-

sons. First, it focuses on a largely neglected and marginalised group, namely non-binary individuals. Second, it broadens the analysis of feminist and queer translation strategies to include fields such as audiovisual and news translation, which have received less attention in the literature (Baer & Kaindl 2017a, von Flotow & Kamal 2020). Concurrently, the study's results reinforce the conceptualisation of translation as a process involving cultural, political, and ethical considerations.

The linguistic representation of non-binary individuals has been underexplored in the field of queer translation. Queer translation has traditionally concentrated on sexual orientation, as exemplified by Harvey (1998), who examined the translation of camp talk – a sociolect specific to gay men. It also has addressed gender identity representation, but primarily from a binary perspective. For example, Rose (2016) investigates how to preserve the gender identity of transgender individuals in the translation of French memoirs into English, where both masculine and feminine gender markers are used for the same transgender person.

The analysis of non-binary GFL in this study, through its translation and PE processes, reveals the necessity of breaking traditional grammatical rules and experimenting with language. This underscores how language and translation can function as queer practices, deconstructing and reconfiguring linguistic boundaries by introducing new linguistic elements, such as neopronouns and neomorphemes, to challenge existing binary gender norms and assumptions. As discussed in the interviews, translators must negotiate various co-textual and contextual factors, including the shifts in meaning induced by GFL usage, target audience considerations, client specifications, and personal and societal views.

Given its controversial and politically/ideologically charged nature, GFL can face harsh criticism and resistance (Vergoossen et al. 2020). Additionally, the unfamiliarity with neopronouns and neomorphemes can impact the readability and comprehensibility of texts. The translation process thus becomes more visible due to the lack of equivalence between language structures. While English is generally neutral, German-speaking language professionals must choose a GFL strategy to avoid producing a misgendering translation.

Feminist and queer translation research has traditionally focused more on literary works, with audiovisual products receiving less attention, and news translation being even more neglected. The concept of social responsibility has been explored in translation and interpreting within sensitive contexts such as medicine, law, and migration (Drugan & Tipton 2017). This study illustrates how gender is a highly sensitive area too, where translators bear the responsibility for a respectful representation of marginalised groups. The increased visibility of non-binary individuals is leading to their representation in popular audiovisual and news

media, which necessitates translation across numerous languages. This can potentially foster social change by normalising the notion that gender is not binary and by reducing gender discrimination.

7.4 A tentative best practice model for GFL in translation and PE

Based on the previous considerations and knowledge gathered by the author while writing this monograph, a simplified and tentative decision tree for selecting the most appropriate GFL strategy is proposed in Figure 7.1. This is by all means an initial proposal that could and should be refined, especially after discussing it with members of the queer and non-binary community and language professionals. The decision tree is discussed below in the context of translations from English into German, thus further commenting on the implications of the results from this monograph.

An important criterion when deciding if and how to use GFL is whether non-binary individuals are referenced. While in the present work it is argued that masculine generics represent a sexist use of language and should be avoided, GFL must actively be used in reference to non-binary people. This is not a question of personal preference and/or ideology. Intentionally assigning the wrong linguistic gender to a person is an act of identity invalidation and misgendering that can have serious negative impacts on the mental health of trans and non-binary people (McLemore 2018).

When a non-binary person is not directly mentioned in the source text, one should consider whether gender plays an important role in the target text. If not, for instance in product instructions, translators could avoid masculine generics by using gender-neutral rewording. If the gender element is important in a text, the strategy selection is dependent, amongst others, on the target audience. In an administrative text, addressing a wide audience, gender-neutral rewording ensures that the text is highly comprehensible and readable. The use of this strategy may also go unnoticed and hence reduce or eliminate perceived ideology in the text.

Texts written using this approach may sound more abstract and/or impersonal, which might not be a desirable solution if the purpose of the source text is to promote gender diversity or to depict non-binary individuals. In this case, other approaches such as gender-inclusive characters and/or neosystems might be more appropriate. Translators could explain the rationale for the strategy selection as a form of awareness raising. Gender star was the most selected gender-inclusive

7.4 A tentative best practice model for GFL in translation and PE

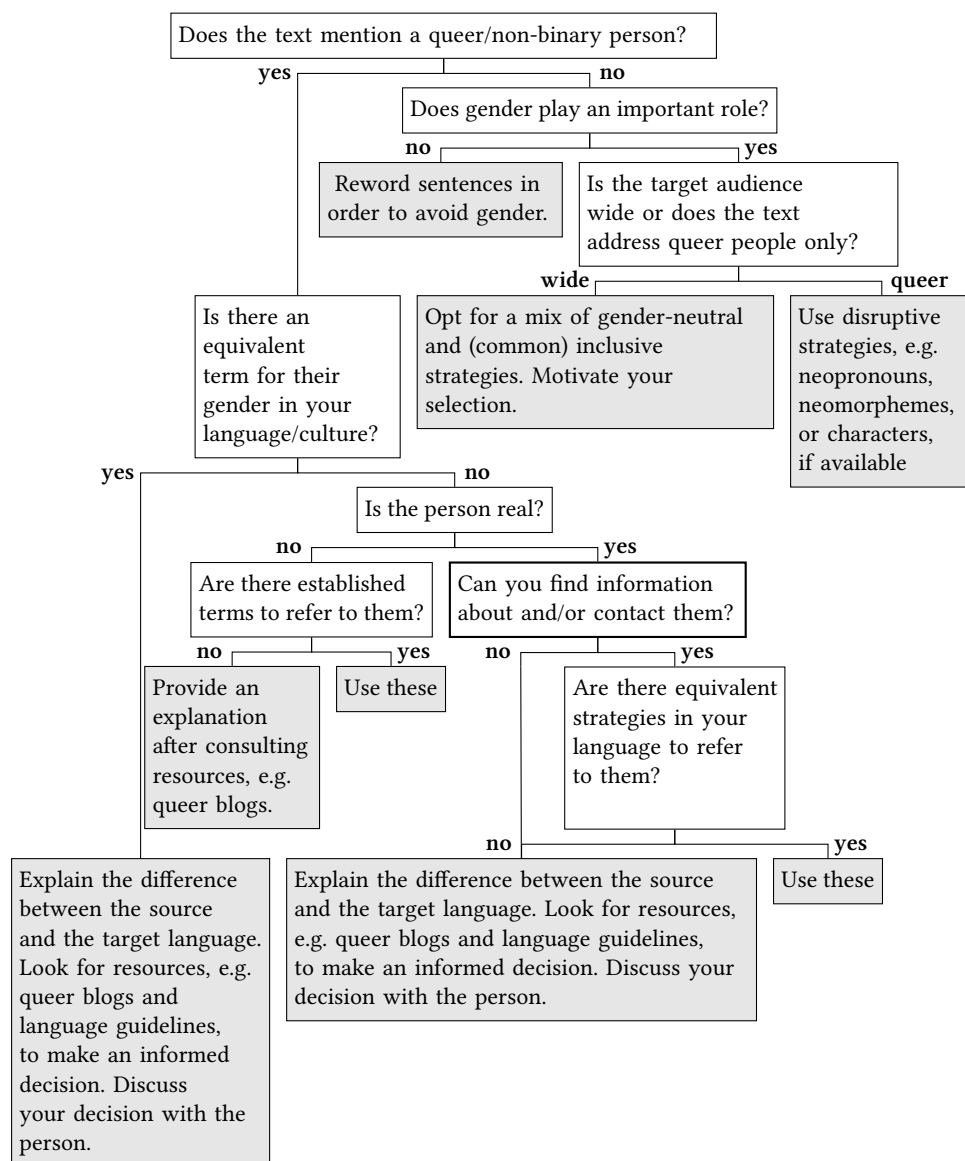


Figure 7.1: Decision tree for the selection of the appropriate GFL strategy

character strategy in the present work, so its use is highly recommended. While combined forms, e.g. “die*r” are a viable solution, these are not always possible, e.g. in the case of possessive pronouns “sein*ihr” (his*hers). If consistency is desired, masculine forms should always be used first, e.g. “der*die Schauspieler*in”, (masculine*feminine article + masculine*feminine noun). Otherwise, the use of feminine articles and pronouns first is recommended as a strategy against androcentrism in language. So, if the word stem changes when adding the feminine suffix “-in” as in “Arzt/Ärztin”, the feminine word stem should be preferred as in “Ärzt*in”. It is also important to stress that gender-inclusive characters should be sparsely used in order not to impact reading flow and readability. A mix of these two approaches was also indicated as the preferred solution by the participants in this study.

In texts written for a queer audience and/or for queer activism purposes, more disruptive strategies, such as neosystems, could be actively used to linguistically include all genders and to question the gender binary. There are many different neosystems, and the selection could be based on their completeness, e.g. the Sylvain system, or inherent ease of use, e.g. the Ens-forms. If possible, it would be best to decide after discussing with members of the queer/non-binary community since other systems may be gaining popularity and/or may be preferred. To the best of my knowledge, neosystems are extremely rare in German-speaking countries and non-binary individuals normally use different neopronouns, e.g. *nin* or *xier*, and gender-inclusive characters to talk and write about themselves.

If a specific person is directly mentioned in the source text, the first important step is to use the correct terminology. The translated identity terms must be current and respectful. If this is a difficult decision, there are plenty of resources at hand, e.g. queer blogs or language guidelines such as APA Style (2019a). If no equivalent terminology is found after consultation, an explanation can be provided together with a tentative translation. In German, English language terms are often borrowed.

If the non-binary person mentioned in the text is real, information on their pronouns could be searched in order to decide on one or more strategies. If no equivalents are found, a good solution would be to explain the difference between the source and the target language, e.g. explaining how the pronoun *they* can be used in the singular to refer to a specific person whereas this is not possible in German, and select a strategy after consulting resources. If possible, the non-binary person mentioned in the text should also be contacted before making a decision. As already mentioned, in German-speaking countries non-binary individuals often use different neopronouns and gender-inclusive characters to talk and write about themselves. This could be a good solution. If neosystems are

used, the translation should be proofread by at least one other person with good knowledge of non-binary GFL in order to avoid mistakes which, in the present study, were very common.

In certain cases, like audiovisual translation and app localisation, there are character limits. Gender-neutral rewording and/or the use of gender-inclusive characters may not always be possible. To overcome these obstacles and avoid misgendering and/or uncommon linguistic strategies, language professionals must be creative. In addition, some markup languages might interpret an asterisk as a command to print the following text in italics. Further research is needed on GFL that takes into account multi-modality and the challenges of audiovisual translation or localisation.

7.5 Implications for MT research

The results of the present study have important implications for further research in MT debiasing and training. These include shifting away from the current one-to-one paradigm in which a single translation is produced for one source text, the need for personalisation, the creation of gender-fair datasets, and constant (re)training as well as the need for participatory research.

Recently, the one-to-one paradigm of MT has been shifted to a one-to-many approach where two or more translations are produced for one single source text. For instance, Google Translate started to show the feminine and masculine translation for gender-ambiguous sentences in 2018 (Kuczmarski 2018) whereas DeepL (2023) allows users to select the tone (informal/formal) of the translation in some languages. This should be similarly implemented for GFL, thus showing different outputs, e.g. a gender-neutral and a gender-inclusive version of the target text.

In order to do so, however, MT systems should be able to recognise terms like singular *they* as such when translating from English into other languages. As seen in this study, singular *they* was always translated as the third-person plural pronoun “sie” in German, thus misrecognising the gender of the referent. Plugins that allow for context disambiguation have been introduced (e.g. Měchura 2022) but still do not work well for non-binary genders and very few studies tried debiasing beyond the binary (Saunders et al. 2020). The lack of gender-fair datasets that could be used during training remains a major problem. Currently, bilingual corpora used for MT training are often composed of speeches from the European Parliament, and analyses of these materials show that most of them are given by men, with only about a third by women (Vanmassenhove et al. 2018). Non-binary people are extremely underrepresented in the training data.

While rewriting models have been proposed to implement GFL in MT systems, these are currently limited to certain strategies and their outputs still need corrections (Amrhein et al. 2023). Another problem of such systems is particular gender-specific terms as shown in the present study. Some nouns are difficult to translate because there is no established gender-neutral alternative or, as in the case of “nanny”, they generally have the masculine or the feminine form only. Therefore, MTPE may at least for the moment be the best solution to produce gender-fair datasets which are inclusive of genders beyond the binary or entirely neutral regarding gender references. Finally, rewriting models reduce representation harms but not the allocative ones: women and non-binary people would still need to generate an automatic translation and then use such models to produce a gender-fair version of the MT outputs, thus making MT less accessible for them.

Another major challenge for MT research is linguistic change. As already mentioned, conceptions of what constitutes GFL shift, and strategies or terms that once were common are no longer used. Now that non-binary people are more visible, many GFL strategies are being proposed and/or redefined. MT systems would thus need constant updates. The present design and training procedures of MT systems make updating existing models quite cost- and resource-intensive and have a considerable carbon footprint (Shterionov & Vanmassenhove 2023). Future research should thus prioritise more efficient methods for training and updating an MT model. Besides, non-binary people use different pronouns and different strategies to talk about themselves. MT systems should allow for personalisation, enabling users to select their preferred strategy.

The above points also require a shift in research on gender (de)bias(ing) in MT. Participatory research approaches (Burtscher et al. 2022, Gromann et al. 2023) are needed to include communities most affected by the use of GFL or lack thereof, i.e. queer and non-binary individuals. When it comes to marginalised communities, their exclusion from decision-making processes could be regarded as a form of epistemic violence (Spivak 1988). Failing to include non-binary people in deciding which GFL strategies are implemented further silences them, depriving them from their right to linguistic self-determination. Also, language professionals who might have different insights into strategy feasibility and comprehensibility should be involved in research to ensure a beneficial exchange between different stakeholders.

7.6 Research process: Positionality and study limitations

The research process involves constant decision-making from the topic selection and the research question to the data collection and analysis. I would like to first describe my positionality because it has an impact, both conscious and unconscious, on the decisions made regarding study design and data analysis. Finally, I will discuss the limitations of the present work.

The use of GFL is often regarded as political and ideological. It also relates to one's beliefs concerning gender and language, mostly linguistic conservatism, e.g. stances on preserving the language. While some publications describe common arguments used against GFL (e.g. Vergoossen et al. 2020), the objective of the present work was to investigate how language professionals apply it. GFL can be regarded as a language variety and as such it is of interest for language professionals. Besides, since the visibility of non-binary people is increasing, translators are and will be increasingly confronted with texts in which non-binary people are mentioned, e.g. news reports, TV series, and literary fiction. Hence, understanding the implications and the process of gender-fair translation is of utmost importance for TS and MT.

In the present work, there is no intention to discuss GFL in a politically and ideologically motivated manner. Nevertheless, it is a fact that refusing to respect a person's gender identity constitutes misgendering and this negatively affects the mental health of transgender people. If misgendering is a form of discrimination and epistemic violence, GFL use is a form of respect. I would argue that refusing to use GFL and intentionally misgendering a person is actually more ideological than the contrary. Language is an instrument to actively construct, negotiate, and represent human identity and social reality, and intentional misgendering aims to invalidate a person's identity.

This view of language reflects my positionality as a researcher and as such my work is politically and ideologically motivated. I refuse essentialist views of gender as innate and biologically determined. I embrace the constructionist view, recognising that gender as well as research are historical, cultural, and social constructs which nevertheless have an impact on how people are perceived and in turn may (or may not) experience forms of discrimination. I also recognise how language is the result of social conventions and how it (re)produces social norms and assumptions. These norms and assumptions, e.g. the gender binary, can also be challenged and changed through language, of course.

The selected method comes with strengths and limitations. To the best of my knowledge, methods from TPR have not yet been applied to research on GFL. As discussed in Lardelli & Gromann (2023a), most of the studies on GFL translation

consist of analyses of translated audiovisual products in which non-binary individuals are featured. Such analyses focus on the use of GFL or lack thereof (López 2022 and Misiek 2020). In some cases, non-binary participants gave interviews and were asked which language strategies they use and/or would like to find in translations (e.g. Šincek 2020). Very little research addresses the translation process, generally involving students who compile questionnaires on the perceived difficulties of different translation tasks that require the use of GFL strategies (Bailach Adsuaara 2021). The selected methodology allows the collection of an interesting and especially multi-faceted dataset, proving this study to be among the few that provide in-depth results on this area of research.

As is common in TPR, data were collected and triangulated using a mixed-method approach. Data from the retrospective interviews, observational protocols and segment-based target text annotation were mainly textual and statistical analyses were performed to summarise the results. Screen recordings allowed translation times to be measured and screen activity tracking data to be analysed. The methodologies adopted allow the analysis of the collected data from different perspectives, ensuring a systematic and valid approach to empirical research.

The selected research design and methodology are also motivated by experimental realism and the sensitivity of the topic. A great variety of studies in the field of TPR adopt keylogging to analyse keyboard activity and eye-tracking to gain insights into the cognitive load. These methodologies are more precise and objective, but they require participants to be in a laboratory and hence impact ecological validity (O'Brien 2009). Study candidates are required to work in a foreign environment with computers and software they are not used to. In eye-tracking studies, participants must always keep a certain distance from the screen for data to be correctly collected. Also, the laboratory setting limits the recruitment of translators to a specific place and prevents researchers from obtaining a larger sample. In this study, it was hard to find the desired number of participants, especially those from marginalised groups such as non-binary individuals. This difficulty is common to research in the field (Burtscher et al. 2022) and reinforced the decision to conduct the translation experiment online.

The analysis of translation and PE times as well as screen activity tracking delivered interesting results and insights into the gender-fair translation and PE process. However, asking participants to work with texts was fundamental to gaining deeper insights into their experience and/or point of view, which was discussed during retrospective interviews. The point of view of professional translators concerning the use of different GFL strategies was of utmost importance because of their cross- and meta-linguistic competence. Besides this, their knowledge is important because translation can change social relations and increase

empowerment. A strict experimental setting was therefore beyond the scope of the present work and many variables were not controlled. As such, the results from this study cannot be generalised.

Furthermore, full experimental realism was not possible despite careful planning because (i) translators were required to work on a single screen in order for the whole translation and PE process to be recorded, and (ii) they received strict requirements concerning GFL, i.e. the use of a single strategy for each text. Limitations such as (i) are usual in similar research settings whereas (ii) pertains to the study objective, namely to investigate the translation and PE process of different strategies and so to allow translators to make informed decisions on how to translate GFL.

The major limitation of the current study is, as explained, the lack of result generalisability. This limitation is common to experimental studies in TPR because funding constraints make it difficult to recruit enough participants, and the triangulation of different methods causes data explosion (O'Brien 2009). Even more so for a PhD project since it has to be carried out in a limited time by a researcher only with the support of their supervisors. As already mentioned, the number of study candidates was selected by reviewing critical literature in the field of TPR (e.g. O'Brien 2009) that illustrates how results obtained from a small number of 12 participants may have significant value although they lack generalisability. Furthermore, the sample of participants, though small, was relatively heterogeneous. Ideally, the group would be more homogeneous in order to control variables such as typing speed and work experience. Also, other languages should be investigated, possibly without English in the combination.

Another limitation concerns the comparability of the texts and hence of the cognitive processes analysed. To ensure comparability of estimated translation and PE times, texts used in the experiment were controlled for length, i.e. each was about 150 words, and, in part, for difficulty by measuring their readability with the Flesch-Kincaid readability test (Kincaid et al. 1975) as in Rojo López & Caro (2016). The test takes into account the number and length of words but ignores semantics. This issue was partially addressed by asking for translators' opinions on the source texts' difficulty. In an ideal scenario, the same text could have been translated or post-edited with different strategies by more language professionals. This would require a higher number of participants than available within the scope of this study. In addition, the assignment order was not changed among participants and the most challenging approach to GFL, i.e. neosystems, was applied in the third text. Longer translation and PE times could hence be caused not only by the specific strategy applied but also by participants' tiredness. Furthermore, the impact of participant familiarity with the texts and po-

tential learning effects on assignments was not considered. Although the author did not specifically ask participants during the interview whether they were familiar with the TV series described in the assignments, most of them revealed that they were not. In addition, gender references in each text slightly differed in their number (nine, ten, and twelve respectively) and difficulty, (e.g. “nanny” in the third text for which German equivalents are usually of feminine gender). Some terms caused difficulties because there is no equivalent or they are not so common in German (e.g. “recurring guest star”). In replicating the study, terminology from the field of cinema and TV series could be provided so as to distinguish more clearly where the effort lies, i.e. in translating a term or in using a gender-fair strategy. Finally, translation and PE screen activity tracking can only be compared if changes are not counted. Screen recordings are of interest though, because they show which gender-specific terms are particularly challenging for translators and possibly for MT debiasing.

Although the study focuses on the translation and PE of GFL beyond the binary, only one non-binary translator participated in the study. To ensure their inclusion in the study, non-binary individuals were directly contacted. While numerous people were found, they had no translation background or translation experience. They were therefore excluded from the study. This may be considered as a major limitation of the present work. However, its scope is not to prescribe the use of a specific GFL strategy. The main objectives are to further expand the field of queer TS and also understand the language professionals’ perceptions and challenges in the use of GFL to better support them in their work.

People who decided to participate in the study had usually positive attitudes toward GFL whilst others were interested in the topic because it is becoming increasingly relevant to their work life. This may have had an impact on the results. Effort may be higher in translators who do not or refuse to use GFL. This is quite plausible since the only translator in the study who did not use GFL in their work life was also one of the participants who faced the highest effort.

The identification of gender phrases for the target text analysis was particularly challenging. While they can be easily identified in the source text, there is no one-to-one correspondence among languages and translation. This may also be one of the reasons why participants used rewording in the second and third assignments as well. During the annotation process, the author read the target texts and identified chunks containing GFL. In this work, the term *gender phrase* was used as it refers to a group of words standing together as a conceptual unit. A similar concept, *translation unit*, is often used in TPR. Since its conceptualisation is widely debated (e.g. Carl & Kay 2011), it was not adopted in the present study. The process of deciding which grouping of words constituted a gendered

phrase is therefore subjective and other researchers could opt for a different categorisation.

As discussed in Subsection 3.1.1, the present work was developed prior to the rise in interest on LLMs, hence the potential of generative AI for personalised, bias-mitigated translation was not addressed and this study only addresses limitations of “classical” NMT systems.

In a nutshell, though the present study has several limitations, it is a first step towards including non-binary language in (queer) TS. A similar setup can be used in future research to replicate this work, possibly with a larger sample of participants, and to investigate the gender-fair translation and PE process in other languages. Finally, other methodologies such as keylogging and eye-tracking could be used in order to gather more objective and precise data on cognitive processes.

The minimal and not statistically significant difference in translation and PE times among assignments may also indicate that language professionals can adapt to different requirements such as the use of a specific language variant without an impact on their speed and productivity. Experienced professionals are expected to have this skill, but it would need to be tested explicitly by translating with and without gendering. However, the resulting cognitive effort seems relatively high and future research could and should clarify whether this has an impact on the general quality of the target texts. Other populations, e.g. students, might produce different results.

Finally, it would be interesting to test the acceptability of the strategies in end users, e.g. studies with readers, especially focusing on those who are most impacted by the use of GFL, i.e. non-binary individuals. Studies on the readability and comprehensibility of new gender-fair forms are needed as well. A major complication in this respect is the lack of a standard, widely adopted system. Language is constantly evolving and minoritised groups do not necessarily agree on a specific set of forms. Moreover, many people do not know what GFL is and why it is being used. Greater societal awareness is needed for gender-fair forms to be more widely understood and used.

8 Conclusion

Gender is an extremely complex phenomenon in language and translation because of its interplay with both linguistic structures, e.g. word inflection and agreement, and the extralinguistic reality, i.e. the gender identity of a referent as well as societal norms and assumptions. Since linguistic structures as well as views on gender are language-dependent, translating gender is not a trivial undertaking – for humans or machines. While issues of translating gender have been investigated both within TS and in MT, little research has tackled them beyond a binary conception of gender.

The present work thus focused on GFL as a means to refer to and represent non-binary individuals in grammatical gender languages such as German, where only the masculine and feminine gender are used in reference to people. Due to the linguistic and cultural influence of English, translators are increasingly confronted with texts depicting non-binary referents (e.g. TV series such as *One Day at a Time*). While gender neutrality may be quite easily reached in English with the pronoun *they* used for referents in the singular and gender-neutral alternatives to common nouns, this is not the case in German. Different strategies have been proposed which can be subsumed under the term gender-fair and include gender-inclusive and gender-neutral approaches.

This wealth of linguistic strategies is challenging both in human and MT since there is no commonly acknowledged standard for GFL, and strategy selection depends on a variety of factors. For this reason, the present work investigated the translation and PE of GFL beyond the binary in a study that was both process- and product-oriented. Twelve professional translators were recruited and asked to translate or to post-edit three English-language texts into German. The source texts were news reports about non-binary actors joining the cast of various TV series. For each text, a different GFL approach, i.e. (i) gender-neutral rewording, (ii) gender-inclusive characters, and (iii) neosystems, was to be used. To prepare for the study, participants received a handout on GFL, which could also be used during the translation and PE process.

Participants joined a video conference, which was open in the background of their computers and their screen was recorded. They were asked to complete their assignments and then they were interviewed. Cognitive processes were

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analysed by focusing on translation and PE times, screen activity tracking data, and the personal experiences of the study participants. Such data were elicited by means of non-participant observation, screen recordings, and retrospective interviews. Product data were analysed by annotating target texts for both the type of GFL strategy used as well as the success of its integration in the translation and PE process.

Neosystems were found to be the most difficult approach to GFL. While differences in translation and PE times were not statistically significant, screen activity tracking data (e.g. the number of searches and changes) and mistakes considerably increased when neosystems were used. Neosystems are still very rare and the study participants felt like they had to learn a foreign language, often using the provided handout to be able to use neosystems.

Participants expressed a clear preference for gender-inclusive characters, especially the gender star, combined with gender-neutral rewording. Gender-inclusive characters are a quite straightforward approach to reach gender fairness in language and their use is quite common. However, they might impact readability if frequently used within the same text. In a similar case, gender-neutral rewording is deemed a good alternative. Neutral formulations should be used carefully in reference to a single person because the target texts might sound quite unnatural or impersonal.

The target text analysis shows that, amongst other things, gender-inclusive characters are applied differently, with language professionals wishing for clearer rules and/or standards for their use. Participants are very creative in gender-neutral rewording, often reformulating whole phrases to avoid gender. When using neosystems, participants often made mistakes, suggesting higher cognitive load and the need for further training to correctly apply them.

PE was significantly faster than translation in the first two assignments. When using neosystems, the speed difference between tasks was not statistically significant. This seems to suggest that the higher cognitive load experienced because of the use of neosystems reduced the advantages in speed brought about by the automated translation. Participants who received the PE task also made fewer mistakes. It was also found that some gender-specific terms are more challenging than others and therefore the application of a strategy is not always straightforward or possible.

The present study has important implications for both TS and professionals. A single source text might be translated differently, depending on a large variety of factors. Such factors may include the client's specifications, target audience and culture as well as personal and societal beliefs. This also influences the selection of a specific GFL strategy, since the study shows how different strategies

carry different implications. The result is that a one-size-fits-all approach to the representation of non-binary individuals is neither possible nor desirable.

GFL is constantly evolving and there are neither norms nor consensus on which and how a specific strategy is to be used; constant training and contact with the queer and non-binary community are needed. This also applies to the technology design process in which relevant stakeholders, e.g. translation professionals and non-binary people, need to be involved. Finally, current MT paradigms need to shift to allow one-to-many translations, personalisation, and strategy selection. Theories and perspectives from TS study might therefore inform the strategy selection as translation scholars and professionals are best equipped to understand the connotations of different GFL strategies in cross-linguistic scenarios.

Finally, non-binary people are still a marginalised group exposed to daily discrimination. The translation of GFL is therefore a highly sensitive area, where translators bear the responsibility for a respectful representation of marginalised groups. However, since GFL is still controversial and politically/ideologically charged, translation is best conceptualised as a process of cross-linguistic, cultural, political, and ideological negotiation.

The present study is a first step towards tackling issues of non-binary representation in languages with grammatical gender. Future studies should involve a higher number of participants and also apply more precise and objective methodologies such as keylogging and eye tracking. A greater involvement of non-binary people in the research process would also be beneficial. Furthermore, future studies should investigate GFL in other language pairs, also without English as the source language and culture. Psycholinguistic studies should also shed light on the readability and comprehensibility of GFL for different groups. Finally, further exchanges on the topic between TS and MT might be beneficial to both fields, promoting multidisciplinary collaboration which should also extend to other disciplines such as sociolinguistics, human-computer interaction, philosophy (e.g. linguistic injustice), ethics and law (e.g. legal definition of discrimination and bias).

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Gender-fair language in translation and post-editing

Machine Translation (MT) models knowingly suffer from gender bias, especially for genders beyond the binary. Since issues of non-binary representation and language use are still often neglected in both Translation Studies (TS) and MT, this book investigates the translation and post-editing of gender-fair language beyond the binary in a process and product-oriented study. Twelve language professionals were recruited and asked to either translate or post-edit three brief English texts into German. In each text, they had to use a different gender-fair language approach, i.e. (i) gender-neutral rewording, (ii) gender-inclusive characters, and (iii) neosystems.

Data on the translation and post-editing process were collected by combining non-participant observation, screen recordings, and interviews. The produced target texts were annotated to elicit product data. The focus of the study was not on translation quality but rather on the ease and success of integrating gender-fair language into the translation and post-editing process. Findings show that post-editing is generally faster than translation, and that the use of neosystems increases screen activity as well as perceived cognitive effort, which in turn reduces success in using gender-fair language. Finally, participants indicated a clear preference for a combination of gender-neutral and gender-inclusive strategie